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M.Sc Electrodynamics

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The current maintainer of this work along with the primary lecturer

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Part I

Prologue

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But still try, for who knows what is possible . . .

(Michael Faraday, in The Life and Letters of Faraday (1870) Vol. II, edited by Henry Bence Jones, p. 483)

Chapter 1

The Purpose of Electromagnetism

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1.1 Mechanics in Four Different Views

Newtonian mechanics is generally sufficient enough for most **everyday** applications ranging from calculating the motion an object falling from a building to predicting the motion of planets. When objects start to move at high speeds,[1] however, the *Newtonian model* fails to predict accurately and was replaced by special relativity, which was introduced by *Einstein* in 1905 [1]. For objects which are **extremely small**, sizes comparable to that of atoms, then the theory of relativity fails for different reasons, and is superseded by quantum mechanics.[2]

[1] as in speeds comparable to the speed of light.

[2] Developed by Bohr, Schrodinger, Heisenberg, and many others, in the 1920's, mostly.

Finally, for objects which are both very fast and very small,[3] a mechanics which combines both relativity and quantum principles is needed:

> This relativistic quantum mechanics is known as **quantum field theory**. Even today this new model cannot claim to be a completely satisfactory system.

[3] As is common in modern particle physics.

Luckily for us, in **M.Sc Electrodynamics**, we will **NOT** look at these models in detail but look at a specific subset of it called **electrodynamics** which is generally in the domain of classical mechanics.[4]

[4] which can be extended to other domains if needed.

1.1.1 Electromagnetism and Relativity

Interestingly, electromagnetism was one of the primary catalyst for developing **general relativity**.

Historically, the *Maxwell equations* were discovered before the theory of special relativity. It was thought light waves we derived above must be oscillations of some substance which fills all of space. This was dubbed the aether.[5] The idea was that Maxwell's equations only hold in the frame in which the aether is at rest; light should then travel at speed c relative to the aether.

[5] the postulated medium for the propagation of light [2]. It was originally used to explain the ability of the apparently wave-based light to propagate through empty space, something waves should **NOT** be able to do. The assumption of a spatial plenum of luminiferous aether, rather than a spatial vacuum, provided the theoretical medium that was required by wave theories of light.

We now know that the **concept of the aether is unnecessary**. Instead, *Maxwell's equations* hold in all inertial frames and are the first equations of physics which are consistent with the laws of special relativity. Ultimately, it was by studying the Maxwell equations that Lorentz was able to determine the form of the Lorentz transformations which subsequently laid the foundation for Einstein's vision of space and time [3].

1.1.2 Four Kinds of Forces

Classical Mechanics tells us how behavior is when subjected to a given force. There are four (**4**) basic forces known to physics. Listing them in the order of decreasing strength:

Strong

Holds protons and neutrons together in the atomic nucleus, have extremely short range, so we do not "feel" them, in spite of the fact that they are a hundred times more powerful than electrical forces.

Weak

Accounts for certain kinds of radioactive decay, are also of short range, and they are far weaker than electromagnetic forces.

Gravitational

it is very weak[6] that it is only in scale of huge mass concentrations, for example the earth and the sun, that we ever notice it at all.

[6] compared to all of the others

Electromagnetic

It is the interaction which occurs between particles with electric charge through electromagnetic fields.

> For reference, the electrical repulsion between two (2) electrons is 1×10^{42} times as large as their gravitational attraction.

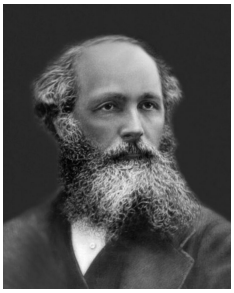
Not only are electromagnetic forces overwhelmingly dominant in everyday life, they are also, the only ones which are completely understood.

There is, of course, a classical theory of gravity and a relativistic one, but no entirely satisfactory quantum mechanical theory of gravity has been constructed. At the present time there is a very successful theory for the weak interactions, and a strikingly attractive candidate, called chromodynamics, for the strong interactions.

Traits	Strong Force	Electromagnetism	Weak Force	Gravity
Affected Particles	quarks[1] and gluons[2]	electrically charged	quarks and leptons[3]	all particles with mass
Force Carrying Particle	gluon (g)	photon[4] (γ)	W and Z bosons (W^+, W^-, Z^0)	graviton (unobserved)
Acting Range	short ($\sim 1\,\text{fm}$)[5]	∞	short	∞
Strength	1	$1/137$[6]	1×10^{-6}	6×10^{-39}

[1] Elementary particle responsible for making **protons** and **neutron**.
[2] Elementary particle acting as exchange particle for the **strong force** between **quarks**.
[3] Elementary particle affected by the **weak force** but not by the **strong force**.
[4] Elementary particle that is a **quantum** of the **electromagnetic** field, such as light and radio waves.
[5] 1 femtometre ($1\,\text{fm} = 1 \times 10^{15}\,\text{m}$). i.e., the gold nucleus radius is approx. 8,45 fm.
[6] Known as the **fine structure constant.**

Table 1.1
The four fundamental forces and their respective properties.



[7] **James Clerk Maxwell**
FRS FRSE
(1831 - 1879)

A Scottish physicist and mathematician who was responsible for the classical theory of electromagnetic radiation, which was the first theory to describe electricity, magnetism and light as different manifestations of the same phenomenon. Maxwell's equations for electromagnetism achieved the second great unification in physics, where the first one had been realised by Isaac Newton. Maxwell was also key in the creation of statistical mechanics.

With the publication of *A Dynamical Theory of the Electromagnetic Field* in 1865, Maxwell demonstrated electric and magnetic fields travel through space as waves moving at the speed of light. He proposed light is an undulation in the same medium that is the cause of electric and magnetic phenomena. The unification of light and electrical phenomena led to his prediction of radio waves. As a result of his equations, and other contributions such as introducing an effective method to deal with network problems and linear conductors, he is regarded as a founder of the modern field of electrical engineering.

All these theories draw their inspiration from electrodynamics. None can claim conclusive experimental verification at this stage. So electrodynamics, a complete and successful theory, has become a pedestal for physicists:

> An ideal model that other theories emulate.

The laws of classical electrodynamics were discovered in bits and pieces by Franklin, Coulomb, Ampere, Faraday, and others, but the person who completed the job, and packaged it all in the compact and consistent form it has today, was James Clerk Maxwell.[7] The theory is now about 150 years old.

1.1.3 Unifying Physical Theories

In the beginning, electricity and magnetism were entirely separate subjects. The one dealt with glass rods and cat's fur, batteries, currents, electrolysis, and lightning, whereas the other with bar magnets, iron filings, compass needles, and the North Pole. However, in 1820 Ørsted noticed that an electric current could deflect a magnetic compass needle [4]. Soon afterwards, Ampére correctly postulated all magnetic phenomena are due to electric charges in motion. Then, in 1831, Faraday discovered that a moving magnet generates an electric current [5]. By the time Maxwell and Lorentz put the finishing touches on the theory, electricity and magnetism were inextricably intertwined.

They could no longer be regarded as separate subjects, but rather as two aspects of a single subject: electromagnetism.

Faraday speculated that light, too, is electrical in nature [6]. Maxwell's theory provided justification for this hypothesis, and soon optics, which is the study of lenses, mirrors, prisms, interference, and diffraction, was incorporated into electromagnetism. Hertz, who presented the decisive experimental confirmation for Maxwell's theory in 1888, said [7]:

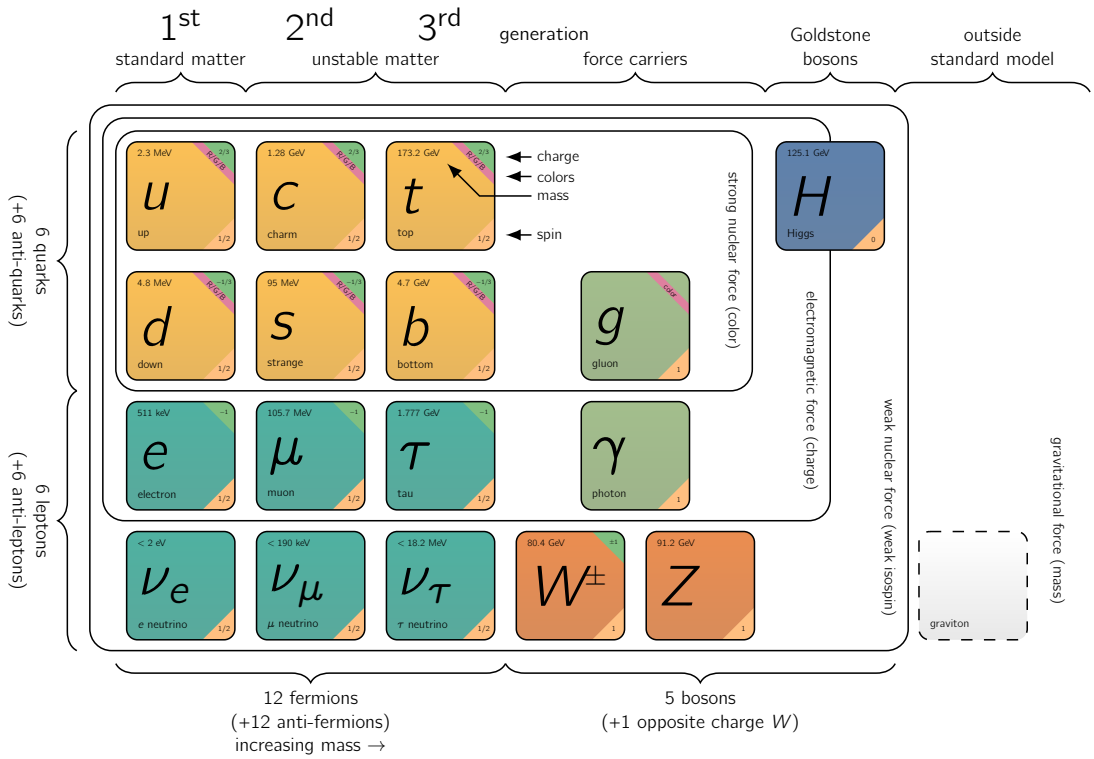


Figure 1.1 **The Standard Model of particle physics is the theory describing three of the four known fundamental forces (electromagnetic, weak and strong interactions - excluding gravity) in the universe and classifying all known elementary particles. It was developed in stages throughout the latter half of the 20th century, through the work of many scientists worldwide, with the current formulation being finalized in the mid-1970s upon experimental confirmation of the existence of quarks.**

> The connection between light and electricity is now established . . . In every flame, in every luminous particle, we see an electrical process . . . Thus, the domain of electricity extends over the whole of nature. It even affects ourselves intimately: we perceive that we possess . . . an electrical organ-the eye.

By 1900, three (**3**) great branches of physics: **electricity**, **magnetism**, and **optics** had merged into a single unified theory.

> It became apparent that visible light represents only a tiny "window" in the vast spectrum of electromagnetic radiation, from radio through microwaves, infrared and ultraviolet . . .

Einstein worked on a further unification, which would combine gravity and electrodynamics, in much the same way as electricity and magnetism had been combined a century earlier. His unified field theory was not particularly successful, but in recent years the same impulse has spawned a hierarchy of increasingly ambitious unification schemes, beginning in the 1960s with the electroweak theory of Glashow, Weinberg, and Salam, which joins the weak and electromagnetic forces [8], and culminating in the 1980s with the superstring theory.[8]

At each step in this hierarchy, the mathematical difficulties mount, and the gap between inspired conjecture and experimental test widens; nevertheless, it is clear the unification of forces initiated by electrodynamics has become a major theme in the progress of physics.

[8] an attempt to explain all of the particles and fundamental forces of nature in one theory by modeling them as vibrations of tiny supersymmetric strings.

1.1.4 The Fields of Electrodynamics

The essential problem the theory of electromagnetism hopes to solve is:

> If there exists a bunch of electric charges here, what happens to some other charge, over another place?

The classical solution takes the form of a **field theory**:

We say that the space around an electric charge is permeated by **electric** and **magnetic** fields. A second charge, in the presence of these fields, experiences a force. The fields, then, transmit the influence from one charge to the other-they "mediate" the interaction.

When a charge undergoes acceleration, a portion of the field **detaches** itself, in sense, and travels off at the speed of light, carrying with it energy, momentum, and angular momentum. We call this **electromagnetic radiation**. Its existence invites[9] us to regard the fields as independent dynamical entities in their own right, every bit as **real** as atoms or baseballs.

[9] if not compels

Our interest accordingly shifts from the study of forces between charges to the theory of the fields themselves. But it takes a charge to produce an electromagnetic field, and it takes another charge to detect one, so we had best begin by reviewing the essential properties of electric charge.

1.1.5 Electric Charge

Charge comes in two varieties

We call **plus** and **minus**, as their effects tend to cancel.[10] The interesting point here is that plus and minus charges occur in exactly equal amounts, to high precision, in bulk matter, so that their effects are almost completely neutralised.

[10] If we have +q and q at the same point, electrically it is the same as having no charge there at all.

Charge is conserved

Charge cannot be created or destroyed [9]. What there is now has always been. A plus charge can **annihilate** an equal minus charge, but a plus charge cannot simply disappear by itself, something must pick up that electric charge. So the total charge of the universe is fixed for all time.

This is called **global conservation of charge**. Charges also cannot disappear and then reappear in some other place, the charge, if moved must follow a continuous path. This is called **local conservation of charge**.

Charge is quantised

While there is no restrictions in classical electrodynamics which require it to be, electric charge comes only in **integer multiples** of the basic unit of charge. If we call the charge on the proton $+e$, then the electron carries charge e.[11]

[11] It is never $7.392e$, or $1/2e$.

1.1.6 Interpreting Conservation of Charge

Charge conservation does **NOT** mean that individual positive and negative charges cannot be created or destroyed. Electric charge is carried by subatomic particles such as **electrons** and **protons**. Charged particles can be created and destroyed in elementary particle reactions.

In particle physics, charge conservation means, in reactions that create charged particles, equal numbers of positive and negative particles are always created, keeping the net amount of charge unchanged. Similarly, when particles are destroyed, equal numbers of positive and negative charges are destroyed.

This property is supported without exception by all empirical observations so far [9].

Even though the conservation of charge requires that the total quantity of charge in the universe is constant, it leaves open the question of what that quantity is. Most evidence indicates that the net charge in the universe is zero [10]. That is, there are equal quantities of positive and negative charge.

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Chapter 2

Vector Calculus

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To understand how electromagnetism works we need to have a look at how **direction** is defined using mathematical notation. In this chapter we will have a good tour of the field of **vector calculus** and learn all then necessary concepts to tackle problems.

This chapter is treated as a more condensed, and specialized chapter of **M.Sc Higher Mathematics I** chapter of Differential and Integral Calculus and therefore, the student is suggested to look into those chapters to get a more in depth understanding of concepts if it is required.

2.1 Vector Algebra

Walking 5 km north and then 12 km east, we will have gone a total of 17 km, but if we carefully observe our current position, we will realize that we're not 17 km from where we set out, but only 13 km. To describe these quantities, we need a set of mathematics principles, which evidently do **NOT** add in the ordinary way.

2.1.1 Vector Operations

The reason they don't, is **displacements** have *direction* as well as *magnitude*, and it is essential to take both into account when we combine them. Such objects are called **vectors**.

For example: velocity, acceleration, force, momentum ...

By contrast, quantities which have magnitude but posses no direction are called **scalars**.

For example: mass, charge, density, temperature, ...

We will use **boldface** (**A**, **B**, and so on) for vectors and **normal-font** for scalars. The magnitude of a vector **A** is written $|\mathbf{A}|$ or, more simply, A. In diagrams, vectors are denoted by arrows:

the length of the arrow is proportional to the magnitude of the vector, and the arrow indicates its direction.

Minus A $(-\mathbf{A})$ is a vector with the same magnitude as **A** but of opposite direction.

> Vectors have magnitude and direction but **not location**.

Here we will define four (**4**) vector operations: addition and three kinds of multiplication.

Addition of Two Vectors Place the tail of **B** at the head of **A**. Their sum, $\mathbf{A}+\mathbf{B}$, is the vector from the tail of **A** to the head of **B**. Addition is commutative[1]:

$$\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A},$$

5 kilometers east followed by 12 kilometers north gets us to the same place as 12 kilometers north followed by 5 kilometers east. Addition is also associative[2]:

$$(\mathbf{A}+\mathbf{B})+\mathbf{C}=\mathbf{A}+(\mathbf{B}+\mathbf{C})\,.$$

To subtract a vector, add its opposite

$$\mathbf{A}-\mathbf{B}=\mathbf{A}+(-\mathbf{B})\,.$$

Multiplication by a Scalar Value Multiplication of a vector by a positive scalar a multiplies the *magnitude* but leaves the direction **unchanged**. This means if a is negative, the direction is reversed. Scalar multiplication is distributive[3]:

$$a\,(\mathbf{A}+\mathbf{B})=a\mathbf{A}+a\mathbf{B}.$$

[1] Meaning that the order in which elements are combined, swapped, or operated upon does **NOT** change the final result.

[2] Meaning that when we are adding or multiplying three or more numbers, the way we group them with parentheses does not change the result.

[3] Meaning that, multiplying the sum (or difference) of numbers by a factor gives the same result as multiplying each number individually by that factor and then adding (or subtracting) the products.

Dot Product of Two Vectors The dot product of two vectors is defined by:

$$\mathbf{A} \cdot \mathbf{B} \equiv AB \cos\theta$$

where θ is the angle they form when placed tail-to-tail.

> $\mathbf{A} \cdot \mathbf{B}$ is itself a scalar.[4]

[4] which is why its alternative name is **scalar product**.

The dot product is both *commutative* and *distributive*,

$$\text{(Commutative)} \qquad \mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A},$$
$$\text{(Distributive)} \quad \mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C},$$

Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A times the projection of $\mathbf{B}$ along $\mathbf{A}$.[5] If the two (2) vectors are parallel, then $\mathbf{A} \cdot \mathbf{B} = AB$. In particular, for any vector $\mathbf{A}$,

[5] Or the product of B times the projection of A along B.

$$\mathbf{A} \cdot \mathbf{A} = A^2$$

If $\mathbf{A}$ and $\mathbf{B}$ are perpendicular, then $\mathbf{A} \cdot \mathbf{B} = 0$.

Cross Product of Two Vectors The cross product of two (2) vectors is defined by:

$$\mathbf{A} \times \mathbf{B} \equiv AB \sin\theta \hat{\mathbf{n}}$$

where $\hat{\mathbf{n}}$ is a **unit vector**[6] pointing **perpendicular** to the plane of $\mathbf{A}$ and $\mathbf{B}$. Of course, there are *two* directions perpendicular to any plane: **in** and **out**.

[6] Meaning it is a vector with a magnitude of 1.

The ambiguity is resolved by the **right-hand rule**, shown in **Fig.** 2.1: let our fingers point in the direction of the first vector and curl around (via the smaller angle) toward the second; then our thumb indicates the direction of $\hat{\mathbf{n}}$.

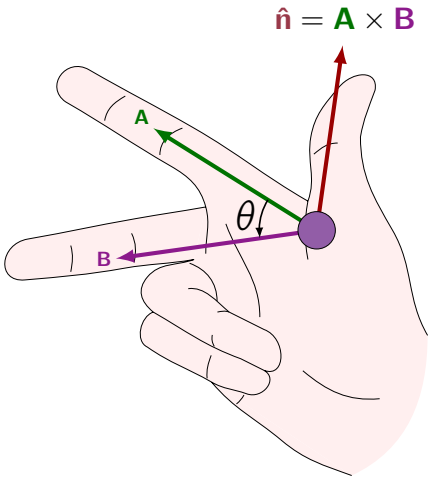


Figure 2.1 **A mnemonic, used to define the orientation of axes in three-dimensional space and to determine the direction of the cross product of two vectors, as well as to establish the direction of the force on a current-carrying conductor in a magnetic field.**

> $\mathbf{A} \times \mathbf{B}$ is itself a *vector* and it is also known as **vector product**.

The cross product is *distributive*,

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$$

but **NOT** commutative:

$$(\mathbf{B} \times \mathbf{A}) = -(\mathbf{A} \times \mathbf{B})$$

If two (2) vectors are parallel, their cross product is zero. In particular,

$$\mathbf{A} \times \mathbf{A} = 0,$$

for any vector $\mathbf{A}$.

2.1.2 Vector Component Forms

Previously, we've defined the four (4) vector operations in abstract form, without reference to any particular coordinate system. In practice, it is often easier to set up **Cartesian coordinates** (x, y, z) and work with vector **components**.

Let $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ be unit vectors parallel to the x, y, and z axes, respectively. An arbitrary vector $\mathbf{A}$ can be expanded in terms of these **basis vectors**:[7]

$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$

The symbols A_x, A_y, A_z, are **the components** of $\mathbf{A}$. In geometrical terms they are the **projections** of $\mathbf{A}$ along the three (3) coordinate axes (i.e., $A_x = \mathbf{A}\cdot\hat{\mathbf{x}}$, $A_y = \mathbf{A}\cdot\hat{\mathbf{y}}$, $A_z = \mathbf{A}\cdot\hat{\mathbf{z}}$).

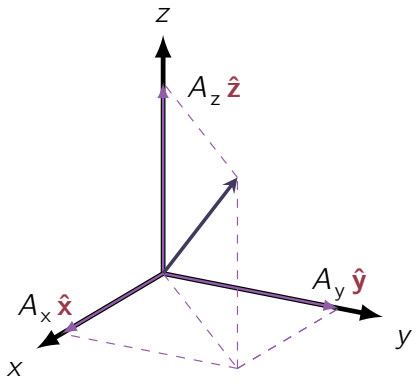


Figure 2.2
The decomposition of the vector A into its components.

[7] One of a specific set of building block vectors that serve as the fundamental coordinate system for a vector space. By scaling and combining these vectors, you can create or describe every other vector in that space.

We can now reformulate each of the four (4) vector operations as a rule for manipulating components:

$$\begin{aligned}\mathbf{A} + \mathbf{B} &= \left(A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}\right) + \left(B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}\right)\\ &= \left(A_x + B_x\right)\hat{\mathbf{x}} + \left(A_y + B_y\right)\hat{\mathbf{y}} + \left(A_z + B_z\right)\hat{\mathbf{z}}\end{aligned}$$

The operation rules are as follows:

i. To add vectors, add like components.

$$a\mathbf{A} = \left(aA_x\right)\hat{\mathbf{x}} + \left(aA_y\right)\hat{\mathbf{y}} + \left(aA_z\right)\hat{\mathbf{z}}$$

ii. To multiply by a scalar, multiply each component. As $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ are mutually perpendicular unit vectors, the following properties are valid:

$$\hat{\mathbf{x}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1 \qquad \hat{\mathbf{x}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{x}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{z}} = 0$$

Accordingly,

$$\mathbf{A}\cdot\mathbf{B} = \left(A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}\right)\cdot\left(B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}\right)$$

iii. To calculate the dot product, multiply like components, and add. In particular,

$$\mathbf{A}\cdot\mathbf{A} = A_x[2] + A_y[2] + A_z[2], \qquad \text{so} \qquad A = \sqrt{A_x[2] + A_y[2] + A_z[2]}$$

Similarly the following relations can be derived:

$$\begin{aligned}\hat{\mathbf{x}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}\times\hat{\mathbf{z}} = 0,\\ \hat{\mathbf{x}}\times\hat{\mathbf{y}} = -\hat{\mathbf{y}}\times\hat{\mathbf{x}} = \hat{\mathbf{z}},\\ \hat{\mathbf{y}}\times\hat{\mathbf{z}} = -\hat{\mathbf{z}}\times\hat{\mathbf{y}} = \hat{\mathbf{x}},\\ \hat{\mathbf{z}}\times\hat{\mathbf{x}} = -\hat{\mathbf{x}}\times\hat{\mathbf{z}} = \hat{\mathbf{y}}.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbf{A}\times\mathbf{B} &= \left(A_x\hat{\mathbf{x}}+A_y\hat{\mathbf{y}}+A_z\hat{\mathbf{z}}\right)\times\left(B_x\hat{\mathbf{x}}+B_y\hat{\mathbf{y}}+B_z\hat{\mathbf{z}}\right)\\ &= \left(A_yB_z-A_zB_y\right)\hat{\mathbf{x}}+\left(A_zB_x-A_xB_z\right)\hat{\mathbf{y}}+\left(A_xB_y-A_yB_x\right)\hat{\mathbf{z}}.\end{aligned}$$

This expression look unruly but we can tidy it up and write it more neatly as a **determinant**:

$$\mathbf{A}\times\mathbf{B}=\begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$$

iv. To calculate the cross product, form the determinant whose first row is $\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}}$, whose second row is $\mathbf{A}$ (in component form), and whose third row is $\mathbf{B}$.

2.1.3 Triple Products

As the cross product of two (2) vectors is itself a vector, it can be dotted or crossed with a 3rd vector to form a **triple** product.

Scalar Triple Product Evidently,

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})=\mathbf{B}\cdot(\mathbf{C}\times\mathbf{A})=\mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}),$$

for they all correspond to the same value.

> The **alphabetical** order is preserved.

Alternatively, the **non-alphabetical** triple products,

$$\mathbf{A}\cdot(\mathbf{C}\times\mathbf{B})=\mathbf{B}\cdot(\mathbf{A}\times\mathbf{C})=\mathbf{C}\cdot(\mathbf{B}\times\mathbf{A}),$$

have the **opposite** sign. In component form,

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})=\begin{vmatrix}A_x & A_y & A_z\\ B_x & B_y & B_z\\ C_x & C_y & C_z\end{vmatrix}.$$

A final point worth mentioning is the dot and cross can be **interchanged**:

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})=(\mathbf{A}\times\mathbf{B})\cdot\mathbf{C},$$

however, it is important to stress it out, the placement of the parentheses is **critical**:

Cross Product of Two Scalars 2.1 Information

$(\mathbf{A}\cdot\mathbf{B})\times\mathbf{C}$ is a meaningless expression. We can't make a cross product from a scalar and a vector.

Vector Triple Product The vector triple product can be simplified by the **BAC-CAB** rule:

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C})=\mathbf{B}\,(\mathbf{A}\cdot\mathbf{C})-\mathbf{C}\,(\mathbf{A}\cdot\mathbf{B})\,.$$

Please observe that the following triple product

$$(\mathbf{A}\times\mathbf{B})\times\mathbf{C}=-\mathbf{C}\times(\mathbf{A}\times\mathbf{B})=-\mathbf{A}(\mathbf{B}\cdot\mathbf{C})+\mathbf{B}(\mathbf{A}\cdot\mathbf{C})$$

is an entirely different vector.[8] All *higher* vector products can be similarly reduced, often by repeated application, so it is never necessary for an expression to contain more than one cross product in any term. For instance,

[8] To reiterate cross-products are not associative.

$$(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D})=(\mathbf{A}\cdot\mathbf{C})\,(\mathbf{B}\cdot\mathbf{D})-(\mathbf{A}\cdot\mathbf{D})\,(\mathbf{B}\cdot\mathbf{C})\,,$$
$$\mathbf{A}\times\big[\mathbf{B}\times(\mathbf{C}\times\mathbf{D})\big]=\mathbf{B}\big[\mathbf{A}\cdot(\mathbf{C}\times\mathbf{D})\big]-(\mathbf{A}\cdot\mathbf{B})\,(\mathbf{C}\times\mathbf{D})\,.$$

2.1.4 Position, Displacement, and Separation Vectors

The location of a point in three dimensions can be described by listing its Cartesian coordinates $(x,\,y,\,z)$. The vector to that point from the origin $(\mathcal{O})$ is called the **position vector**:

$$\mathbf{r}\equiv x\hat{\mathbf{x}}+y\hat{\mathbf{y}}+z\hat{\mathbf{z}}$$

Throughout **M.Sc Electrodynamics**, $\mathbf{r}$ will be used to measure **distance**. Its magnitude:

$$r=\sqrt{x^2+y^2+z^2}$$

is the distance from the origin, and

$$\hat{\mathbf{r}}=\frac{\mathbf{r}}{r}=\frac{(x)\,\hat{\mathbf{x}}+(y)\,\hat{\mathbf{y}}+(z)\,\hat{\mathbf{z}}}{\sqrt{x^2+y^2+z^2}}$$

is a unit vector pointing **radially outward**. The **infinitesimal displacement vector**, from $(x,\,y,\,z)$ to $(x+\mathrm{d}x,\,y+\mathrm{d}y,\,z+\mathrm{d}z)$, is

$$\mathrm{d}\mathbf{l}=(\,\mathrm{d}x)\,\hat{\mathbf{x}}+(\,\mathrm{d}y)\,\hat{\mathbf{y}}+(\,\mathrm{d}z)\,\hat{\mathbf{z}}.$$

In electrodynamics, one frequently encounters problems involving two (**2**) points:

source point $(\mathbf{r}')$, where an electric charge is located, and

field point $(\mathbf{r})$, at which we are calculating the electric or magnetic field.

To make these redundant calculations easier to handle, let's use the following short-hand notation:

$$\{^{\eta}\mathbf{r}\} \equiv \mathbf{r} - \mathbf{r}'$$

Its magnitude is:

$$\{^{\eta}\mathrm{r}\} = \left|\mathbf{r} - \mathbf{r}'\right|$$

and a unit vector in the direction from $\mathbf{r}^r$ to $\mathbf{r}$ is

$$\{^{\eta}\hat{\mathbf{r}}\} = \frac{\{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathrm{r}\}} = \frac{\mathbf{r} - \mathbf{r}^r}{|\mathbf{r} - \mathbf{r}^r|}$$

In Cartesian coordinates our new short-notations would be as following:

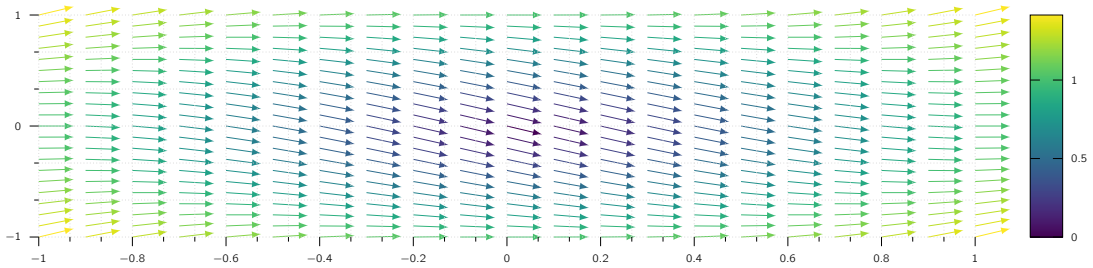


Figure 2.3
An example of a gradient field. Here the field itself is plotted by using arrows to designate the direction of the gradient. To explain the magnitude of the gradient it is either shown by the length of the arrow or by imposing colour on to the plot, which the latter has been used here. Extending this plot to our example, we can see that most of the high temperature resides on the edges of the room whereas the centre remains cool.

2.2 Differential Calculus

In mathematics, differential calculus is a subfield of calculus that studies the rates at which quantities change and therefore plays a vital role in understanding electric and magnetic fields interacting and changing with time. This section will give a broad overview. If you require a detailed look, please consult **M.Sc Higher Mathematics I**.

2.2.1 Ordinary Derivatives

Assume a function of just one (**1**) variable: $f(x)$. Therefore, the question we need to answer is what does the derivative, $\mathrm{d}[/\mathrm{d}f][x]$, do and convey?

It tells us how rapidly the function $f(x)$ **varies** when we change the argument x by an infinitesimal amount, $\mathrm{d}x$:

$$\mathrm{d}f = \left(\frac{\mathrm{d}[\,]}{\mathrm{d}f}[t]\right)\ \mathrm{d}x$$

If we increment x by an infinitesimal amount $\mathrm{d}x$, then f changes by an amount $\mathrm{d}f$.[9]

[9] Here we can think the derivative as the proportionality factor.

> Geometrically, the derivative $\mathrm{d}[/\mathrm{d}f][t]$ is the **slope** of the graph of f versus x.

2.2.2 Gradient

Assume a function which accepts three (**3**) variables. As an example, lets take the temperature $T(x, y, z)$ in the lecture room. Start out in one corner, and set up a system of cardinal directions. Then for each point (x, y, z) in the room, T gives the temperature at that spot. We want to generalise the notion of **derivative** to functions like T, which depend **NOT** on one but on three variables.

A derivative tells us **how fast the function varies**, if we move a little distance. But this time the situation is more complicated, because it depends on what **direction** we move:

- Going straight up, the temperature will probably increase fairly rapidly,

- Moving horizontally, it may not change much at all.

In fact, the question of "How fast does T vary?" can have an infinite number of answers, one for each direction we might choose to explore. Fortunately, the problem is not as bad as it looks. A theorem on partial derivatives states:

$$\mathrm{d}T = \left(\frac{\partial T}{\partial x}\right)\mathrm{d}x + \left(\frac{\partial T}{\partial y}\right)\mathrm{d}y + \left(\frac{\partial T}{\partial z}\right)\mathrm{d}z$$

This tells us how T changes when we alter all three (3) variables by the infinitesimal amounts of $\mathrm{d}x$, $\mathrm{d}y$, $\mathrm{d}z$. We can write the aforementioned equation as a dot product:

$$\mathrm{d}T = \left(\frac{\mathrm{d}[\,]}{\mathrm{d}T}[x]\hat{\mathbf{x}} + \frac{\mathrm{d}[\,]}{\mathrm{d}T}[y]\hat{\mathbf{y}} + \frac{\mathrm{d}[\,]}{\mathrm{d}T}[z]\hat{\mathbf{z}}\right)\cdot\left((\,\mathrm{d}x)\,\hat{\mathbf{x}} + (\,\mathrm{d}y)\,\hat{\mathbf{y}} + (\,\mathrm{d}z)\,\hat{\mathbf{z}}\right) = (\nabla T)\cdot(\,\mathrm{d}\mathbf{l})\,,$$

where

$$\nabla T \equiv \frac{\mathrm{d}[\,]}{\mathrm{d}T}[x]\hat{\mathbf{x}} + \frac{\mathrm{d}[\,]}{\mathrm{d}T}[y]\hat{\mathbf{y}} + \frac{\mathrm{d}[\,]}{\mathrm{d}T}[z]\hat{\mathbf{z}}$$

is the **gradient** of T. Note that ∇T is a **vector quantity**, with three (3) components.[10]

2.2.3 The Del Operator

The gradient has the formal appearance of a vector, (∇),[11] **multiplying** a scalar T:

$$\nabla T = \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right)T$$

The term in parentheses is called **del** operator:

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{y}}\frac{\partial}{\partial z}$$

Del is **NOT** a vector, in the usual sense. It doesn't mean much until we provide it with a function to act upon. Furthermore, it does **NOT** "multiply" T. Rather, it is an instruction to **differentiate** what follows. To be precise, then, we say that ∇ is a vector operator which **acts upon** T, not a vector that multiplies T.

With this qualification, though, ∇ mimics the behaviour of an ordinary vector in virtually every way. Almost anything that can be done with other vectors can also be done with ∇. Now, an ordinary vector $\mathbf{A}$ can multiply in three (3) ways:

1. By a scalar a: $\mathbf{A}a$;
2. By a vector $\mathbf{B}$, via the dot product: $\mathbf{A}\cdot\mathbf{B}$;
3. By a vector $\mathbf{B}$ via the cross product: $\mathbf{A}\times\mathbf{B}$.

Correspondingly, there are three ways the operator ∇ can act:

$$\triangleleft = i\frac{\mathrm{d}}{\mathrm{d}x} + j\frac{\mathrm{d}}{\mathrm{d}y} + k\frac{\mathrm{d}}{\mathrm{d}z},$$

[11] The History of Nabla

The operator (∇), was originally introduced by William Rowan Hamilton (1805-1865). Hamilton wrote the operator as a rotated nabla and it was P. G. Tait who established An upside-down delta as the conventional symbol in *An Elementary Treatise on Quaternions* (1867) [1]. Tait was also responsible for establishing the term *nabla*.

David Wilkins suggests Hamilton may have used the nabla as a general purpose symbol or abbreviation for whatever operator he wanted to introduce at any time. In 1837 Hamilton used the nabla, in its modern orientation, as a symbol for any arbitrary function in *Trans. R. Irish Acad. XVII. 236.* (OED.) [2] He used the nabla to signify a permutation operator in *"On the Argument of Abel, respecting the Impossibility of expressing a Root of any General Equation above the Fourth Degree, by any finite Combination of Radicals and Rational Functions*, (1839).

Hamilton used the rotated nabla ($\triangleleft$), for the vector differential operator in the *Proceedings of the Royal Irish Academy* (1846). Hamilton also used the rotated nabla as the vector differential operator in *On Quaternions; or on a new System of Imaginaries in Algebra*. For more information, please visit here.

[10] This is the generalised derivative we have been looking for.

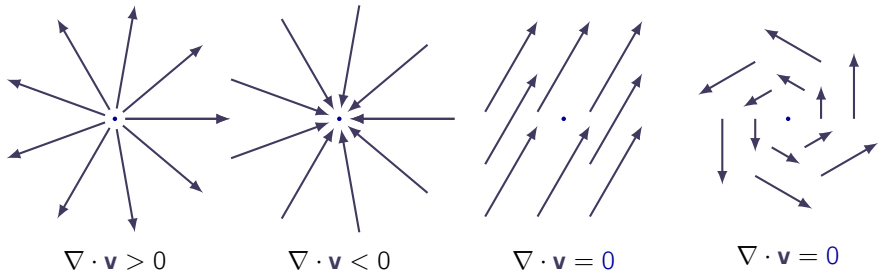


Figure 2.4
Visual description of the divergence operation and their possible outcomes.

1. On a scalar function T : ∇T (the gradient);
2. On a vector function $\mathbf{v}$, via the dot product: $\nabla \cdot \mathbf{v}$ (divergence)
3. On a vector function $\mathbf{v}$, via the cross product: $\nabla \times \mathbf{v}$ (curl).

It is time to examine the other two (**2**) vector derivatives: divergence and curl.

Divergence From the definition of ∇ we construct the divergence as follows:

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \left(\left(\frac{\partial}{\partial x}\right)\hat{\mathbf{x}} + \left(\frac{\partial}{\partial y}\right)\hat{\mathbf{y}} + \left(\frac{\partial}{\partial z}\right)\hat{\mathbf{z}}\right) \cdot \left((v_x)\,\hat{\mathbf{x}} + (v_y)\,\hat{\mathbf{y}} + (v_z)\,\hat{\mathbf{z}}\right) \\ &= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.\end{aligned}$$

Observe that the divergence of a vector function $\mathbf{v}$ is itself a **scalar** $\nabla \cdot \mathbf{v}$.

Let's try to visualise this concept. The name divergence is well chosen, for $\nabla \cdot \mathbf{v}$ is a measure of how much $\mathbf{v}$ spreads out[12] from the initial point.

For example, a vector function in **Fig.** 2.4 has a large (positive) divergence,[13] two functions in **Fig.** 2.4 has zero divergence, and one function has a negative divergence.

As an example, imagine standing at the edge of a pond. Sprinkle some dust on the surface.

1. If the material spreads out, then we dropped it at a point of **positive divergence**;
2. If it collects together, we dropped it at a point of **negative divergence**.

[12] i.e., diverges

[13] if the arrows pointed in, it would be a negative divergence

Curl From the definition of ∇ we construct the curl:

$$\begin{aligned}\mathbf{V} \times \mathbf{v} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} \\ &= \hat{\mathbf{x}}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right).\end{aligned}$$

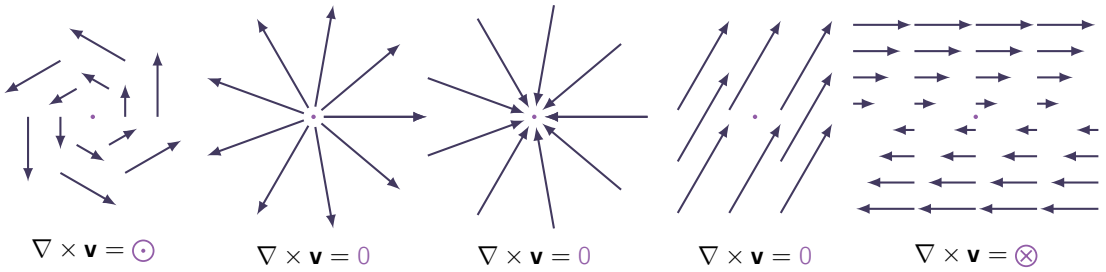


Figure 2.5
Different behaviours of the curl operation.

Similar to divergence, the name curl is also well chosen, for $\nabla \times \mathbf{v}$ is a measure of how much $\mathbf{v}$ swirls around the point in question. Therefore all function except the rightmost in **Fig.** 2.5 all have zero curl, whereas the functions in **Fig.** 2.5 all have a substantial curl sans one, pointing in the $\hat{\mathbf{z}}$ direction, as the natural right-hand rule would suggest.

To finish these two (2) important operations, let's again imagine we are standing at the edge of a pond. Float a small flower, if it starts to rotate, then we placed it at a point of nonzero curl. A whirlpool would be a region of large curl.

2.2.4 Product Rules

The calculation of ordinary derivatives is facilitated by a number of rules which are as follows:

$$\begin{aligned}
\text{Sum Rule} \qquad & \frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx} \\
\text{Constant Multiplication} \qquad & \frac{d}{dx}(kf) = k\frac{df}{dx} \\
\text{Product Rule} \qquad & \frac{d}{dx}(fg) = f\frac{dg}{dx} + g\frac{df}{dx} \\
\text{Quotient Rule} \qquad & \frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\dfrac{df}{dx} - f\dfrac{dg}{dx}}{g^2}
\end{aligned}$$

Similar relations hold for the vector derivatives. Thus,

$$\begin{aligned}
\nabla(f+g) &= \nabla f + \nabla g, & \qquad \nabla(kf) &= k\nabla f, \\
\nabla \cdot (\mathbf{A}+\mathbf{B}) &= (\nabla \cdot \mathbf{A}) + (\nabla \cdot \mathbf{B}), \quad \text{and} & \qquad \nabla \cdot (k\mathbf{A}) &= k(\nabla \cdot \mathbf{A}), \\
\nabla \times (\mathbf{A}+\mathbf{B}) &= (\nabla \times \mathbf{A}) + (\nabla \times \mathbf{B}), & \qquad \nabla \times (k\mathbf{A}) &= k(\nabla \times \mathbf{A}),
\end{aligned}$$

The product rules, on the other hand, are not quite so simple. There are two (2) ways to construct a scalar as the product of two functions:

$$\begin{aligned}
fg \qquad & \text{(product of two scalar functions)}, \\
\mathbf{A} \cdot \mathbf{B} \qquad & \text{(dot product of two vector functions)},
\end{aligned}$$

and two ways to make a vector:

$$\begin{aligned}
f\mathbf{A} \qquad & \text{(scalar times vector)}, \\
\mathbf{A} \times \mathbf{B} \qquad & \text{(cross product of two vectors)}.
\end{aligned}$$

Accordingly, there are six (**6**) product rules, which are as follows:

$$\begin{aligned}
\text{Gradient I} \quad & \nabla(fg) = f\nabla g + g\nabla f \\
\text{Gradient II} \quad & \nabla(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\times(\nabla\times\mathbf{B}) + \mathbf{B}\times(\nabla\times\mathbf{A}) + (\mathbf{A}\cdot\nabla)\mathbf{B} + (\mathbf{B}\cdot\nabla)\mathbf{A} \\
\text{Divergence I} \quad & \nabla\cdot(f\mathbf{A}) = f(\nabla\cdot\mathbf{A}) + \mathbf{A}\cdot(\nabla f) \\
\text{Divergence II} \quad & \nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B}) \\
\text{Curl I} \quad & \nabla\times(f\mathbf{A}) = f(\nabla\times\mathbf{A}) - \mathbf{A}\times(\nabla f), \\
\text{Curl II} \quad & \nabla\times(\mathbf{A}\times\mathbf{B}) = (\mathbf{B}\cdot\nabla)\mathbf{A} - (\mathbf{A}\cdot\nabla)\mathbf{B} + \mathbf{A}(\nabla\cdot\mathbf{B}) - \mathbf{B}(\nabla\cdot\mathbf{A})
\end{aligned}$$

The proofs for rules come straight from the product rule for ordinary derivatives. As an example:

$$\begin{aligned}
\nabla\cdot(f\mathbf{A}) &= \frac{\partial}{\partial x}(fA_x) + \frac{\partial}{\partial y}(fA_y) + \frac{\partial}{\partial z}(fA_z) \\
&= \left(\frac{\partial f}{\partial x}A_x + f\frac{\partial A_x}{\partial x}\right) + \left(\frac{\partial f}{\partial y}A_y + f\frac{\partial A_y}{\partial y}\right) + \left(\frac{\partial f}{\partial z}A_z + f\frac{\partial A_z}{\partial z}\right) \\
&= (\nabla f)\cdot\mathbf{A} + f(\nabla\cdot\mathbf{A}).
\end{aligned}$$

It is also possible to formulate three quotient rules:

$$\begin{aligned}
\nabla\left(\frac{f}{g}\right) &= \frac{g\nabla f - f\nabla g}{g^2}, \\
\nabla\cdot\left(\frac{\mathbf{A}}{g}\right) &= \frac{g(\nabla\cdot\mathbf{A}) - \mathbf{A}\cdot(\nabla g)}{g^2}, \\
\nabla\times\left(\frac{\mathbf{A}}{g}\right) &= \frac{g(\nabla\times\mathbf{A}) + \mathbf{A}\times(\nabla g)}{g^2}.
\end{aligned}$$

However, given these can be obtained quickly from the previously mentioned product rules, there is no point in listing them separately and is left for the reader as exercise.

2.2.5 Second Derivatives

The gradient, the divergence, and the curl are the only first derivatives we can make with $\nabla\cdot\mathbf{v}$ by applying ∇ twice, we can construct five (**5**) types of 2nd derivatives.

The gradient ∇T is a vector, so we can take the divergence and curl of it:

1. Divergence of gradient: $\nabla\cdot(\nabla T)$.
2. Curl of gradient: $\nabla\times(\nabla T)$.

The divergence $\nabla\cdot\mathbf{v}$ is a scalar, therefore all we can do is take its gradient:

3. Gradient of divergence: $\nabla(\nabla\cdot\mathbf{v})$.

The curl $\nabla\times\mathbf{v}$ is a vector, so we can take its divergence and curl:

4. Divergence of curl: $\nabla\cdot(\nabla\times\mathbf{v})$.

5. Curl of curl: $\nabla \times (\nabla \times \mathbf{v})$.

This exhausts the possible combinations, and in fact **NOT** all of them give anything new. Let's consider them one at a time:

Divergence of a Gradient

$$\begin{aligned}\nabla \cdot (\nabla T) &= \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right) \cdot \left(\frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}\right) \\ &= \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}.\end{aligned}$$

This object, which we write as $\nabla^2 T$ for short, is called the Laplacian[14] of T, which will be our focus later.

> The Laplacian of a scalar T is a scalar value.

Occasionally, we will use the Laplacian of a vector, $\nabla^2 \mathbf{v}$. By this we mean a **vector** quantity whose x-component is the Laplacian of v_x, and so on.

$$\nabla^2 \mathbf{v} \equiv \left(\nabla^2 v_x\right)\hat{\mathbf{x}} + \left(\nabla^2 v_y\right)\hat{\mathbf{y}} + \left(\nabla^2 v_z\right)\hat{\mathbf{z}}$$

This is nothing more than a convenient extension of the meaning of ∇^2.

Curl of a Gradient

The curl of a gradient is **always** zero:

$$\nabla \times (\nabla T)$$

This is an **important fact**, which will be used repeatedly. Without going into too much detail into the proof, it relies on the following relation:

$$\frac{\partial}{\partial x}\left(\frac{\partial T}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial T}{\partial x}\right)$$

Gradient of Divergence

This operation rarely occurs in physical applications, and it has not been given any special name of its own. Notice that $\nabla(\nabla \cdot \mathbf{v})$ is **NOT** the same as the Laplacian of a vector:

$$\nabla^2 = (\nabla \cdot \nabla) \neq \nabla(\nabla \cdot \mathbf{v})$$

Divergence of a Curl

Similar to the curl of a gradient, it is always zero:

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0.$$

[14] A differential operator given by the divergence of the gradient of a scalar function on Euclidean space. It is usually denoted by the symbols $\nabla \cdot \nabla$, ∇^2 or Δ. In a Cartesian coordinate system, the Laplacian is given by the sum of second partial derivatives of the function with respect to each independent variable.

The Laplace operator is named after the French mathematician *Pierre-Simon de Laplace (1749-1827)*, who first applied the operator to the study of celestial mechanics: the Laplacian of the gravitational potential due to a given mass density distribution is a constant multiple of that density distribution. Solutions of Laplace's equation $\Delta f = 0$ are called harmonic functions and represent the possible gravitational potentials in regions of vacuum.

The Laplacian occurs in many differential equations describing physical phenomena. Poisson's equation describes electric and gravitational potentials; the diffusion equation describes heat and fluid flow; the wave equation describes wave propagation; and the Schrödinger equation describes the wave function in quantum mechanics. In image processing and computer vision, the Laplacian operator has been used for various tasks, such as blob and edge detection.

Curl of a Curl

As we can check from the definition of ∇:

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}.$$

So curl-of-curl gives nothing new as the first term is just the divergence of a curl, and the second is the Laplacian. To put it short, then, there are just two kinds of second derivatives:[15]

1. Laplacian,
2. Gradient of divergence.

[15] It is possible to work out 3rd derivatives, but fortunately second derivatives suffice for practically all physical applications.

2.3 Integral Calculus

Integral calculus is a branch of mathematics focused on integration. It is the inverse of differentiation and is primarily used to calculate the accumulation of quantities, find the area under curves, determine volumes of 3D objects, and solve real-world problems in physics and engineering. As with previous section, this is just an overview. For a detailed look, please consult **M.Sc Higher Mathematics** chapter of the same name.

2.3.1 Line, Surface, and Volume Integrals

In electrodynamics, we encounter several different kinds of integrals, among which the most important are **line** (or **path**) **integrals**, **surface integrals** (or **flux**), and **volume integrals**, which will be the focus of this section.

Line Integrals Has an expression of the form:

$$\int_a^b \mathbf{v} \cdot \mathrm{d}\mathbf{l}$$

where $\mathbf{v}$ is a vector function, $d\mathbf{l}$ is the infinitesimal displacement vector, and the integral is to be carried out along a prescribed path $\mathcal{P}$ from point a to point b. If the path forms a closed loop,[16]

[16] i.e., if b = a.

We put a circle on the integral sign:

$$\oint \mathbf{v} \cdot \mathrm{d}\mathbf{l}$$

At each point on the path, we take the dot product of $\mathbf{v}$ (evaluated at that point) with the displacement $d\mathbf{l}$ to the next point on the path.

> A good example of a line integral is the work done by a force $\mathbf{F}$:
>
> $$W = \int \mathbf{F} \cdot \mathrm{d}\mathbf{l}$$

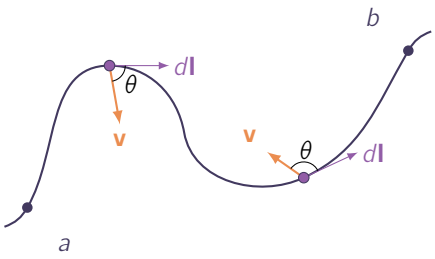


Figure 2.6 The method in which line integral is calculated. At each point the dot product of the vector (v) is taken with the length vector (dl) which is always tangential to the point in which the integration is taken.

Ordinarily, the value of a line integral depends critically on the path taken from **a** to **b**, but there is an important special class of vector functions for which the line integral is independent of path and is determined entirely by the end points.

It will be our business in due course to characterise this special class of vectors.

> A force which has this property is called **conservative**.[17]

[17] A conservative vector field is a vector field that is the gradient of some function [3] and has the property that its line integral is **path independent**.

Surface Integrals A surface integral is an expression of the form:

$$\int_{\mathcal{S}} \mathbf{v} \cdot \mathrm{d}\mathbf{a}$$

where $\mathbf{v}$ is a vector function, and the integral is over a specified surface $\mathcal{S}$. Here $\mathrm{d}a$ is an infinitesimal patch of area, with direction **perpendicular to the surface**. There are, two (2) directions perpendicular to any surface, so the **sign** of a surface integral is intrinsically ambiguous.

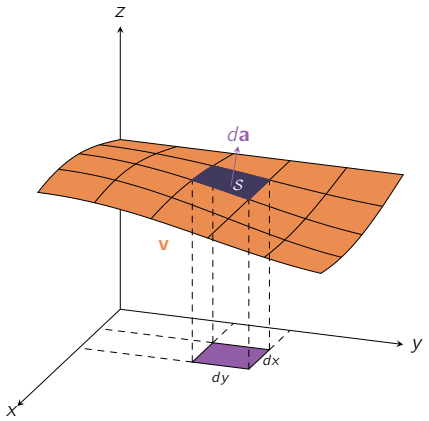


Figure 2.7: An visual description of the surface integral.

If the surface is closed,[18] we put a circle on the integral sign:

$$\oint \mathbf{v} \cdot \mathrm{d}\mathbf{a}$$

[18] imagine it forming a balloon.

> Tradition dictates that **outward** is positive, but for open surfaces it's arbitrary.

As an example, if $\mathbf{v}$ describes the flow of a fluid,[19] then $\int \mathbf{v} \cdot \mathrm{d}\mathbf{a}$ represents the total mass per unit time passing through the surface.

[19] Measured in mass per unit area per unit time.

Ordinarily, the value of a surface integral depends on the particular surface chosen, but there is a special class of vector functions for which it is **independent** of the surface and is determined entirely by the boundary line. An important task will be to characterise this special class of functions.

Volume Integrals A volume integral is an expression of the form:

$$\int_{\mathcal{V}} T \, \mathrm{d}\tau$$

where T is a scalar function and $d\tau$ is an infinitesimal volume element. In Cartesian coordinates,

$$\mathrm{d}\tau = \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$$

As an example, if T is the density of a substance,[20] then the volume integral would give the total mass.

[20] This might vary from point to point.

Occasionally we shall encounter volume integrals of vector functions:

$$\int \mathbf{v} \, \mathrm{d}\tau = \int \left(v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}} \right) \, \mathrm{d}\tau = \hat{\mathbf{x}} \int v_x \, \mathrm{d}\tau + \hat{\mathbf{y}} \int v_y \, \mathrm{d}\tau + \hat{\mathbf{z}} \int v_z \, \mathrm{d}\tau.$$

> As the unit vectors $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$ are constants, they can be taken outside the integral.

2.3.2 The Fundamental Theorems of Vector Calculus

Assume $f(x)$ is a function of one (1) variable. Based on this, the fundamental theorem of calculus says:

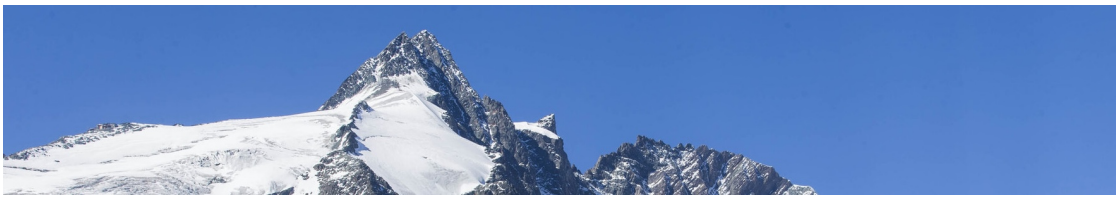


Figure 2.8
To measure the height of a mountain, it doesn't matter what way we take, as long as we know the base and the top, we will know the height .

Calculus Theorem — 2.1 Theorem

The **integral** of a **derivative** over some **region** is given by the **value of the function** at the end points (**boundaries**)

$$\int_a^b \left(\frac{\mathrm{d}[\,]}{\mathrm{d}f}[x]\right) \mathrm{d}x = f(x) - f(a) \qquad \text{or} \qquad \int_a^b F(x)\ \mathrm{d}x = f(x) - f(a)$$

In vector calculus there are three (3) species of derivative[21] and each has its own "fundamental theorem" with essentially the same format. Our purpose here is to not prove these theorems here, but rather, understand what they **mean**.

[21] These are gradient, divergence, and curl

The Fundamental Theorem for Gradients Suppose we have a scalar function of three (3) variables $T(x, y, z)$. Starting at point **a**, move a small distance $\mathrm{d}\mathbf{l}_1$. The function T will change by an amount:

$$\mathrm{d}T = (\nabla T) \cdot \mathrm{d}\mathbf{l}_1$$

Now let's move an additional small displacement $\mathrm{d}\mathbf{l}_2$. The incremental change in T will be now:

$$\mathrm{d}T = (\nabla T) \cdot \mathrm{d}\mathbf{l}_2$$

In this manner, proceeding by infinitesimal steps, we make the journey to point **b**. At each step we compute the gradient of T, at that point, and dot it into the displacement $\mathrm{d}\mathbf{l}$. . . this gives us the change in T.

Gradient Theorem — 2.2 Theorem

The total change in T in going from **a** to **b** (along the path selected) is:

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

Similar to "ordinary" fundamental theorem, this theorem says the integral[22] of a derivative, which here the gradient, is given by the value of the function at the boundaries which are **a** and **b** respectively.

[22] In this case it is a line integral

As an example, assume we want to measure the height of GrossGlockner. We could climb the mountain from base, or take the high alpine road, or take a helicopter ride all the way up to top. Regardless or the options we take, we should get the same answer either way.[23]

[23] This is the essence of the fundamental theorem

Line Independence of Gradient — 2.3 Theorem

Gradients have the special property that their line integrals are path independent:

- $\int_a^b (\nabla T) \cdot d\mathbf{l}$ is independent of the path taken from **a** to **b**.
- $\oint (\nabla T) \cdot d\mathbf{l} = 0$, since the beginning and end points are identical, and hence $T(\mathbf{b}) - T(\mathbf{a}) = 0$.

The Fundamental Theorem for Divergences

This theorem has at least three (3) special names depending on the literature source:

1. Gauss's theorem,
2. Green's theorem,
3. Divergence theorem.

The fundamental theorem for divergences states:

Divergence Theorem — 2.4 Theorem

The **integral** of a **derivative** (the **divergence**) over a **region** (in this case the **volume**, $\mathcal{V}$) is equal to the value of the function at the **boundary** (in this case the **surface** $\mathcal{S}$ bounding the volume).

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v}) \, d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}.$$

The boundary term is itself an integral, more specifically, a surface integral. This is reasonable: the "boundary" of a line is just two end points, but the boundary of a volume is a closed surface. To create an analogy, if **v** represents the flow of an incompressible fluid, then the flux **v** is the total amount of fluid passing out through the surface, per unit time. Now, the divergence measures the *spreading out* of the vectors from a point, a place of high divergence is like a tap, pouring out liquid. If we have a bunch of tap in a region filled with incompressible fluid, an equal amount of liquid will be forced out through the boundaries of the region. In fact, there are two (2) ways we could determine how much is being produced:

i. we could count up all the faucets, recording how much each puts out

ii. we could go around the boundary, measuring the flow at each point, and add it all up

We get the same answer either way:

$$\int (\text{faucets within the volume}) = \oint (\text{flow out through the surface})$$

2.3.3 The Fundamental Theorem for Curls

The fundamental theorem for curls, also known as **Stokes' theorem**, states:[24]

Stokes' Theorem — 2.5 Theorem

Integral of a **derivative** over a **region** ($\mathcal{S}$) is equal to the value of the function at **boundary** ($\mathcal{P}$).

$$\int_{\mathcal{S}} (\nabla \times \mathbf{v}) \cdot \mathrm{d}\mathbf{a} = \oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l}.$$

Similar to the divergence theorem, the boundary term is itself an integral. Specifically, a *closed line integral*.

Remember the curl measures the *twist* of the vectors $\mathbf{v}$. Think of a region of high curl as a whirlpool, where if we put a wheel there, it will rotate. Now, the integral of the curl over some surface (or, more precisely, the *flux* of the curl through the surface) represents the *total amount of swirl*, and we can determine that just as well by going around the edge and finding how much the flow is following the boundary.

> $\oint \mathbf{v} \cdot d\mathbf{l}$ is sometimes called the **circulation** of $\mathbf{v}$ [4].

There seems to be an ambiguity in *Stokes' theorem* concerning the boundary line integral:

Which way are we supposed to go around?[25]

The answer is that it doesn't matter which way we go **as long as we are consistent**, for there is an additional sign ambiguity in the surface integral:

Which way does $\mathrm{d}\mathbf{a}$ point?

For a closed surface,[26] $\mathrm{d}\mathbf{a}$ points in the direction of the outward normal. But for an open surface, which way would be defined as out? Consistency in Stokes' theorem is given by the right-hand rule. If our rings point in the direction of the line integral, then our thumb fixes the direction of $\mathrm{d}\mathbf{a}$.

Ordinary, a flux integral depends critically on what surface we integrate over, but this is **NOT** the case with curls. For Stokes' theorem says that $\int(\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ is equal to the line integral of $\mathbf{v}$ around the boundary, and the latter makes no reference to the specific surface we choose.

Preposition I $\int(\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary line, not on the particular surface used.

Preposition II $\oint(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$ for any closed surface.



[24] **Sir George Gabriel Stokes, 1st Baronet,** *(1819 - 1903)* was an Irish mathematician and physicist. Born in County Sligo, Ireland, Stokes spent his entire career at the University of Cambridge, where he served as the Lucasian Professor of Mathematics for 54 years, from 1849 until his death in 1903, the longest tenure held by the Lucasian Professor. As a physicist, Stokes made seminal contributions to fluid mechanics, including the Navier-Stokes equations; and to physical optics, with notable works on polarisation and fluorescence. As a mathematician, he popularised "Stokes' theorem" in vector calculus and contributed to the theory of asymptotic expansions. Stokes, along with Felix Hoppe-Seyler, first demonstrated the oxygen transport function of haemoglobin, and showed colour changes produced by the aeration of haemoglobin solutions.

[25] I.e., counterclockwise.

[26] I.e., the divergence theorem.

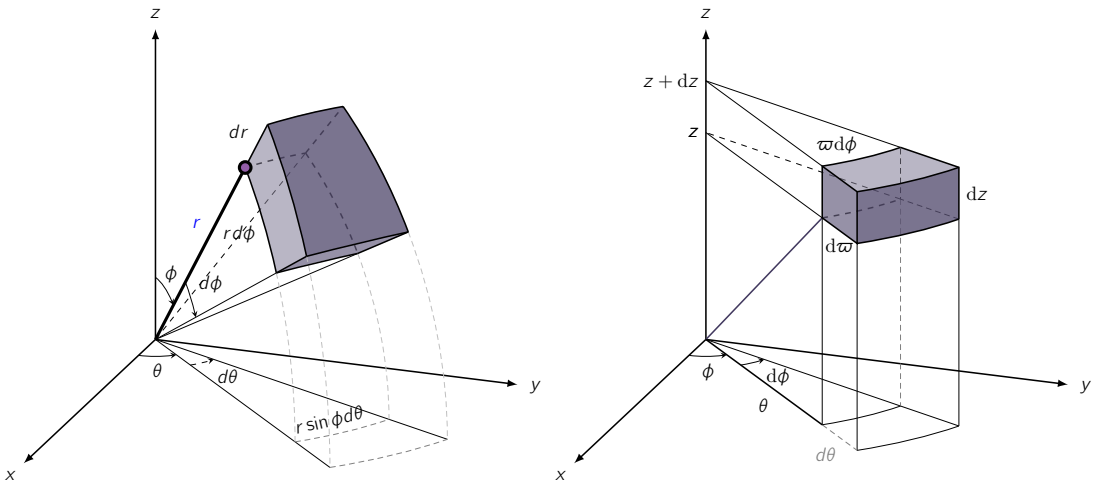


Figure 2.9 The two types of coordinate systems in question. (a) Spherical coordinate system (b) Spherical coordinate system.

2.4 Curvilinear Coordinates

Curvilinear coordinates are a generalized coordinate system in which coordinate lines can be curved. Unlike Cartesian systems that use straight, fixed axes, curvilinear coordinates feature basis vectors that change in direction or magnitude from point to point. Here we will have a look at the two most important ones which have numerous applications in engineering and physics problems.

2.4.1 Spherical Coordinate System

It is possible to label a point P in Cartesian coordinates (x, y, z), but sometimes it is more convenient to use **spherical** coordinates (r, θ, ϕ); r is the distance from the origin, θ is called the **polar angle**, and ϕ is the **azimuthal angle**. Their relation to Cartesian coordinates can be read from **Fig.** 2.9a.

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta.$$

Fig. 2.9a also shows three unit vectors, $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$, pointing in the direction of increase of the corresponding coordinates.

They constitute an **orthogonal** basis set,[27] similar to $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$, and any vector $\mathbf{A}$ can be expressed in terms of them, in the usual way:

[27] meaning mutually perpendicular.

$$\mathbf{A} = (A_r)\,\hat{\mathbf{r}} + (A_\theta)\,\hat{\boldsymbol{\theta}} + (A_\phi)\,\hat{\boldsymbol{\phi}}$$

Here, A_r, A_θ, A_ϕ are the radial, polar, and azimuthal components of vector $\mathbf{A}$. In terms of the Cartesian unit vectors:

$$\begin{aligned}
\hat{\mathbf{r}} &= (\sin\theta\cos\phi)\,\hat{\mathbf{r}} + (\sin\theta\sin\phi)\,\hat{\boldsymbol{\theta}} + (\cos\theta)\,\hat{\boldsymbol{\phi}},\\
\hat{\boldsymbol{\theta}} &= (\cos\theta\cos\phi)\,\hat{\mathbf{r}} + (\cos\theta\sin\phi)\,\hat{\boldsymbol{\theta}} + (-\sin\theta)\,\hat{\boldsymbol{\phi}},\\
\hat{\boldsymbol{\phi}} &= (-\sin\phi)\,\hat{\mathbf{r}} + (\cos\phi)\,\hat{\boldsymbol{\theta}} + (0)\,\hat{\boldsymbol{\phi}}.
\end{aligned}$$

An infinitesimal displacement in the $\hat{\mathbf{r}}$ direction is simply $\mathrm{d}r$, just as an infinitesimal element of length in the $\hat{\mathbf{x}}$ direction is $\mathrm{d}x$:

$$\mathrm{d}l_r = \mathrm{d}r$$

On the other hand, an infinitesimal element of length in the $\hat{\ }$ direction is **NOT** just $\mathrm{d}\theta$ but rather,

$$\mathrm{d}l_\theta = r\,\mathrm{d}\theta$$

Similarly, an infinitesimal element of length in the $\hat{\mathbf{\Phi}}$ direction is

$$\mathrm{d}l_\phi = r\sin\theta\,\mathrm{d}\phi$$

Thus the general infinitesimal displacement $\mathrm{d}\mathbf{l}$ is:

$$\mathrm{d}\mathbf{l} = (\mathrm{d}r)\,\hat{\mathbf{r}} + (r\,\mathrm{d}\theta)\,\hat{\ } + (r\sin\theta)\,\hat{\mathbf{\Phi}}$$

This plays the role $\mathrm{d}\mathbf{l} = (\mathrm{d}x)\,\hat{\mathbf{x}} + (\mathrm{d}y)\,\hat{\mathbf{y}} + (\mathrm{d}z)\,\hat{\mathbf{z}}$ plays in Cartesian coordinates. The infinitesimal volume element $d\tau$, in spherical coordinates, is the product of the three (3) infinitesimal displacements:

$$\mathrm{d}\tau = \mathrm{d}l_r\,\mathrm{d}l_\theta\,\mathrm{d}l_\phi = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi.$$

> It is not possible to give a general expression for **surface** elements $\mathrm{d}\mathbf{a}$, since these depend on the orientation of the surface. We simply have to analyse the geometry for any given case, which goes for **Cartesian** and **curvilinear** coordinates.

Integrating over the surface of a sphere, for instance, makes r constant, whereas θ and ϕ change:

$$\mathrm{d}\mathbf{a}_1 = \mathrm{d}l_\theta\,\mathrm{d}l_\phi\hat{\mathbf{r}} = r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi\hat{\mathbf{r}}$$

On the other hand, if the surface lies in the xy plane, making θ is constant, while r and ϕ vary:

$$\mathrm{d}\mathbf{a}_2 = \mathrm{d}l_r\,\mathrm{d}l_\phi\,\hat{\ } = r\,\mathrm{d}r\,\mathrm{d}\phi\,\hat{\ }$$

Finally: r ranges from 0 to ∞, ϕ from 0 to 2π, and θ from 0 to π.

2.4.2 Cylindrical Coordinates

The cylindrical coordinates (s, ϕ, z) of a point P are defined in **Fig.** 2.9b. Observe that ϕ has the same meaning as in spherical coordinates, and z is the same as Cartesian; s is the distance to P from the z axis, whereas the spherical coordinate r is the distance from the origin. The relation to Cartesian coordinates is:

$$x = s\cos\phi \qquad y = s\sin\phi \qquad z = z.$$

The unit vectors are:

$$\hat{\mathbf{s}} = \cos\phi\hat{\mathbf{x}} + \sin\phi\hat{\mathbf{y}}$$
$$\hat{\mathbf{\Phi}} = -\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}}$$
$$\hat{\mathbf{z}} = \hat{\mathbf{z}}$$

The infinitesimal displacements are

$$dl_s = ds \qquad dl_\phi = s\,d\phi, \qquad dl_z = dz$$

which makes:

$$d\mathbf{l} = ds\hat{\mathbf{s}} + s\,d\phi\hat{\mathbf{phi}} + dz\hat{\mathbf{z}}.$$

and the volume element is

$$d\tau\prime = s\,ds\,d\phi\,dz$$

The range of s is $(0, \infty)$, ϕ is from 0 to $2pi$ and z is from $-\infty$ to $+\infty$

2.5 Dirac Delta Function

2.5.1 A Mathematical Anomaly

Consider the following vector function:

$$\mathbf{v} = \frac{1}{r^2}\hat{\mathbf{r}}$$

At every location, $\mathbf{v}$ is directed **radially outward** which we can see in **Fig.** 2.10. Let's calculate its divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{1}{r^2}\right) = \frac{1}{r^2}\frac{\partial}{\partial r}(1) = 0$$

This is interesting as this calculation gives us an unforeseen solution. Let's look at it closer. Suppose we integrate over a sphere of radius R, centred at the origin. The surface integral therefore is

$$\oint \mathbf{v} \cdot \mathrm{d}\mathbf{a} = \int \left(\frac{1}{R^2}\hat{\mathbf{r}}\right) \cdot \left(R^2 \sin\theta \,\mathrm{d}\theta \,\mathrm{d}\phi \hat{\mathbf{r}}\right) = \left(\int_0^{\pi} \sin\theta \,\mathrm{d}\theta\right)\left(\int_0^{2\pi} \mathrm{d}\phi\right) = 4\pi$$

But the volume integral ($\int \nabla \cdot \mathbf{v} \,\mathrm{d}\tau$), should be zero (**0**) if we assume the aforementioned calculation to be true.

Does this mean that the divergence theorem is false?

The source of the problem lies at the point $r = 0$, where **v blows up**. It is true that $\nabla \cdot \mathbf{v} = 0$ everywhere **except** the origin, but right at the origin is the situation gets a little bit complicated.

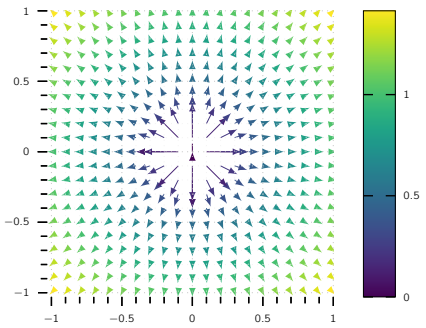


Figure 2.10
The vector plot of the "divergence problem".

Observe, the surface integral is **independent** of R. If the divergence theorem is to be true, we should expect

$$\int \nabla \cdot \mathbf{v} \,\mathrm{d}\tau = 4\pi,$$

for any non-zero vector and the origin.

This means the value of 4π must be coming from the point $r = 0$. Therefore, $\nabla \cdot \mathbf{v}$ has the unique property that it vanishes everywhere except at one point, and yet its **integral** is 4π.

No normal function behaves like that.

2.2 Information

An Analogy with Density

To wrap our heads around this property think of **density**.

The density of a point particle is zero except at the exact location of the particle, and yet its **integral** is finite. Namely, the mass of the particle.

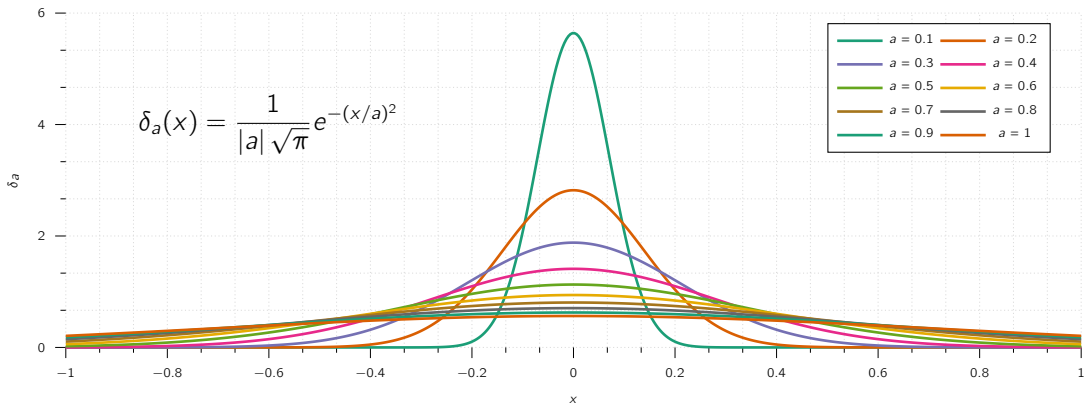


Figure 2.11
A visual representation of a 1D Dirac Delta Function. Think of it as a distribution function being squeezed to an infinitely small width.

What we have stumbled upon is called the **Dirac delta function**. It arises in numerous branches of theoretical physics and plays a central role in the theory of electrodynamics.

2.5.2 The 1D Dirac Delta Function

The one-dimensional Dirac delta function ($\delta(x)$), can be pictured as an infinitely high, infinitesimally narrow "spike", with area 1.

That is to say:

$$\delta(x) = \begin{cases} 0, & \text{if} \quad x \neq 0 \\ \infty, & \text{if} \quad x = 0 \end{cases}$$

and,

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1$$

In a strict sense of definition, $\delta(x)$ is **NOT** a function at all, as its value is not finite at $x = 0$. In literature it is known as a generalised function.[28]

If $f(x)$ is some "ordinary" function, then the product $(f(x)\delta(x))$ is zero everywhere except at $x = 0$. It follows that:

$$f(x)\delta(x) = f(0)\delta(x). \tag{2.1}$$

> The product is zero anyway except at $x = 0$. Based on this property, we may as well replace $f(x)$ by the value it assumes at the origin.

Particularly

$$\int_{-\infty}^{\infty} f(x)\delta(x)\, dx = f(0) \int_{-\infty}^{\infty} \delta(x)\, dx = f(0). \tag{2.2}$$

[28] Objects extending the notion of functions on real or complex numbers. There is more than one recognised theory, for example the theory of distributions. Generalised functions are especially useful for treating discontinuous functions more like smooth functions, and describing discrete physical phenomena such as point charges.

Under an integral, then, the delta function "picks out" the value of $f(x)$ at $x = 0$.[29] Of course, we can shift the spike from $x = 0$ to some other point, $x = a$:

$$\delta(x-a) = \begin{cases} 0, & \text{if} x \neq a \\ \infty, & \text{if } x = a \end{cases} \qquad \text{with} \qquad \int_{-\infty}^{\infty} \delta(x-a)\,dx = 1.$$

[29] Here and below, the integral need not run from $-\infty$ to $+\infty$; it is sufficient that the domain extend across the delta function, and $-\epsilon$ to $+\epsilon$ would do as well.

which turns Eq. (**2.1**) into:

$$f(x)\delta(x-a) = f(a)\delta(x-a),$$

and Eq. (**2.2**) generalises to:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)\,dx = f(a).$$

While δ itself is not a proper function, integrals over δ are perfectly acceptable. In fact, it's best to think of the delta function as something that is always intended for use under an integral sign. In particular, two expressions involving delta functions. say $D_1(x)$ and $D_2(x)$, are considered equal if:

$$\int_{-\infty}^{\infty} f(x)D_1(x)\,dx = \int_{-\infty}^{\infty} f(x)D_2(x)\,dx,$$

for all **ordinary** functions $f(x)$.

2.5.3 The 3D Dirac Delta Function

Once we have defined the 1D Dirac, it is simple to generalise it to 3D with the following:

$$\delta^3(\mathbf{r}) = \delta(x)\,\delta(y)\,\delta(z)\,,$$

As we can see, it similar to 1D, where 3D Dirac is zero everywhere except at (0, 0, 0), where it blows up. Its volume integral is 1:

$$\int_{\text{all space}} \delta^3(\mathbf{r})d\tau = \iiint_{-\infty}^{\infty} \delta(x)\delta(y)\delta(z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = 1$$

And, the general form is:

$$\int_{\text{all space}} f(\mathbf{r})\delta^3(\mathbf{r}-\mathbf{a})\,\mathrm{d}\tau = f(\mathbf{a}).$$

As in the 1D case, integration with δ picks out the value of the function f at the location of the spike.

We can fix the paradox introduced in Section 2.5. Remember, the divergence of $\hat{\mathbf{r}}/r^2$ is zero everywhere except at the origin, however, its integral over any volume containing the origin is a constant.

These are precisely the defining conditions for the Dirac delta function; evidently

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2}\right) = 4\pi\delta^3(\mathbf{r})$$

Or in a more general fashion:

$$\nabla \cdot \left(\frac{\{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathbf{r}\}^2} \right) = 4\pi\delta^3\left(\{^{\eta}\hat{\mathbf{r}}\}\right)$$

Differentiation here is with respect to $\mathbf{r}$, while $\mathbf{r}^r$ is held constant.

2.6 Vector Field Theory

2.6.1 Helmholtz Theorem

Electricity and magnetism are generally expressed as **electric and magnetic fields**, **E** and **B** and like many physical laws, these are most compactly expressed as **differential equations**.

As **E** and **B** are **vectors**, the differential equations naturally involve vector derivatives: *divergence* and *curl*. Maxwell reduced the entire theory to four (**4**) fundamental equations, specifying respectively the divergence and the curl of **E** and **B**.

This formulation raises an interesting question:

> To what extent is a vector function determined by its divergence and curl?

To study this case lets assume a vector of **F**. If the divergence of **F** is a specified scalar function D,

$$\nabla \cdot \mathbf{F} = D,$$

and the curl of **F** is a specified vector function **C**,

$$\nabla \times \mathbf{F} = \mathbf{C},$$

and for consistency, we assume **C** to have **NO** divergence,

$$\nabla \cdot \mathbf{C} = 0,$$

> Remember, the divergence of a curl is **ALWAYS** zero.

Using this knowledge, is it possible to determine the function **F**?

Without knowing more information, it is not really possible. There are many functions whose divergence and curl are both zero everywhere.

Some examples are:

$$\mathbf{F} = 0,$$
$$\mathbf{F} = (y)\,\hat{\mathbf{x}} + (zx)\,\hat{\mathbf{y}} + (xy)\,\hat{\mathbf{z}},$$
$$\mathbf{F} = (\sin x \cosh y)\,\hat{\mathbf{x}} + (-\cos x \sinh y)\,\hat{\mathbf{y}} + (.)\,\hat{\mathbf{z}}$$

If we recall **Higher Mathematics I**, to solve a differential equation with a particular solution, we must also be supplied with appropriate **boundary conditions**.

In electrodynamics we typically require the fields go to zero at infinity. With that extra information, the **Helmholtz theorem**[30] guarantees the field is uniquely determined by its divergence and curl.



[30] **Hermann Ludwig Ferdinand von Helmholtz** *(1821 - 1894)*

was a German physicist and physician who made significant contributions in several scientific fields, particularly hydrodynamic stability. The Helmholtz Association, the largest German association of research institutions, was named in his honour.

In physics, he is known for his theories on the conservation of energy and on the electrical double layer, work in electrodynamics, chemical thermodynamics, and on a mechanical foundation of thermodynamics. Although credit is shared with Julius von Mayer, James Joule, and Daniel Bernoulli-among others-for the energy conservation principles that eventually led to the first law of thermodynamics, he is credited with the first formulation of the energy conservation principle in its maximally general form.

2.6.2 Potentials

If the curl of a vector field ($\mathbf{F}$) vanishes everywhere, then $\mathbf{F}$ can be written as the **gradient of a scalar potential** (V):

$$\nabla \times \mathbf{F} = 0 \iff \mathbf{F} = -\nabla V$$

> The minus sign is purely conventional.

That's the essential idea of the following theorem:

Zero Curl Fields — 2.6 Theorem

The following conditions are **equivalent**.

i. $\nabla \times \mathbf{F} = 0$ everywhere,

ii. $\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path, for any given end points,

iii. $\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop,

iv. $\mathbf{F}$ is the gradient of some scalar function: $\mathbf{F} = -\nabla V$.

The potential is **NOT** unique as any constant can be added to V, since this will not affect its gradient.

If the divergence of a vector field ($\mathbf{F}$) vanishes everywhere, then $\mathbf{F}$ can be expressed as the curl of a **vector potential** ($\mathbf{A}$):

$$\nabla \cdot \mathbf{F} = 0 \iff \mathbf{F} = \nabla \times \mathbf{A}$$

That's the main conclusion of the following theorem:

Zero Divergence Fields — 2.7 Theorem

The following conditions are **equivalent**:

i. $\nabla \cdot \mathbf{F} = 0$ everywhere.

ii. $f\mathbf{F} \cdot d\mathbf{a}$ is independent of surface, for any given boundary line.

iii. $f\mathbf{F} \cdot d\mathbf{a} = 0$ for any closed surface.

iv. $\mathbf{F}$ is the curl of some vector function: $\mathbf{F} = \nabla \times \mathbf{A}$.

The vector potential is **NOT** unique as the gradient of any scalar function can be added to $\mathbf{A}$ without affecting the curl, given the curl of a gradient is zero.

Incidentally, in all cases, a vector field $\mathbf{F}$ can be written as the gradient of a scalar plus the curl of a vector.

$$\mathbf{F} = -\nabla V + \nabla \times \mathbf{A}$$

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Part II

Electric Fields

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Wahrlich es ist nicht das Wissen, sondern das Lernen, nicht das Besitzen sondern das Erwerben, nicht das Da-Seyn, sondern das Hinkommen, was den grössten Genuss gewährt.

It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.

(Carl Friedrich Gauss in Letter to Farkas Bolyai (2 September 1808))

Chapter 3

Electrostatics

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3.1 The Electric Field

In essence, problem of electrodynamics is to solve:

> We have some electric charges, q_1, q_2, $q_3, \ldots$, let's call them **source charges**. What force do they exert on another charge (i.e., **test charge**), Q?

The positions of the source charges are given[1] and the trajectory of the test particle is to be calculated. For most realistic cases, both the **source charges** and the **test charge** are in motion.

[1] usually as functions of time

The solution to this problem is solved with the **principle of superposition**, which states:

> The interaction between any two (**2**) charges is **unaffected** by the presence of others.

This means to determine the force on Q, we can first calculate the force $\vec{F}_1$, caused by q_1 alone, then we calculate the force $\vec{F}_2$, and then q_2 alone and so on.

Finally, we take the vector sum of all these individual forces:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots$$

Therefore, if we can find the force on Q caused by a single source charge q, we are, in principle, done.[2]

It seems easy, right? Why don't we just write down the formula for the force on Q directed **towards** q, and calculate it?

We could, but some technicalities limit us, for not only does the force on Q depend on the separation distance $\{^{\eta}\mathbf{r}\}$ between the charges, but also depends on *both* their velocities and on the *acceleration* of q.

Moreover, it is not the position, velocity, and acceleration of q **right now** that matter:

> electromagnetic particles/waves travels at the speed of light, so what concerns Q is the position, velocity, and acceleration q *had* at some earlier time, when the message left.

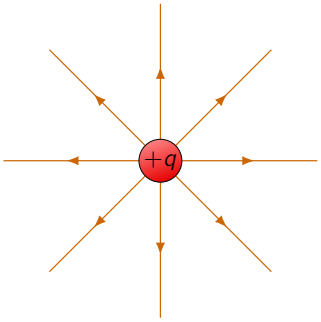


Figure 3.1 **An example of a singular charge in space. As can be seen a charge with a plus value will generate a repulsive force.**

[2] The principle of superposition may seem "obvious" to us, but it did **NOT** have to be simple. If the electromagnetic force were proportional to the square of the total source charge, the principle of superposition would not hold, as $\left(q_1+q_2\right)^2 \neq q_1^2+q_2^2$, as there would be "cross terms" to consider. Superposition is **NOT** a logical necessity, but an experimental fact.

Therefore, in spite of the fact that the basic question is easy to state[3], therefore we shall not answer this with a **direct** answer now but go at it by stages.

[3] Most complicated problem are funny enough relatively easy to state (see Collatz conjecture)

To simplify this problem, we assume time to be **irrelevant**, making the special case of **electrostatics** in which all the *source charges are stationary* whereas the test charge may be in motion.

3.1.1 Coulomb's Law

What is the force on a test charge Q due to a single point charge q, that is at *rest* a distance $\{^{\eta}\mathbf{r}\}$ away? The answer is given by Coulomb's law[4]:

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{\{^{\eta}\mathbf{r}\}^2}\{^{\eta}\mathbf{r}\}^2 \tag{3.1}$$

The constant ε_0 is the **permittivity of free space**. In SI units, where force is in newtons (N) distance in meters (m), and charge in coulombs (C).

$$\varepsilon_0 8.85 \times 10^{-12} \quad \mathrm{C}^2 \cdot \mathrm{N}^{-1} \cdot \mathrm{m}^2$$

In other words,

> the force is proportional to the product of the charges and inversely proportional to the square of the separation distance.

Remember, $\{^{\eta}\mathbf{r}\}$ is the separation vector from $\vec{r}$, (the location of q) to $\vec{r}$ (the location of Q):

$$\{^{\eta}\mathbf{r}\} = \vec{r} - \vec{r},$$

$\{^{\eta}\mathbf{r}\}$ is its magnitude, and $\{^{\eta}\hat{\mathbf{r}}\}$ is its direction. The force points **along the line** from q to Q;



[4] Charles-Augustin de Coulomb (1736 - 1806) was a French officer, engineer, physicist and the discoverer of what is now called Coulomb's law, the description of the electrostatic force of attraction and repulsion. He also did important work on friction, and his work on earth pressure formed the basis for the later development of much of the science of soil mechanics. The SI unit of electric charge, the **coulomb**, was named in his honor in 1880.

- it is repulsive if q and Q have the same sign,
- it is attractive if their signs are opposite

Coulomb's law and the principle of superposition form the physical foundation for electrostatics with the rest, except for some special properties of matter, is **mathematical derivation** of these fundamental rules.

3.1.2 The Electric Field

If we have *several* point charges q_1, q_2, ..., q_n, at distances $\{^{\eta}\mathrm{r}\}_1$, $\{^{\eta}\mathrm{r}\}_2$, ..., $\{^{\eta}\mathrm{r}\}_n$ from Q, the total force on Q is evidently

$$\begin{aligned}\vec{F} = \vec{F}_1 + \vec{F}_2 + \ldots &= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 Q}{\{^{\eta}\mathrm{r}\}_1^2}\{^{\eta}\hat{\mathbf{r}}\}_1 + \frac{q_2 Q}{\{^{\eta}\mathrm{r}\}_1^2}\{^{\eta}\hat{\mathbf{r}}\}_2 + \ldots\right) \\ &= Q\frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{\{^{\eta}\mathrm{r}\}_1^2}\{^{\eta}\hat{\mathbf{r}}\}_1 + \frac{q_2}{\{^{\eta}\mathrm{r}\}_2^2}\{^{\eta}\hat{\mathbf{r}}\}_2 + \ldots\right)\end{aligned}$$

or in a more cleaner way:

$$F = Q\vec{E}$$

where

$$\vec{E}\left(\vec{r}\right) \equiv \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{\{^{\eta}\mathrm{r}\}_i^2}\{^{\eta}\hat{\mathbf{r}}\}_i \tag{3.2}$$

$\vec{E}$ is called the **electric field** of the source charges. It is important to mention, it is a function of the position $(\vec{r})$, as the separation vectors $\{^{\eta}\hat{\mathbf{r}}\}_i$ depend on the location of the **field point** P.

But it makes no reference to the test charge Q.

The electric field is a **vector quantity** that varies from point to point and is determined by the configuration of source charges. Physically, $\vec{E}\left(\vec{r}\right)$ is the force per unit charge that would be exerted on a test charge, if you were to place one at P.

Electric Field at a Distance — 3.1 Exercise

Find the $\vec{E}$ field a distance z above the midpoint between two (**2**) equal charges (q), a distance d apart:

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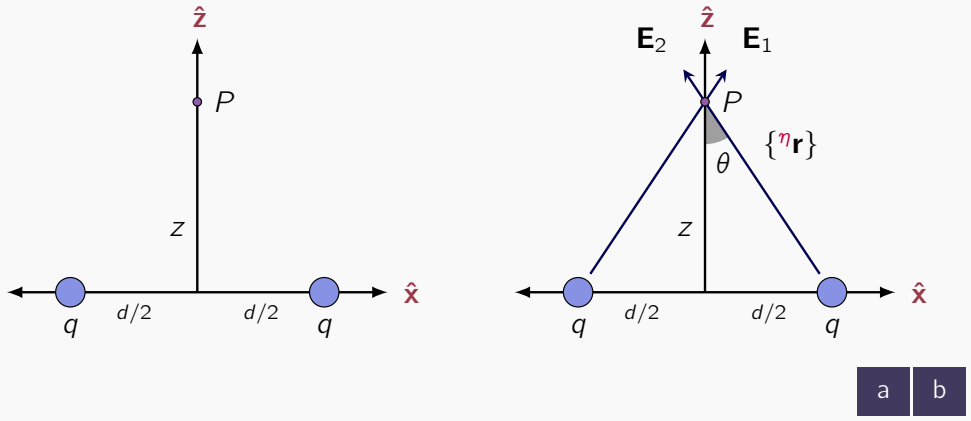


Figure 3.2: The diagram for the question "Electric Field at a Distance" (a) The geometrical description (b) Additional annotations added.

Let $\vec{E}_1$ be the field of the left charge, and $\vec{E}_2$ that of the right charge alone. Adding them, the horizontal components cancel and the vertical components add to:

$$E_z = 2\frac{1}{4\pi\varepsilon_0}\frac{q}{\{^{\eta}\mathbf{r}\}^2}\cos\theta.$$

Here $\{^{\eta}\mathbf{r}\} = \sqrt{z^2 + (d/2)^2}$ and $\cos\theta = z/\{^{\eta}\mathbf{r}\}$, therefore:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{2qz}{\left[z^2 + (d/2)^2\right]^{3/2}}\hat{\mathbf{z}} \quad \blacksquare$$

3.1.3 Continuous Charge Distributions

Our definition of the electric field Eq. (**3.2**) assumes the field source is a **collection** of discrete point charges q_i. If, the charge is distributed continuously over some region, the sum becomes an integral:

$$\vec{E}\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0}\int \frac{1}{\{^{\eta}\mathbf{r}\}^2}\{^{\eta}\hat{\mathbf{r}}\}\,\mathrm{d}q$$

1. If charge is spread out along a **line**, with charge-per-unit-length λ, then $\mathrm{d}q = \lambda\,\mathrm{d}l'$, where $\mathrm{d}l'$ is an element of length along the line.

2. If charge is smeared out over a **surface**, with charge-per-unit-area σ, then $\mathrm{d}q = \sigma\,\mathrm{d}a'$, where da' is an element of area on the surface.

3. If charge fills a **volume** with charge-per-unit-volume ρ, then $\mathrm{d}q = \rho\,\mathrm{d}\tau'$, where $\mathrm{d}\tau'$ is an element of volume.

These aforementioned volumetric definitions can be rewritten as:

$$\mathrm{d}q \rightarrow \lambda\,\mathrm{d}l' \sim \sigma\,\mathrm{d}a' \sim \rho\,\mathrm{d}\tau'.$$

Therefore the electric field of a line charge is,

$$\vec{E}\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda\left(\vec{r}\,'\right)}{\{^{\eta}\mathbf{r}\}^2}\{^{\eta}\hat{\mathbf{r}}\}\,\mathrm{d}l',$$

for a surface charge,

$$\vec{E}\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma\left(\vec{r}\,'\right)}{\{^{\eta}\mathbf{r}\}^2}\{^{\eta}\hat{\mathbf{r}}\}\,\mathrm{d}a',$$

and for a volume charge,

$$\vec{E}\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho\left(\vec{r}\,'\right)}{\{^{\eta}\mathbf{r}\}^2}\{^{\eta}\hat{\mathbf{r}}\}\,\mathrm{d}\tau'. \tag{3.3}$$

Eq. (**3.3**) is often referred to as **Coulomb's law**, as it is such a short step from the original shown in Eq. (**3.1**), and because a volume charge is in a sense the most general and realistic case.

3.2 Divergence and Curl of Electrostatic Fields

As we have seen by now, the integrals involved in computing $\vec{E}$ can be complicated and often laborious, even for reasonably simple charge distributions.

> Therefore, rest of electrostatics from this point onwards is devoted to assembling methods for **avoiding these integrals**.

It all begins with the **divergence** and **curl** of $\vec{E}$. We shall calculate the divergence of $\vec{E}$ directly from Eq. (**3.3**), but first we should develop an intuitive view of $\vec{E}$.

Let's begin with the simplest possible case:

> A singular point charge (q), situated at the **origin**:

$$\vec{E}\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$

As the equation tells, the field falls off with a factor of $1/r^2$, the vectors get shorter as you go farther away from the origin and **they always point radially outward**. But there is a nicer way to represent this field, and that's to connect up the arrows, to form **field lines**. The magnitude of the field is indicated by the **density** of the field lines:

> It's strong near the center where the field lines are close together, and weak farther out, where they are relatively far apart.

In truth, the field-line diagram can be deceptive, as when we draw it on a two-dimensional surface, for the density of lines passing through a circle of radius r is the total number divided by the circumference $\left(n/2\pi r\right)$, which goes at a rate of $\left(1/r\right)$, and not $1/r^2$.

But if were to imagine the model in three dimensions then the density of lines is the total number divided by the area of the sphere $\left(n/4\pi r^2\right)$, which **does** change with a factor of $1/r^2$.

Such diagrams are also convenient for representing more complicated fields. Field lines begin on positive charges and end on negative ones and they cannot simply terminate in midair, though they may extend out to infinity.

> If they were to terminate, the divergence of $\vec{E}$ would not be zero, and that cannot happen in empty space. In addition, field lines can never cross at the intersection,

With all this in mind, it is easy to sketch the field of any simple configuration of point charges.

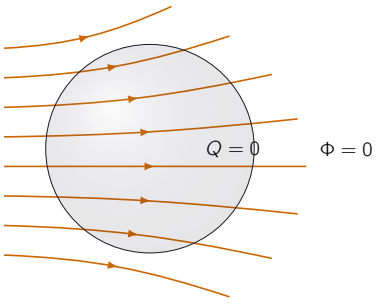


Figure 3.3
An example of imagining electric field as a collection of lines passing through. Here we can see the flux lines (i.e., the arrows) passing through an object.

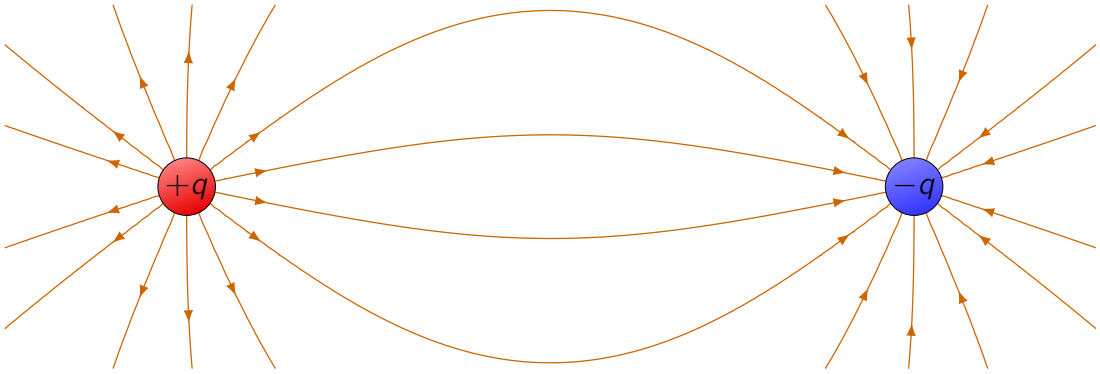


Figure 3.4
An example interaction of tow charges interacting with one another with their field lines visible. Please observe that the field line spread away from the plus charge and are collected by the minus charge.

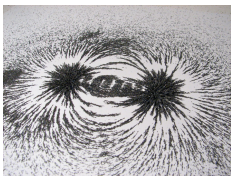
In this model, the **flux** of $\vec{E}$ through a surface $\mathcal{S}$,

$$\Phi_E = \int_{\mathcal{S}} \vec{E} \cdot \mathrm{d}\vec{a}.$$

is a measure of **the number of field lines** passing through $\mathcal{S}$. It is worth mentioning here that it is only a representation of the field lines,[5] as the **total** number would be infinite. But **for a given sampling rate** the flux is *proportional* to the number of lines drawn, because the field strength, is proportional to the density of field lines, and therefore $\vec{E} \cdot \mathrm{d}\vec{a}$, is proportional to the number of lines passing through the infinitesimal area $\mathrm{d}\vec{a}$.

This means the flux through any **closed** surface is a measure of the total mass inside the surface. For the field lines that originate on a positive charge must either pass through the surface or else terminate on a negative charge inside. On the other hand, a charge *outside* the surface will contribute nothing to the total flux, since its field lines pass in one side and out the other.

This is the basic idea behind Gauss' law.



[5] In the history of physics, a line of force in Michael Faraday's extended sense is synonymous with James Clerk Maxwell's line of induction. According to J.J. Thomson, Faraday usually discusses lines of force as chains of polarized particles in a dielectric, yet sometimes Faraday discusses them as having an existence all their own as in stretching across a vacuum. From the 20th century perspective, lines of force are energy linkages embedded in a 19th-century field theory that led to more mathematically and experimentally sophisticated concepts and theories, including Maxwell's equations and Albert Einstein's theory of relativity.

3.2.1 Quantification

In the case of a point charge q at the origin, the flux of $\vec{E}$ through a spherical surface of radius (r) is:

$$\oint \vec{E} \cdot \mathrm{d}\vec{a} = \int \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2}\hat{\mathbf{r}}\right) \cdot \left(r^2 \sin\theta \, \mathrm{d}\theta d\phi \hat{\mathbf{r}}\right) = \frac{1}{\varepsilon_0} q$$

Observe that the radius of the sphere **cancels out**, as the surface area goes **up** as r^2, the field goes *down* as $1/r^2$, so the product is **constant**. In terms of the field-line picture, this makes sense, given the same number of field lines pass through any sphere centered at the origin, regardless of its size.

In fact, it doesn't have to be a sphere. **Any** closed surface, whatever is shape, would be pierced by the same number of field lines. Evidently the flux through any surface enclosing the charge is q/ε_0.

Let's extend this idea a bit further and assume instead of a single charge at the origin, we have a bunch of charges scattered about.

According to the principle of superposition, the total field is the vector sum of all the individual fields:

$$\vec{E} = \sum_{i=1}^{n} \vec{E}_{\mathrm{i}}$$

The flux through a surface that encloses them all is:

$$\oint \vec{E} \cdot \mathrm{d}a = \sum_{i=1}^{n} \left(\oint \vec{E}_{\mathrm{i}} \cdot \mathrm{d}\vec{a} \right) = \sum_{i=1}^{n} \left(\frac{1}{\varepsilon_0} q_{\mathrm{i}} \right)$$

For any closed surface, then,

$$\oint \vec{E} \cdot \mathrm{d}\vec{a} = \frac{1}{\varepsilon_0} Q_{\mathrm{enc}} \tag{3.4}$$

where Q_{enc} is the total charge enclosed within the surface. This is the quantitative statement of Gauss's law. As it stands, Gauss's law is an *integral* equation, but we can easily turn it into a *differential* one, by applying the divergence theorem:

$$\oint_S \vec{E} \cdot \mathrm{d}\vec{a} = \int_{\mathcal{V}} \left(\nabla \cdot \vec{E} \right) \mathrm{d}\tau$$

Rewriting Q_{enc} in terms of the charge density ρ, we have

$$Q_{\mathrm{enc}} = \int_{\mathcal{V}} \rho \, \mathrm{d}\tau$$

So Gauss's law becomes

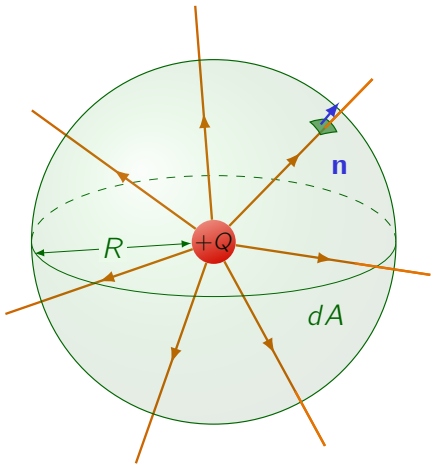


Figure 3.5: A visual representation of the gauss' law.

$$\int_{\mathcal{V}} \left(\nabla \cdot \vec{E} \right) \mathrm{d}\tau = \int_{\mathcal{V}} \left(\frac{\rho}{\varepsilon_0} \right) \mathrm{d}\tau$$

And since this holds for *any* volume, the integrands must be equal:

$$\nabla \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho \tag{3.5}$$

Eq. (3.5) carries the same message as Eq. (3.4) as it is **Gauss's law in differential form**. The differential version is idler, but the integral form has the advantage in that it accommodates point, line, and surface charges more naturally.

3.2.2 Divergence of E

Let's go back, now, and calculate the divergence of $\vec{E}$ directly from Eq. (3.3):

$$\vec{E}\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0}\int\limits_{\text{all space}} \frac{\{^{\eta}\mathbf{r}\}}{\{^{\eta}\mathbf{r}\}^2}\rho\left(\mathbf{r}'\right)\,\mathrm{d}\tau'$$

While the interation is originally defined over the volume occupied by the charge, it is much easier to encompass all the space as $\rho = 0$ outside the volume regardless.

Noting that the $\vec{r}$-dependence of the aforementioned equation is contained in the following:

$$\{^{\eta}\mathbf{r}\} = \vec{r} - \mathbf{r}', \qquad \text{where we have} \qquad \nabla\cdot\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \nabla\cdot\left(\frac{\{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathbf{r}\}^2}\right)\rho\left(\mathbf{r}'\right)\,\mathrm{d}\tau'$$

This is precisely the divergence we calculated in the previous chapter regarding the divergence anomaly.

$$\left(\frac{\{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathbf{r}\}^2}\right) = 4\pi\delta^2\left(\{^{\eta}\mathbf{r}\}\right)$$

Using this relation we can arrive at the following:

$$\nabla\cdot\vec{E} = \frac{1}{4\pi\varepsilon_0}\int 4\pi\delta^3\left(\vec{r}-\mathbf{r}'\right)\rho\left(\mathbf{r}'\right)\,\mathrm{d}\tau' = \frac{1}{\varepsilon_0}\rho\left(\vec{r}\right)$$

which is Gauss's law in differential form given in Eq. (**3.5**). To recover the integral form in Eq. (**3.4**), we run the previous argument **in reverse**, integrate over a volume and apply the divergence theorem:

$$\int_{\mathcal{V}} \nabla\cdot\vec{E}\,\mathrm{d}\tau = \oint_{\mathcal{S}} \vec{E}\cdot\mathrm{d}\vec{a} = \frac{1}{\varepsilon_0}\int_{\mathcal{V}}\rho\,\mathrm{d}\tau = \frac{1}{\varepsilon_0}Q_{\text{enc}}.$$

3.2.3 Applications of Gauss's Law

Let us look at some of the use cases of Gauss's law, in integral form. When the problem allows symmetry, it allows us the **quickest** and **easiest** way of computing electric fields. Let's understand this statement with a series of examples.

A Gaussian Sphere 3.2 Exercise

Find the field outside a uniformly charged solid sphere of radius R and total charge q.

Imagine a spherical surface at radius $r > R$, shown below. This is called a Gaussian surface.

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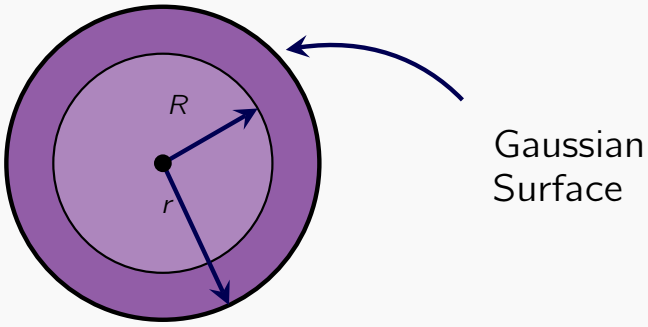


Figure 3.6: An imaginary Gaussian sphere.

Gauss's law says that:

$$\oint_{\mathcal{S}} \vec{E} \cdot \mathrm{d}\vec{a} = \frac{1}{\varepsilon_0} Q_{\text{enc}},$$

For this case it is $Q_{\text{enc}} = q$. At first glance this doesn't seem to get us very far, as the quantity we want (E) is buried inside the surface integral. Luckily, symmetry allows us to extract E from under the integral sign: E certainly points radially outward,5 as does da, so we can drop the dot product,

$$\int_{\mathcal{S}} \vec{E} \cdot \mathrm{d}\vec{a} = \int_{\mathcal{S}} \left|\vec{E}\right| \mathrm{d}a$$

and the magnitude of E is constant over the Gaussian surface, so it comes outside the integral

$$\int_{\mathcal{S}} \left|\vec{E}\right| \mathrm{d}a = \left|\vec{E}\right| \int_{\mathcal{S}} \mathrm{d}a = |E| 4\pi r^2$$

Therefore:

$$\left|\vec{E}\right| 4\pi r^2 = \frac{1}{\varepsilon_0} q \qquad \text{or} \qquad \vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \hat{\mathrm{r}} \quad \blacksquare$$

Gauss's law is always **true**, however it may **NOT** always be useful. If ρ had not been uniform,[6] or if we had chosen some other shape for our Gaussian surface, it would still have been true that the flux of $\vec{E}$ is q/ε_0, but $\vec{E}$ would not have pointed in the same direction as $\mathrm{d}\vec{a}$, and its magnitude would not have been constant over the surface, and without that we cannot get $\left|\vec{E}\right|$ outside of the integral.

[6] or spherically symmetrical.

> Symmetry is crucial to this application of Gauss's law.

There are three (**3**) kinds of symmetry that work:

Spherical it is a concentric sphere,

Cylindrical it is a coaxial cylinder,

Plane A pillbox which straddles the surface.

While cylindrical and plane symmetries technically require infinitely long cylinders, and planes are assumed to extend to infinity, we shall often use them to get approximate answers for "long" cylinders or "large" planes, at points far from the edges.

Although the direct use of Gauss's law to compute electric fields is limited to cases of spherical, cylindrical, and planar symmetry, we can put together **combinations** of objects possessing such symmetry, even though the arrangement as a whole is not symmetrical. For example, invoking the principle of superposition, we could find the field in the vicinity of two uniformly charged parallel cylinders, or a sphere near an infinite charged plane.

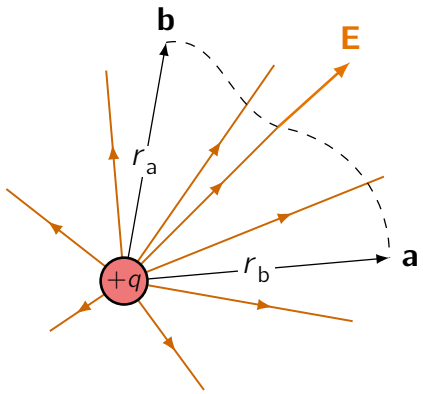


Figure 3.7

3.2.4 Curl of E

It is time to calculate the curl of $\vec{E}$, as we did the divergence. As usual, to derive this behaviour, we shall use the simplest case:

a point charge at the origin.

In this case we write down:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}.$$

To get a feel of the curve of $\vec{E}$, let us calculate the line integral of this field from some point $\vec{a}$ to some other point $\vec{b}$:

$$\int_{\vec{a}}^{\vec{b}} \vec{E}\cdot \mathrm{d}\vec{l}$$

As we are working in spherical coordinates, $(\mathrm{d}r)\,\hat{\mathbf{r}} + (r\,\mathrm{d}\theta)\,\hat{\theta} + (r\sin\theta d\phi)\,\hat{\boldsymbol{\phi}}$, so

$$\vec{E}\cdot \mathrm{d}\vec{l} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\,\mathrm{d}r$$

Therefore

$$\int_{\vec{a}}^{\vec{b}} \vec{E}\cdot \mathrm{d}\vec{l} = \frac{1}{4\pi\varepsilon_0}\int_a^b \frac{q}{r^2}\,\mathrm{d}r = -\frac{1}{4\pi\varepsilon_0}\frac{q}{r}\Bigg|_{r_a}^{r_b} = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_a} - \frac{q}{r_b}\right)$$

where r_a is the distance from the origin to the point $\vec{a}$ (i.e., $r_a = \left|\vec{a} - \mathcal{O}\right|$) and r_b is the distance to $\vec{b}$ (i.e., $r_b = \left|\vec{b} - \mathcal{O}\right|$). The integral around a closed path is evidently zero (for then $r_a = r_b$):

$$\oint \vec{E}\cdot \mathrm{d}\vec{l} = 0 \qquad (3.6)$$

And through the use of **Stokes' Theorem** one could arrive its differential form:

$$\nabla \times \vec{E} = 0 \tag{3.7}$$

Now, we proved Eq. (**3.6**) and Eq. (**3.7**) only for the field of a single point charge at the origin, but these results make no reference to what is, a perfectly arbitrary choice of coordinates.

They hold no matter where the charge is located. Moreover, if we have many charges, the principle of superposition states that the total field is a vector sum of their individual fields:

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$$

which means

$$\nabla \times \vec{E} = \nabla \times \left(\vec{E}_1 + \vec{E}_2 + \dots \right)$$

Therefore, Eq. (**3.6**) and Eq. (**3.7**) hold for any static charge distribution.

3.3 Electric Potential

Now we look at another important concept in electric fields which is generally used as a mathematical concept but has significant use nonetheless.

3.3.1 Defining the Potential

The electric field $\vec{E}$ is a **special kind** of vector function, one whose curl is zero.

> For example, $\vec{E} = y\hat{\mathbf{x}}$, could **NOT** possibly be an electrostatic field as **NO** set of charges, regardless of their sizes and positions, could ever produce such a field.

We're going to exploit this special property of electric fields to reduce a **vector** problem (finding $\vec{E}$) to a **scalar** problem. We know from the previous chapter that any vector whose curl is zero is equal to the gradient of some scalar. What we are going to do now amounts to a proof in the context of electrostatics.

As $\nabla \times \vec{E} = 0$, the line integral of $\vec{E}$ around any closed loop is zero.[7] As $\oint \vec{E} \cdot \mathrm{d}l = 0$, the line integral of $\vec{E}$ from point $\vec{a}$ to point $\vec{b}$ is the same for all paths.

[7] This is a consequence of the Stokes' Theorem.

Because the line integral is independent of path, we can define a function

$$V\left(\vec{r}\right) \equiv -\int_{\mathcal{O}}^{\vec{r}} \vec{E} \cdot \mathrm{d}\vec{l} \tag{3.8}$$

Here $\mathcal{O}$ is some standard reference point on which we have agreed beforehand. V then depends only on the point $\vec{r}$. We will call this the **electric potential**.

The potential difference between two (2) points $\vec{a}$ and $\vec{b}$ is

$$\begin{aligned} V\left(\vec{b}\right) - V\left(\vec{a}\right) &= -\int_{\mathcal{O}}^{\vec{b}} \vec{E} \cdot \mathrm{d}\vec{l} + \int_{\mathcal{O}}^{\vec{a}} \vec{E} \cdot \mathrm{d}\vec{l} \\ &= -\int_{\mathcal{O}}^{\vec{b}} \vec{E} \cdot \mathrm{d}\vec{l} - \int_{\vec{a}}^{\mathcal{O}} \vec{E} \cdot \mathrm{d}\vec{l} = -\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot \mathrm{d}\vec{l}. \end{aligned} \tag{3.9}$$

Now, the fundamental theorem for gradients states the following:

$$V(\mathbf{b}) - V(\mathbf{a}) = \int_{\mathrm{a}}^{\mathrm{b}} (\nabla V) \cdot \mathrm{d}\vec{l}, \qquad \text{and} \qquad \int_{\mathrm{a}}^{\mathrm{b}} (\nabla V) \cdot \mathrm{d}\vec{l} = -\int_{\mathrm{a}}^{\mathrm{b}} \vec{E} \cdot \mathrm{d}\vec{l}.$$

Since, finally, this is true for any points $\vec{a}$ and $\vec{b}$, the integrands must be equal:

$$\vec{E} = -\nabla V \tag{3.10}$$

Eq. (3.10) is the differential version of Eq. (3.8) which says that the electric field is the gradient of a scalar potential, which is what we set out to prove.

Notice the subtle but crucial role played by path independence[8] in this argument. If the line integral of $\vec{E}$ depended on the path taken, then the "definition" of V, Eq. (3.8), would **NOT** be useful as it simply would **NOT** define a function, since changing the path would alter the value of $V\left(\vec{r}\right)$.

[8] or, equivalently, the fact that $\nabla \times \vec{E} = 0$

The minus sign in $\vec{E} = -\nabla V$ is there due to historical conventions.

3.1 Information

An Important Distinction

It is easy to be confused about the term **electrical potential** and a **voltage source** we encounter when we analyse a normal circuit. However there is a significant difference between them.

Electrical potential is the energy per unit charge gained or lost when a charge is moved from some reference point at which the potential is defined to be zero. **Voltage**, on the other hand, is the difference in potential between two (2) arbitrary points at which the potential is **NOT** necessarily zero.

For example, the potential at point **A** relative to a defined reference point[9] might be 100 V, and the potential at point **B** might be 10 V volts. Then the voltage between **B** and **A** is 90 V volts.

[9] sometimes an infinite distance away

3.3.2 Some Digressions on Potential

Naming

The naming of V as **potential** is unfortunate as it inevitably reminds us of potential *energy*. This is particularly problematic, as there is a connection between **potential** and **potential energy**. Incidentally, a surface over which the potential is constant is called an **equipotential**.

Advantages of the Potential

If we know V, we can easily get $\vec{E}$. It's simple, just take the gradient: $\vec{E} = -\nabla\vec{V}$. This is due to the three (3) components of $\vec{E}$ not being independent. In fact, they are **explicitly** interrelated by the very condition we started with, $\nabla \times \vec{E} = 0$.

In terms of components,

$$\frac{\partial[}{\partial E}[x]][y] = \frac{\partial[}{\partial E}[y]][x], \qquad \frac{\partial[}{\partial E}[z]][y] = \frac{\partial[}{\partial E}[y]][z], \qquad \frac{\partial[}{\partial E}[x]][z] = \frac{\partial[}{\partial E}[z]][x].$$

The Reference Point

There is a **necessary** ambiguity in the definition of potential, as the choice of reference point $\mathcal{O}$ was arbitrary. Changing reference points amounts to adding a constant K to the potential:

$$V'\left(\vec{r}\right) = -\int_{\mathcal{O}}^{\vec{r}} \vec{E}\cdot \mathrm{d}\vec{l} = -\int_{\mathcal{O}'}^{\mathcal{O}} \vec{E}\cdot \mathrm{d}\vec{l} - \int_{\mathcal{O}}^{\vec{r}} \vec{E}\cdot \mathrm{d}\vec{l} = K + V\left(\vec{r}\right)$$

where K is the line integral of $\vec{E}$ from the old reference point $\mathcal{O}$ to the new one $\mathcal{O}'$. Of course, adding a constant to V will **NOT** affect the potential **difference** between two (**2**) points:

$$V'(\mathbf{b}) - V'(\mathbf{a}) = V(\mathbf{b}) - V(\mathbf{a}),$$

since the K's cancel out.[10] Nor does the ambiguity affect the gradient of V:

$$\nabla V' = \nabla V,$$

since the derivative of a constant is zero (**0**). That's why all such V's, differing only in their choice of reference point, correspond to the same field $\vec{E}$.

[10] Actually, it was already clear from Eq. (**3.9**) that the potential difference is independent of $\mathcal{O}$, as it can be written as the line integral of $\vec{E}$ from $\vec{a}$ to $\vec{a}$, with no reference to $\mathcal{O}$.

> Potential as such carries no real physical significance, for at any given point we can adjust its value at will by a suitable relocation of $\mathcal{O}$.

However, therefore V's a "reference level to use for $\mathcal{O}$ in electrostatics –almost-one-like equations to sea level for altitude–and that is a point infinitely far from the charge. Ordinarily, then, we "set the zero of potential at infinity." (Since $V(\mathcal{O}) = 0$, choosing a reference point is equivalent to selecting a place where V is to be zero.) But I must warm you that there is one special circumstance in which this convention fails: when the charge distribution itself extends to infinity. The symptom of trouble, in such cases, is that the potential blows up. For instance, the field of a uniformly charged plane is ($\sigma/2\varepsilon_0$

$$V(z) = -\int_{\infty}^{z} \frac{1}{2\varepsilon_0}\sigma\, dz = -\frac{1}{2\varepsilon_0}\sigma(z-\infty).$$

The remedy is simply to choose some other reference point (in this example you might use a point on the plane). Notice that the difficulty occurs only in textbook problems; in "real life" there is no such thing as a charge distribution that goes on forever, and we can always use infinity as our reference point.

Superposition Principle

The original superposition principle pertains to the force on a test charge (Q). It says that the total force on Q is the **vector sum** of the forces attributable to the source charges individually:

$$\vec{F} = \vec{F}_1 + \vec{F}_1 + \vec{F}_2 \ldots$$

Dividing through by Q, we see that the electric field, too, obeys the superposition principle:

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \ldots$$

Integrating from the common reference point to $\vec{r}$, it follows that the potential also satisfies such a principle:

$$V = V_1 + V_2 + \dots$$

That is, the potential at any given point is the sum of the potentials due to all the source charges separately. Only this time it is an ordinary sum, not a vector sum, which makes it a lot easier to work with.

Units of Potential

For our system of units, SI, force is measured in newtons (N) and charge in coulombs (C), so electric fields are in newtons per coulomb ($N \cdot C^{-1}$).

Accordingly, potential is known-meters per coulomb ($m \cdot C^{-1}$), or joules per coulomb ($J \cdot C^{-1}$).

> A joule per coulomb is a volt.

3.3.3 Poisson's Equation and Laplace's Equation

We now know the electric field ($\vec{E}$) can be written as the **gradient of a scalar potential**:

$$\vec{E} = -\nabla V$$

The question arises:

> *How are divergence and curl of $\vec{E}$ related to our new definition of electric potential, V?*

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \qquad \text{and} \qquad \nabla \times \vec{E} = 0$$

Well, we can write the aforementioned equations in the following form:

$$\nabla \cdot \vec{E} = \nabla \cdot -\nabla V = -\nabla^2 V$$

so, apart from that persistent minus sign, the divergence of $\vec{E}$ is the **Laplacian** of V. Gauss's law, then, says:

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}$$

This is known as **Poisson's equation**. In regions where there is no charge, so $\rho = 0$, Poisson's equation reduces to **Laplace's equation**:

$$\nabla^2 V = 0$$

We'll explore this in Chapter 3 when we work with on **Advanced Mathematical Methods**.

What about the curl? Let's do the math and write the following form:

$$\nabla \times \vec{E} = \nabla \times (-\nabla V) = 0$$

But that's no condition on V as curl of gradient is always zero.

Of course, here, we used curl to show $\vec{E}$ could be expressed as the gradient of a scalar, so it's not really surprising that this works out. It takes only one (**1**) Partial Differential Equation (PDE) (Poisson's) to determine V, as V is a scalar; for $\vec{E}$ we needed two (**2**), the divergence and the curl.[11]

[11] This comes from *Helmholtz theorem*, which we discussed previously.

3.3.4 The Potential of a Localised Charge Distribution

Previously, we have defined V in terms of $\vec{E}$ as shown in Eq. (**3.8**).

Ordinarily, though, it's E that we're looking for.[12] The idea is that it might be easier to get V first, and then calculate $\vec{E}$ by taking the gradient. Typically, then, we know where the charge is,[13] and we want to find V.

[12] if we already knew $\vec{E}$, there wouldn't be much point in calculating V

[13] that is, we know where the charge density (ρ) is.

Now, Poisson's equation relates V and ρ, but unfortunately it's "the wrong way around":

> it would give us ρ, if we knew V, whereas we want V, knowing ρ.

What we need to do is **invert** Poisson's equation. That's the idea for this section, although we shall do it by roundabout means, beginning, as always, with a point charge at the origin.

The electric field is:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\left(\frac{1}{r^2}\right)\hat{\mathbf{r}},$$

so

$$\vec{E}\cdot \mathrm{d}\vec{l} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\,\mathrm{d}r \qquad \text{where} \qquad \mathrm{d}\vec{l} = \mathrm{d}r\hat{\mathbf{r}} + r\,\mathrm{d}\theta\hat{\boldsymbol{\theta}} + r\sin\theta\,\mathrm{d}\phi\hat{\boldsymbol{\phi}}.$$

Setting the reference point at a place in infinity,[14] the potential of a point charge q at the origin is

$$V(r) = -\int_{\infty}^{\vec{r}} \vec{E}\cdot \mathrm{d}\vec{l} = \frac{1}{4\pi\varepsilon_0}\int_{\infty}^{\vec{r}} \frac{q}{r'^2}\,\mathrm{d}r' = \frac{1}{4\pi\varepsilon_0}\frac{q}{r'}\Big|_{\infty}^{r} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}.$$

[14] We will see here the advantage of using infinity for the reference point as it kills the lower limit on the integral.

Please observe the sign of V; presumably the conventional minus sign in the definition Eq. (**3.8**) was chosen to make the potential of a positive charge come out positive.

Think Topographically — 3.2 Information

It is useful to remember that regions of positive charge are potential "hills", regions of negative charge are potential "valleys", and the electric field points "downhill", from plus toward minus.

In general, the potential of a point charge q is

$$V\left(\overrightarrow{r}\right) = \frac{1}{4\pi\varepsilon_0}\frac{q}{\{^{\eta}\mathrm{r}\}},$$

where $\{^{\eta}\mathrm{r}\}$, as always, is the distance from q to $\overrightarrow{r}$. Invoking the superposition principle, then, the potential of a collection of charges is

$$V\left(\overrightarrow{r}\right) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_{\mathrm{i}}}{\{^{\eta}\mathrm{r}\}_i},$$

or, for a continuous distribution,

$$fnrV[\overrightarrow{r}] = \frac{1}{4\pi\varepsilon_0}\int\frac{1}{t}\,\mathrm{d}q.$$

In particular, for a volume charge, it's

$$fnrV[\overrightarrow{r}] = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\mathbf{r}')}{\{^{\eta}\mathrm{r}\}}\,\mathrm{d}\tau'. \tag{3.11}$$

This is the equation we were looking for, telling us how to compute V when we know ρ. It is useful to think of this equation as the "solution" to Poisson's equation, for a localized charge distribution.

Let's compare Eq. (3.11) with the corresponding formula for the electric field in terms of ρ Eq. (3.3):

$$\overrightarrow{E}\left(\overrightarrow{r}\right) = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\mathbf{r}')}{\{^{\eta}\mathrm{r}\}^2}\{^{\eta}\hat{\mathbf{r}}\}\,\mathrm{d}\tau'$$

The main point to notice is that the unit vector ($\{^{\eta}\hat{\mathbf{r}}\}$) is gone, so there is no need to fuss with components. The potentials of line and surface charges are

$$V = \frac{1}{4\pi\varepsilon_0}\int\frac{\lambda\left(\overrightarrow{r}'\right)}{\{^{\eta}\mathrm{r}\}}\,\mathrm{d}l' \quad \text{and} \quad V = \frac{1}{4\pi\varepsilon_0}\int\frac{\sigma\left(\overrightarrow{r}'\right)}{\{^{\eta}\mathrm{r}\}}\,\mathrm{d}a' \quad \text{and} \tag{3.12}$$

It is worth stressing that everything in this section is predicated on the assumption that the reference point is at **infinity**. This is hardly apparent in Eq. (3.11), but remember that we got that equation from the potential of a point charge at the origin, $(1/4\pi\varepsilon_0)(q/r)$, which is valid only when $\mathcal{O} = \infty$.

> If we try to apply these formulas to one of those artificial problems in which the charge itself extends to infinity, the integral will diverge.

3.3.5 Boundary Conditions

In the typical electrostatic problem we are given a source charge distribution (ρ), and we want to find the electric field ($\vec{E}$) it produces.

> Unless the symmetry of the problem allows a solution by Gauss's law, it is generally to our advantage to calculate the potential first, as an intermediate step.

These are the three fundamental quantities of electrostatics:

ρ, $\vec{E}$, and V.

We have, in the course of our discussion, derived all six (**6**) formulas interrelating them. These equations are neatly summarized in **Fig.** 3.8. We began with just two (**2**) experimental observations:

1. The principle of superposition-a broad general rule applying to **all** electromagnetic forces, and
2. Coulomb's law-the fundamental law of electrostatics.

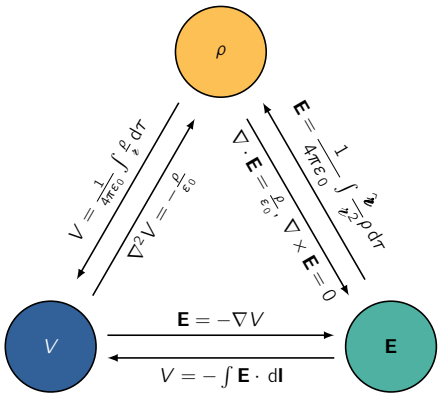


Figure 3.8 The relation between three (3) major variables used in the study of electrostatics.

From these, all else are derived. You may have noticed that the electric field always undergoes a discontinuity when you cross a surface charge σ. In fact, it is a simple matter to find the **amount** by which $\vec{E}$ changes at such a boundary. Suppose we draw a wafer-thin Gaussian pillbox, extending just barely over the edge in each direction as seen in **Fig.** 3.9. Gauss's law says that

$$\oint \vec{E}\cdot \mathrm{d}\vec{a} = \frac{1}{\varepsilon_0}Q_{\text{enc}} = \frac{1}{\varepsilon_0}\sigma A,$$

where A is the area of the pillbox lid (m^2).[15] Now, the sides of the pillbox contribute **nothing** to the flux, in the limit as the thickness ϵ goes to zero, so we are left with:

$$E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}} = \frac{1}{\varepsilon_0}\sigma,$$

where $E^{\perp}_{\text{above}}$ denotes the component of $\vec{E}$ **perpendicular** to the surface immediately above, and $E^{\perp}_{\text{below}}$ is the same, only just below the surface. For consistency, we let "upward" be the positive direction for both.

[15] If σ varies from point to point or the surface is curved, we need to make sure A to be extremely small.

Based on this model we can say normal component of $\vec{E}$ (for clarity it is, $\vec{E}\cdot\hat{\mathbf{n}}$) is discontinuous by an amount σ/ε_0 at any boundary. In particular, where there is **no** surface charge, $E^{\perp}$ is continuous, as for instance at the surface of a uniformly charged solid sphere. The tangential component of vE, by contrast,

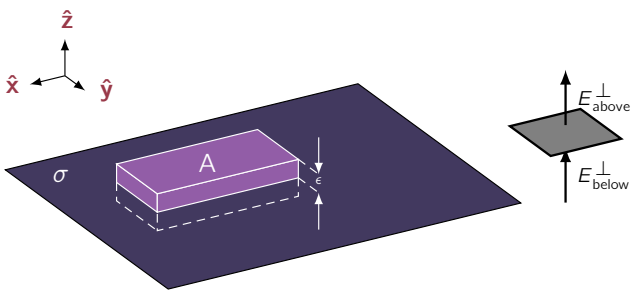


Figure 3.9 A thought experiment on how to figure out the discrepancy when a boundary is changed.

is always continuous.

For if we apply:

$$\oint \vec{E} \cdot \mathrm{d}\vec{l} = 0$$

to the thin rectangular loop of Fig. 2.37, the ends give nothing,[16] and the sides give $(E^{\parallel}_{\text{above}} l - E^{\parallel}_{\text{below}} l)$, so

$$\vec{E}^{\,\parallel}_{\text{above}} = \vec{E}^{\,\parallel}_{\text{below}}$$

where $\vec{E}^{\parallel}$ stands for the components of $\vec{E}$ parallel to the surface. The boundary conditions on E (Eqs. 2.31 and 2.32) can be combined into a single formula:

$$\vec{E}_{\text{above}} - \vec{E}_{\text{below}} = \frac{\sigma}{\varepsilon_0}\hat{\mathbf{n}}, \tag{3.13}$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the surface, pointing from "below" to "above."8 The potential, meanwhile, is continuous across any boundary (Fig. 2.38), since $V_{\text{above}} - V_{\text{below}} = -\int_0^b \vec{E} \cdot \mathrm{d}\vec{l}$; as the path length shrinks to zero, so too does the integral:

$$V_{above} = V_{below}.$$

However, the **gradient** of V inherits the discontinuity in $\vec{E}$ given and following this $\vec{E} = -\nabla V$, which Eq. (**3.13**) implies that:

$$\nabla V_{\text{above}} - \nabla V_{\text{below}} = -\frac{1}{\varepsilon_0}\sigma\hat{\mathbf{n}},$$

or, more conveniently,

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\varepsilon_0}\sigma, \tag{3.14}$$

where

$$\frac{\partial V}{\partial n} = \nabla V \cdot \hat{\mathbf{n}}$$

denotes the **normal derivative** of V.[17] Please note that these boundary conditions relate the fields and potentials just above and **just** below the surface. For example, the derivatives in Eq. (**3.14**) are the **limiting** values as we approach the surface from either side.[18]

[16] as $\epsilon \to 0$.

[17] That is, the rate of change in the direction perpendicular to the surface

[18] Observe that it doesn't matter which side you call **above** and which **below**, given the reversal would switch the direction of $\hat{n}$.

Incidentally, if we're only interested in the field due to the essentially flat local patch of surface charge itself, the answer is $(\sigma/2\varepsilon_0)\,\hat{n}$ immediately above the surface, and $-(\sigma/2\varepsilon_0)\,\hat{n}$ immediately below. Evidently the entire discontinuity in $\vec{E}$ is attributable to this local patch of surface charge.

3.4 Work and Energy

3.4.1 To Move a Charge

To understand how to define work, Let us assume that we have a stationary configuration of source charges, and we want to move a test charge (Q) from point $\vec{a}$ to point $\vec{b}$. Here we need to ask the following question:

How much work will we have to do?

At any point along the path, the electric force on Q is $\vec{F} = Q\vec{E}$; the force we must exert, in opposition to this electrical force, is $-Q\vec{E}$. The work we do is therefore:

$$W = \int_0^b \vec{F} \cdot \mathrm{d}\vec{l} = -Q \int_0^b \vec{E} \cdot \mathrm{d}\vec{l} = Q\left[V\left(\vec{b}\right) - V\left(\vec{a}\right)\right].$$

Please observe the answer is **independent of the path** we take from $\vec{a}$ to $\vec{b}$;

In mechanics, we call the electrostatic force "conservative".[19]

[19] A force with the property which the total work done by the force in moving a particle between two (2) points is independent of the path taken. Equivalently, if a particle travels in a closed loop, the total work done by a conservative force is zero.

Dividing through by Q, we have:

$$V\left(\vec{b}\right) - V\left(\vec{a}\right) = \frac{W}{Q}. \tag{3.15}$$

In other words, the potential difference between points $\vec{a}$ and $\vec{b}$ is equal to the work per unit charge required to carry a particle from $\vec{a}$ to $\vec{b}$. In particular, if we want to bring Q in from far away and stick it at point $\vec{r}$, the work we must do is

$$W = Q\left[V\left(\vec{r}\right) - V(\infty)\right],$$

so, if we have set the reference point at infinity,

$$W = QV(r) \tag{3.16}$$

Here, **potential** is potential **energy**, the work it takes to create the system, per unit charge, just as the field is the force per unit charge.

3.4.2 The Energy of a Point Charge Distribution

How much work would it take to assemble an entire **collection** of point charges? Imagine bringing in the charges, one by one, from far away (Fig. 2.40). The first charge, q_1, takes no work, since there is no field yet to fight against. Now bring in q_2. According to Eq. (**3.16**), this will cost you $q_2 V_1(\mathbf{r}_2)$, where V_1 is the potential due to q_1, and $\mathbf{r}_2$ is the place we're putting q_2:

$$W = \frac{1}{4\pi\varepsilon_0} q_2 \left(\frac{q_1}{q_1 2}\right)$$

(λ_{12}) is the distance between q_1 and q_2 once they are in position). As you bring in each charge, nail it down in its final location, so it doesn't move when you bring in the next charge. Now bring in q_3; this requires work $q_3 V_{1,2}(\mathbf{r}_3)$, where $V_{1,2}$ is the potential due to charges q_1 and q_2, namely, $(1/4\pi\varepsilon_0)(q_1/\nu_{13} + q_2/\nu_{23})$. Thus

$$W_3 = \frac{1}{4\pi\varepsilon_0} q_3 \left(\frac{q_1}{q_{13}} + \frac{q_2}{q_{23}} \right).$$

Similarly, the extra work to bring in q_4 will be

$$W_4 = \frac{1}{4\pi\varepsilon_0} q_4 \left(\frac{q_1}{\lambda_{14}} + \frac{q_2}{\lambda_{24}} + \frac{q_3}{\lambda_{34}} \right).$$

The total work necessary to assemble the first four charges, then, is

$$W = \frac{1}{4\pi\varepsilon_0} \left(\frac{q_1 q_2}{\lambda_{12}} + \frac{q_1 q_3}{\lambda_{13}} + \frac{q_1 q_4}{\lambda_{14}} + \frac{q_2 q_3}{\lambda_{23}} + \frac{q_2 q_4}{\lambda_{24}} + \frac{q_3 q_4}{\lambda_{34}} \right).$$

You see the general rule: Take the product of each pair of charges, divide by their separation distance, and add it all up:

$$W = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{n} \sum_{j>i}^{n} \frac{q_i q_j}{n_{ij}}.$$

The stipulation $j > i$ is to remind you not to count the same pair twice. A nicer way to accomplish this is **intentionally** to count each pair twice, and then divide by 2:

$$W = \frac{1}{8\pi\varepsilon_0} \sum_{i=1}^{n} \sum_{j=i}^{n} \frac{q_i q_j}{n_{ij}}$$

(we must still avoid $i = j$, of course). Notice that in this form the answer plainly does not depend on the **order** in which you assemble the charges, since every pair occurs in the sum. Finally, let's pull out the factor q_i:

$$W = \frac{1}{2} \sum_{i=1}^{n} q_i \left(\sum_{i \neq i}^{n} \frac{1}{4\pi\varepsilon_0} \frac{q_i}{\nu_{ij}} \right).$$

The term in parentheses is the potential at point $\mathbf{r}_i$ (the position of q_i) due to all the **other** charges-all of them, now, not just the ones that were present at some stage during the assembly. Thus,

$$W = \frac{1}{2} \sum_{i=1}^{n} q_i V(\mathbf{r}_i). \tag{3.17}$$

That's how much work it takes to assemble a configuration of point charges; it's also the amount of work you'd get back if you dismantled the system. In the meantime, it represents energy stored in the configuration ("potential" energy, if you insist, though for obvious reasons I prefer to avoid that word in this context).

3.4.3 The Energy of a Continuous Charge Distribution

For a volume charge density (ρ), Eq. (3.17) becomes:[20]

$$W = \frac{1}{2} \int \rho V \, d\tau \tag{3.18}$$

[20] The corresponding integrals for line and surface charges would be $\int \lambda V \, dl$ and $\int \sigma V \, da$.

There is a tidy way to rewrite this result, in which ρ and V are eliminated in favor of $\vec{E}$. First use Gauss's law to express ρ in terms of $\vec{E}$:

$$\rho = \varepsilon_0 \nabla \cdot \vec{E} \qquad \text{so} \qquad W = \frac{\varepsilon_0}{2} \int \left(\nabla \cdot \vec{E} \right) V \, \mathrm{d}\tau$$

Now use integration by parts, to transfer the derivative from $\vec{E}$ to V:

$$W = \frac{\varepsilon_0}{2} \left[- \int \vec{E} \cdot (\nabla V) \, \mathrm{d}\tau + \oint V \vec{E} \cdot \mathrm{d}\vec{a} \right].$$

But we know $\nabla V = -\vec{E}$, so

$$W = \frac{\varepsilon_0}{2} \left(\int_{\mathcal{V}} E^2 \, \mathrm{d}\tau + \oint_{\mathcal{S}} V \vec{E} \cdot \mathrm{d}\vec{a} \right) \tag{3.19}$$

But what volume is this we're integrating over? Let's go back to the formula we started with, Eq. (3.18). From its derivation, it is clear that we should integrate over the region where the charge is located. But actually, any larger volume would do just as well:

> The "extra" territory we throw in will contribute nothing to the integral, since $\rho = 0$ out there.

With this in mind, we return to Eq. (3.19). What happens here, as we enlarge the volume beyond the minimum necessary to trap all the charge? Well, the integral of E^2 can only increase;[21] evidently the surface integral must decrease correspondingly to leave the sum intact.[22]

[21] the integrand being positive.

It is now worth stressing out that Eq. (3.19) gives us the correct energy W, whatever volume we use,[23] but the contribution from the volume integral goes up, and that of the surface integral goes down, as you take larger and larger volumes.

In particular, why not integrate over all space?

[22] In fact, at large distances from the charge, $\vec{E}$ goes $1/r^2$ and V $1/r$, while the surface area grows like r^2; roughly speaking, then, the surface integral goes down like $1/r$.

Then the surface integral goes to zero, and we are left with:

$$W = \frac{\varepsilon_0}{2} \int E^2 \, \mathrm{d}\tau \tag{3.20}$$

[23] as long as it encloses all the charge

3.4.4 Comments on Electrostatic Energy

An Interesting Inconsistency

We can see Eq. (3.20) implies the energy of a stationary charge distribution is always **positive**. On the other hand, Eq. (3.17) which we derived Eq. (3.20), can be either **positive** or **negative**.

For instance, according to Eq. (3.17), the energy of two (2) equal but **opposite** charges a distance $\{^{\eta}\mathrm{r}\}$ apart is $-(1/4\pi\varepsilon_0)(q^2/\{^{\eta}\mathrm{r}\})$.

What's gone wrong? Which equation is correct?

The answer is that **both** are correct, but they speak to slightly different questions. Eq. (**3.17**) does **NOT** take into account the work necessary to make the point charges in the first place; we **started** with point charges and simply found the work required to bring them together. This is wise strategy, given Eq. (**3.20**) indicates that the energy of a point charge is in fact **infinite**:

$$W = \frac{\varepsilon_0}{2\left(4\pi\varepsilon_0\right)^2}\int\left(\frac{q^2}{r^4}\right)\left(r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi\right) = \frac{q^2}{8\pi\varepsilon_0}\int_0^\infty \frac{1}{r^2}\,\mathrm{d}r = \infty.$$

Eq. (**3.20**) is more **complete**, in the sense that it tells you the **total** energy stored in a charge configuration, but Eq. (**3.17**) is more appropriate when we're dealing with point charges, because we prefer to leave out that portion of the total energy that is attributable to the fabrication of the point charges themselves.

In practice, after all, the point charges are given to us ready-made; all we do is move them around. Since we did not put them together, and we cannot take them apart, it is immaterial how much work the process would involve.[24]

[24] Still, the infinite energy of a point charge is a recurring source of embarrassment for electromagnetic theory, afflicting the quantum version as well as the classical.

Now, it is a valid question to ask when the inconsistency crept into an apparently watertight derivation. The "flaw" lies between Eq. (**3.17**) and Eq. (**3.18**):

In the former, $V\left(\vec{r}_{\,i}\right)$ represents the potential due to all the **other** charges **but not** q_i, whereas in the latter, $V\left(\vec{r}\right)$ is the **full** potential. For a **continuous** distribution, there is no distinction, given the amount of charge **right at the point** $\vec{r}$ is vanishingly small, and its contribution to the potential is zero.

> In the presence of point charges we should better stick with Eq. (**3.17**).

The Place where Energy is Stored

Eq. (**3.18**) and Eq. (**3.20**) offer two (**2**) different ways of calculating the same thing.

1. The first is an integral over the charge distribution;
2. The second is an integral over the field.

These can involve completely different regions. For instance, in the case of the spherical shell, the charge is confined to the surface, whereas the electric field is everywhere **outside** this surface.

Where is the energy, then?

Is it stored in the field, as Eq. (**3.20**) seems to suggest, or is it stored in the charge, as Eq. (**3.18**) implies? At the present stage this is simply an unanswerable question: I can tell you what the total energy is, and I can provide you with several different ways to compute it, but it is impertinent to worry about where the energy is located. In the context of radiation theory it is useful to regard the

energy as stored in the field, with a density:

$$\frac{\varepsilon_0}{2}E^2 = \text{energy per unit volume.}$$

But in electrostatics one could just as well say it is stored in the charge, with a density $\frac{1}{2}\rho V$.

> The difference is purely a matter of bookkeeping.

Principle of Superposition

Because electrostatic energy is quadratic in the fields, it does **NOT** obey a superposition principle. The energy of a compound system is **NOT** the sum of the energies of its parts considered separately as there are also "cross terms" to take into account:

$$\begin{aligned} W_{\text{tot}} &= \frac{\varepsilon_0}{2}\int E^2\,\mathrm{d}\tau' = \frac{\varepsilon_0}{2}\int \left(\vec{E}_1 + \vec{E}_2\right)^2\,\mathrm{d}\tau' \\ &= \frac{\varepsilon_0}{2}\int \left(E_1^2 + E_2^2 + 2\vec{E}_1\cdot\vec{E}_2\right)\,\mathrm{d}\tau \\ &= W_1 + W_2 + \varepsilon_0\int \vec{E}_1\cdot\vec{E}_2\,\mathrm{d}\tau. \end{aligned}$$

For example, if you double the charge everywhere, you **quadruple** the total energy.

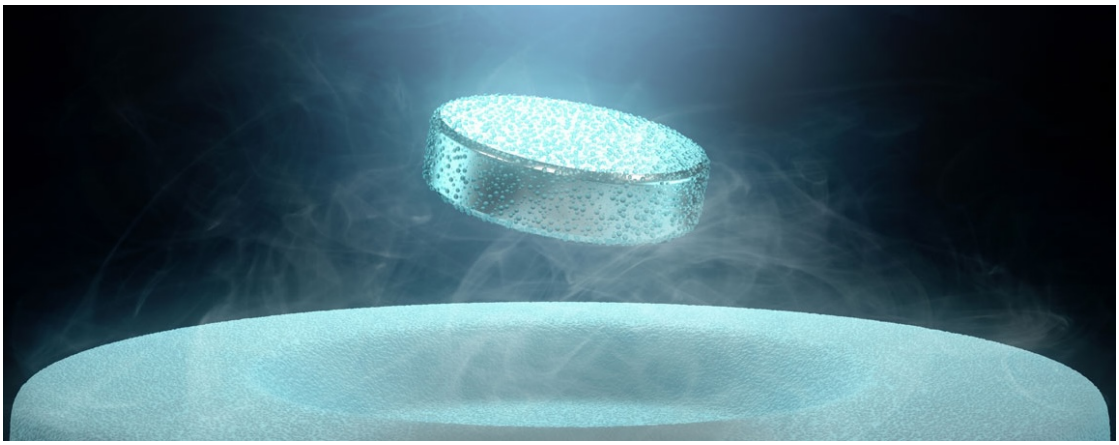


Figure 3.10
Some may say that superconductors can be considered as "perfect conductors", however, they exhibit quantum effects such as the Meissner effect and quantization of magnetic flux. In perfect conductors, the interior magnetic field must remain fixed but can have a zero or nonzero value [1]. In real superconductors, all magnetic flux is expelled during the phase transition to superconductivity (the Meissner effect), and the magnetic field is always zero within the bulk of the superconductor [2].

3.5 Conductors

3.5.1 Properties

In an insulator, for example glass or rubber, each electron is on a short leash, as it is attached to a particular atom. In a metallic **conductor**, by contrast, one or more electrons per atom are free to roam.[25] A **perfect** conductor would contain an **unlimited** supply of free charges.

> In real life there are no such things perfect conductors, but metals come pretty close, for most purposes and therefore we can assume them to be perfect.

From this aforementioned definition, the basic electrostatic properties of ideal conductors immediately follow:

i. No Electric Field Inside a Conductor

Let's have a think about this statement. If there **were** any field, those free charges would **move**, and it wouldn't be electrostatics any more. To give a more concrete explanation, we better examine what happens when we put a conductor into an external electric field $\vec{E}_0$, seen in **Fig.** 3.12.

Initially, the field will drive any free positive charges to the right, and negative ones to the left:[26] When they come to the edge of the material, the charges pile up: plus on the right side, minus on the left. Now, these induced charges produce a field of their own ($\vec{E}_1$), which, as we can see from the figure, is in the **opposite direction** to $\vec{E}_0$.

That's the crucial point, for it means the field of the induced charges tends to cancel the original field. Charge will continue to flow until this cancellation is complete, and the resultant field inside the conductor is precisely zero.[27] The whole process is practically instantaneous.

[25] In liquid conductors such as salt water, it is jons that do the moving. However, for now we shall only focus on solid materials.

[26] In practice, it's the electrons that do the moving, but when they depart, the right side is left with a net positive charge, so it doesn't really matter which charges move; the effect is the same.

[27] Outside the conductor the field is not zero, for here $\vec{E}_0$ and $\vec{E}_1$ do **NOT** tend to cancel.

ii. Charge Density is Zero inside a Conductor

This follows from Gauss's law:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

If $\vec{E}$ is zero, then by extension, so is ρ. There is still charge around, but exactly as much plus as minus, so the **net** charge density in the interior is zero.

iii. Any Net Charge is on the Surface

That's the only place left.

iv. A Conductor is an Equipotential

For if $\vec{a}$ and $\vec{b}$ are any two (**2**) points within (or at the surface of) a given conductor,

$$V\left(\vec{b}\right) - V\left(\vec{a}\right) = -\int_a^b \vec{E} \cdot \mathrm{d}\vec{l} = 0, \quad \text{therefore} \quad V\left(\vec{b}\right) = V\left(\vec{a}\right).$$

v. E is Perpendicular to the Surface

Otherwise, charge will immediately flow around the surface until it kills off the tangential component which are shown in **Fig.** 3.11 .[28]

[28] Perpendicular to the surface, charge cannot flow, of course, since it is confined to the conducting object.

It is an astonishing fact the charge on a conductor flows to the surface. Given their mutual repulsion, the charges naturally **spread out** as much as possible, but for all of them to go to the surface seems like a waste of the interior space. Surely we could do better, from the point of view of making each charge as far as possible from its neighbors, to sprinkle **some** of them throughout the volume Well, it simply is not so. You do best to put all the charge on the surface, and this is true regardless of the size or shape of the conductor.

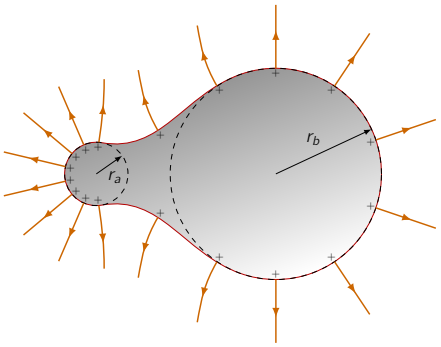


Figure 3.11
A visual representation of treating a non-standard body and the generated electric field. Please observe that all lines flowing from the body is perpendicular.

The problem can also be phrased in terms of **energy**.

Like any other free dynamical system, the charge on a conductor will seek the configuration that minimizes its potential energy. What property (**iii**) asserts is that the electrostatic energy of a solid object[29] is a minimum when charge is spread over the surface.

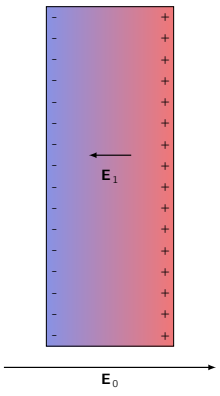


Figure 3.12
A visual representation of the electric field inside a conductor.

[29] with specified shape and total charge

For instance, the energy of a sphere is

$$(1/8\pi\varepsilon_0)(q^2/R)$$

if the charge is uniformly distributed over the surface, but it is,

$$(3/20\pi\varepsilon_0)(q^2/R)$$

if the charge is uniformly distributed throughout the volume

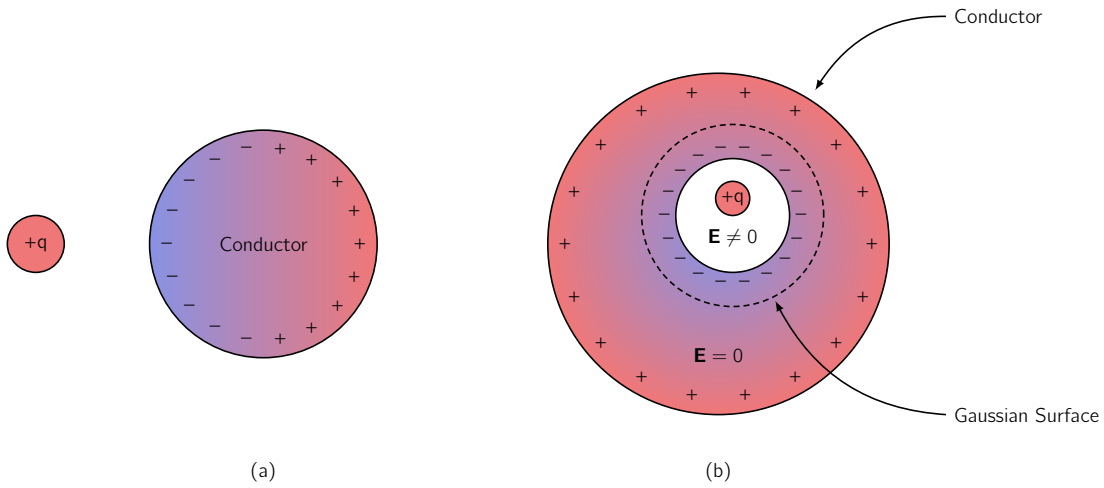


Figure 3.13
Charges induced by a point charge in a conductor (a) When a charge approaches a conductor the charges will start to move where opposite charges will move towards the point charge and same charges will move away. While there is an imbalance now, the overall charge of the conductor remains the same (b) When a charge is given inside the cavity, again similar behaviour is to be expected inside the conductor.

3.5.2 Induced Charges

If we were to hold a charge $(+q)$ near an **uncharged** conductor seen in **Fig.** 3.13a, the two (**2**) will attract one another. The reason for this is that $+q$ will pull minus charges $(-q)$ over to the near side and repel plus charges to the far side.[30] Since the negative induced charge is closer to q, there is a net force of attraction.

[30] Another way to think of it is that the charge moves around in such a way as to kill off the field of q for points inside the conductor, where the total field must be zero.

When we speak of the field, charge, or potential "inside" a conductor, we should think of the "meat" of the conductor; if there is some hollow **cavity** in the conductor, and within that cavity we put some charge, then the field in the cavity will **NOT** be zero. But in a remarkable way the cavity and its contents are electrically isolated from the outside world by the surrounding conductor, shown in **Fig.** 3.13b.

No external fields penetrate the conductor; they are cancelled at the outer surface by the induced charge there. Similarly, the field due to charges within the cavity is cancelled, for all exterior points, by the induced charge on the inner surface. However, the compensating charge left over on the outer surface of the conductor effectively "communicates" the presence of q to the outside world.

The total charge induced on the cavity wall is equal and opposite to the charge inside, for if we surround the cavity with a Gaussian surface, all points of which are in the conductor (**Fig.** 3.13b), $\oint \vec{E} \cdot \mathrm{d}\vec{a} = 0$, and therefore with Gauss's law, the net enclosed charge must be zero (**0**). But $Q_{\text{enc}} = q + q_{\text{induced}}$, so $q_{\text{induced}} = -q$. Then if the conductor as a whole is electrically neutral, there must be a charge $+q$ on its outer surface.

If a cavity surrounded by conducting material is itself empty of charge, then the field within the cavity is zero. For any field line would have to begin and end on the cavity wall, going from a plus charge to a minus charge, shown in **Fig.** 3.14.

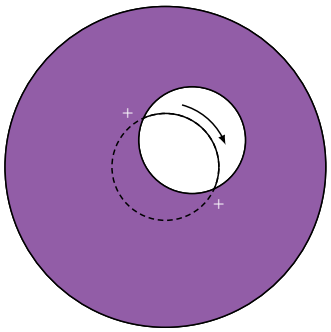


Figure 3.14
Visual description of an electric field inside a cavity. As can be seen all the lines generated by the positive side of the edge is cancelled by the negative side with it absorbs making the inside of the cavity have 0 field.

Letting that field line be part of a closed loop, the rest of which is entirely inside the conductor (where $\vec{E} = 0$), the integral $\oint \vec{E} \cdot d\vec{l}$ is distinctly **positive**, in violation of $\oint \vec{E} \cdot d\vec{l} = 0$. It follows that $\vec{E} = 0$ within an **empty** cavity, and there is in fact **no** charge on the surface of the cavity.[31]



[31] This is why we are **relatively safe** inside a metal car during a thunderstorm-you may get cooked, if lightning strikes, but we will **NOT** be **electrocuted**. The same principle applies to the placement of sensitive apparatus inside a grounded Faraday cage, to shield out stray electric fields. In practice, the enclosure doesn't even have to be solid conductor-chicken wire will often suffice [3].

3.5.3 Surface Charge and the Force on a Conductor

Because the field inside a conductor is zero (**0**), boundary condition 2.33 requires the field immediately **outside** is:

$$E = \frac{\sigma}{\varepsilon_0}\hat{\mathbf{n}}, \tag{3.21}$$

consistent with our earlier conclusion that the field is **normal to the surface**. In terms of potential, Eq. (**3.14**) gives us:

$$\sigma = -\varepsilon_0 \frac{\partial[\,]}{\partial V}][n]$$

These equations enable us to calculate the surface charge on a conductor, if we can determine $\vec{E}$ or V.[32] In the presence of an electric field, a surface charge will experience a force; the force per unit area ($\vec{f}$), is $\sigma\vec{E}$.

[32] we shall use them frequently in the next chapter.

But there's a problem here:

> The electric field is **discontinuous** at a surface charge.

So what are we supposed to use: E_{above}, E_{below}, or something in between? The answer is that we should use the average of the two:

$$\vec{f} = \sigma\vec{E}_{\text{average}} = \frac{1}{2}\sigma\left(\vec{E}_{\text{above}} + \vec{E}_{\text{below}}\right).$$

You may ask:, why the average?

The reason is very simple, though the telling makes it sound complicated: Let's focus our attention on a tiny patch of surface surrounding the point in question (Fig. 2.50).[33] The total field consists of two parts-that attributable to the patch itself, and that due to everything else (other regions of the surface, as well as any external sources that may be present): $\vec{E} = \vec{E}_{\text{patch}} + \vec{E}_{\text{other}}$. Now, the patch cannot exert a force on itself, any more than you can lift yourself by standing in a basket and pulling up on the handles. The force on the patch, then, is due exclusively to E_{other}, and this suffers no discontinuity (if we removed the patch, the field in the "hole" would be perfectly smooth). The discontinuity is due entirely to the charge on the patch, which puts out a field $(\sigma/2\varepsilon_0)$ on either side, pointing away from the surface. Thus,

[33] Make it small enough so it is essentially flat and the surface charge on it is essentially constant.

$$\vec{E}_{\text{above}} = \vec{E}_{\text{other}} + \frac{\sigma}{2\varepsilon_0}\hat{\mathbf{n}}, \qquad \vec{E}_{\text{below}} = \vec{E}_{\text{other}} - \frac{\sigma}{2\varepsilon_0}\hat{\mathbf{n}}. \qquad \text{therefore,} \qquad \vec{E}_{\text{other}} = \frac{1}{2}(\vec{E}_{\text{above}} + \vec{E}_{\text{below}}) = \vec{E}_{\text{average}}.$$

Averaging is really just a device for removing the contribution of the patch itself. That argument applies to **any** surface charge; in the particular case of a conductor, the field is zero inside and

(σ/ε_0) m outside as shown in Eq. (3.21), so the average is $(\sigma/2\varepsilon_0)\hat{\mathbf{n}}$, and the force per unit area is

$$\vec{f} = \frac{1}{2\varepsilon_0}\sigma^2\hat{\mathbf{n}}.$$

This amounts to an outward **electrostatic pressure** on the surface, tending to draw the conductor into the field, regardless of the sign of σ. Expressing the pressure in terms of the field just outside the surface,

$$P = \frac{\varepsilon_0}{2}E^2.$$

3.5.4 Capacitors

Suppose we have two (2) conductors, and we put charge $+Q$ on one and $-Q$ on the other as shown in **Fig.** 3.15. Since V is **constant** over a conductor, we can speak unambiguously of the potential difference between them:

$$V = V_+ - V_- = -\int_-^+ \vec{E}\cdot \mathrm{d}\vec{l}.$$

We don't know how the charge distributes itself over the two conductors, and calculating the field would be a nightmare, if their shapes are complicated, but this much we do know:

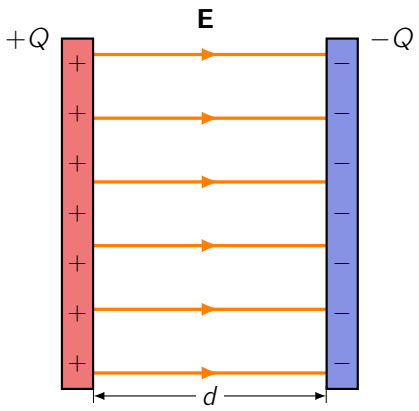


Figure 3.15
A parallel-plate capacitor, as shown in part A, consists of two flat conducting plates. Here one plate is positively charged and the other one is charged negatively. At a distance of d apart, an electric field ($\vec{E}$) is generated between the two (2) plates.

E is proportional to Q.

For $\vec{E}$ is given by Coulomb's law:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho}{\lambda^2}\hat{\mathbf{z}}\,\mathrm{d}\tau, \quad \mathrm{V\cdot m^{-1}}$$

so if we double ρ, we double $\vec{E}$.[34] Since $\vec{E}$ is proportional to Q, so also is $\vec{V}$. The constant of proportionality is called the **capacitance** of the arrangement:

$$C \equiv \frac{Q}{V} \qquad [\mathrm{V\cdot C^{-1}\ or\ F}] \quad (3.22)$$

Capacitance is a **purely geometrical quantity**, determined by the sizes, shapes, and separation of the two (2) conductors. In SI units, C is measured in farad (F); a farad is a coulomb-per-volt. This turns out to be inconveniently large; more practical units are the microfarad (1×10^{-6} F) and the picofarad (1×10^{-12} F).

Notice V is, by definition, the potential of the **positive** conductor less that of the negative one; likewise, Q is the charge of the **positive** conductor. Accordingly, capacitance is an intrinsically positive quantity.[35]

[34] Doubling Q doubles ρ everywhere, it doesn't shift the charge around.

[35] We may occasionally hear someone speak of the capacitance of a single conductor. In this case the "second conductor", with the negative charge, is an imaginary spherical shell of infinite radius surrounding the one conductor. It contributes **nothing** to the field, so the capacitance is given by Eq. (3.22), where V is the potential with infinity as the reference point.

Charging a Capacitor

To "charge up" a capacitor, so to say, we have to remove electrons from the positive plate and carry them to the negative plate. By doing so, we fight against the electric field, which is pulling them back toward the positive conductor and pushing them away from the negative one.

How much work does it take, then, to charge the capacitor up to a final amount Q?

Suppose that at some intermediate stage in the process the charge on the positive plate is q, so the potential difference is q/C. According to Eq. (**3.15**), the work we need do to transport the next piece of charge, $\mathrm{d}q$, is:

$$\mathrm{d}W = \left(\frac{q}{C}\right)\mathrm{d}q.$$

The total work necessary, then, to go from $q = 0$ to $q = Q$, is

$$W = \int_0^Q \left(\frac{q}{C}\right)\mathrm{d}q = \frac{1}{2}\frac{Q^2}{C},$$

or, since $Q = CV$,

$$W = \frac{1}{2}CV^2$$

where V is the final potential of the capacitor.



Part III

Magnetic Fields

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The experimental investigation by which Ampère established the law of the mechanical action between electric currents is one of the most brilliant achievements in science.

The whole, theory and experiment, seems as if it had leaped, full grown and full armed, from the brain of the "Newton of electricity". It is perfect in form, and unassailable in accuracy, and it is summed up in a formula from which all the phenomena may be deduced, and which must always remain the cardinal formula of electro-dynamics.

(James Clerk Maxwell in A Treatise on Electricity and Magnetism (1873) quoted from 3rd edition (1892) Vol. 2, Ch. 3, p. 175.)

Chapter 4

Magnetic Fields in Matter

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4.1 Magnetisation

Magnetism is the class of physical attributes which occur through a **magnetic field**, allowing objects to either **attract** or **repel** each other. Because both electric currents and magnetic moments of elementary particles give rise to a magnetic field, magnetism is one of two (**2**) aspects of electromagnetism.

In this chapter we will shift our focus into materials[1] which exhibit magnetism through various forms.



[1] As we now, bar magnets are generally attributed to exhibiting magnetism, but nature has more variety.

4.1.1 Diamagnets, Paramagnets, and Ferromagnets

If the topic of **magnetism** ever comes to play during lunch break,[2] you may think about refrigerator decorations, compass needles, and the North Pole which initially seems to have no obvious connection with moving charges or current-carrying wires.

[2] Perhaps when you are discussing the current events or why the elevator is again under maintenance at MCI IV.

However, it is important to state, all magnetic phenomena are cause by **electric charges in motion**. If we could examine a piece of magnetic material on an atomic scale we would find tiny **currents**:

electrons orbiting around nuclei and spinning about their axes.

For macroscopic domain,[3] these current loops are **minuscule** so we treat them as magnetic dipoles. Ordinarily, they cancel each other out due of the **random orientation of the atoms**. But when a magnetic field is applied, a **net alignment** of these magnetic dipoles occurs, and the medium becomes magnetically polarised, or **magnetised**.

[3] The domain in which most day-to-day events occur.

Unlike electric polarisation we have looked previously, which is **almost** always in the same direction as the **E**-field, some materials acquire a magnetisation **parallel** to **B**, called **paramagnets** and some **opposite** to **B**, called **diamagnets**. A few substances[4] retain their magnetization even after the external field has been removed-for these, the magnetisation is **NOT** determined by the **present** field but by the whole magnetic "history" of the object.

[4] called **ferromagnets**, named after to the most common example, iron (Fe).

Lode Stones — 4.1 Information

Lodestones are naturally magnetized pieces of the mineral magnetite [2, 3]. They are naturally occurring magnets, which can attract iron. The property of magnetism was first discovered in antiquity through lodestones [4]. Pieces of lodestone, suspended so they could turn, were the first magnetic compasses, [4] and their importance to early navigation is indicated by the name lodestone, which in Middle English means "course stone" or "leading stone", from the now-obsolete meaning of lode as "journey, way".

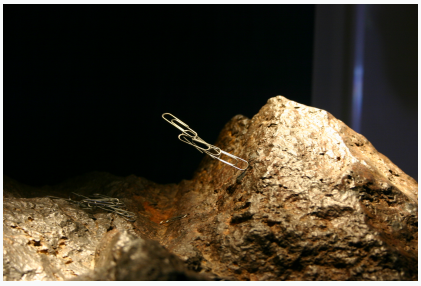


Figure 4.1: Lodestone in the Hall of Gems of the Smithsonian [1].

Lodestone is one of only a very few minerals that is found naturally magnetised.

Permanent Magnet (PM)s made of iron are the most familiar examples of magnetism, but from a theoretical point of view they are the most complicated.

We will have a look at ferromagnetism for the end of the chapter, and begin with qualitative models of paramagnetism and diamagnetism.

4.1.2 Torques and Forces on Magnetic Dipoles

A magnetic dipole experiences a torque under the effect of a magnetic field (**B**), similar to an electric dipole in an electric field.

Let's calculate the torque on a rectangular current loop in a uniform field **B**.[5] To begin, centre the loop at the origin, and tilt it an angle θ from the $\hat{\mathbf{z}}$-axis towards the $\hat{\mathbf{y}}$-axis (Fig. 6.2).

[5] Since any current loop could be built up from infinitesimal rectangles, with all the "internal" sides canceling, as indicated in Fig. 6.1, there is no real loss of generality here.

Let **B** point in the $\hat{\mathbf{z}}$ direction. The forces on the two (2) sloping sides cancel.[6] The forces on the "horizontal" sides are likewise equal and opposite, which makes the net **force** on the loop is zero, but

[6] they tend to **stretch** the loop, but they don't **rotate** it

regardless they generate a torque:

$$\mathbf{N} = aF\sin\theta\hat{\mathbf{x}}.$$

The magnitude of the force on each of these segments is

$$F = IbB,$$

and therefore

$$\mathbf{N} = IabB\sin\theta\hat{\mathbf{x}} = mB\sin\theta\hat{\mathbf{x}} \qquad \text{or} \qquad \mathbf{N} = \mathbf{m}\times\mathbf{B} \tag{4.1}$$

where $m = Iab$ is the magnetic dipole moment of the loop with Eq. (4.1) giving the torque on any localized current distribution, in the presence of a **uniform** field; in a nonuniform field it is the exact torque[7] for a perfect dipole of infinitesimal size.

[7] about the center.

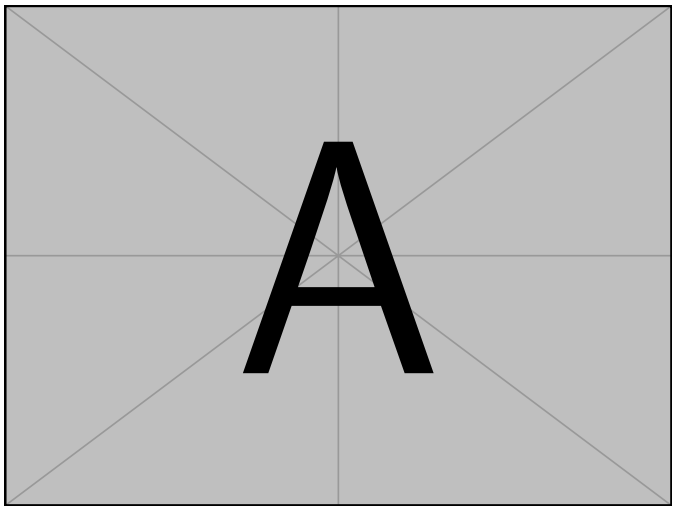


Figure 4.2: FIgure 6.1

Here, Eq. (4.1) is **identical** in form to the electrical analog, Eq. 4.4: $N = p\times E$. In particular, the torque is again in such a direction as to line the dipole up **parallel** to the field.
It is this torque which accounts for **paramagnetism** effect.

Given every electron constitutes a magnetic dipole,[8] we might expect paramagnetism to be a **universal phenomenon**.

[8] picture it, if you wish, as a tiny spinning sphere of charge

Actually, quantum mechanics[9] tends to lock the electrons within a given atom together in pairs with opposing spins,[10] and this effectively **neutralises** the torque on the combination.

[9] specifically, the Pauli exclusion principle.

As a result, paramagnetism most often occurs in atoms or molecules with an **odd number** of electrons, where the "extra" unpaired member is subject to the magnetic torque.

[10] This is **NOT** always true for the outermost electrons in unfilled shells.

Even here, the alignment is far from complete, given random thermal collisions tend to destroy the order.

In a uniform field, the net **force** on a current loop is zero:

$$\mathbf{F} = I \oint (d\mathbf{l} \times \mathbf{B}) = I \left(\oint d\mathbf{l} \right) \times \mathbf{B} = 0;$$

Here, the constant **B** comes **outside the integral**, and the net displacement $\left(\oint d\mathbf{l}\right)$ around a closed loop vanishes. In a **non-uniform** field this is no longer the case.

As an example, assume a circular wire ring of radius of R, carrying a current of I, is suspended above a short solenoid in the "fringing" region (Fig. 6.3). Here **B** has a radial component, and there is a net downward force on the loop (Fig. 6.4):

$$F = 2\pi I R B \cos\theta.$$

For an **infinitesimal** loop, with dipole moment **m**, in a field **B**, the force is:

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$$

Once again the magnetic formula is identical to its electrical "twin," if we write the latter in the form of:

$$\mathbf{F} = \nabla\left(\mathbf{p} \cdot \mathbf{E}\right)$$

4.2 Information

An Unfortunate Analogy

If you're starting to get a sense of deja vu, perhaps you will have more respect for those early physicists who thought magnetic dipoles consisted of positive and negative magnetic "charges" (north and south "poles," they called them), separated by a small distance, just like electric dipoles (Fig. 6.5(a)). They wrote down a "Coulomb's law" for the attraction and repulsion of these poles, and developed the whole of magnetostatics in exact analogy to electrostatics. It's not a bad model, for many purposes-it gives the correct field of a dipole (at least, away from the origin), the right torque on a dipole (at least, on a stationary dipole), and the proper force on a dipole (at least, in the absence of external currents). But it's bad physics, because there's no such thing as a single magnetic north pole or south pole. If you break a bar magnet in half, you don't get a north pole in one hand and a south pole in the other; you get two complete magnets. Magnetism is not due to magnetic monopoles, but rather to moving electric charges; magnetic dipoles are tiny current loops (Fig. 6.5(c)), and it's an extraordinary thing, really, that the formulas involving **m** bear any resemblance to the corresponding formulas for **p**. Sometimes it is easier to think in terms of the "Gilbert" model of a magnetic dipole (separated monopoles), instead of the physically correct "Ampere" model (current loop).

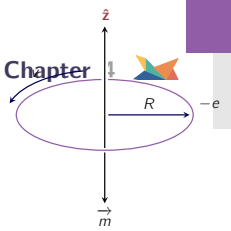
Indeed, this picture occasionally offers a quick and clever solution to an otherwise cumbersome problem

We just copy the corresponding result from electrostatics, changing:

$$p \to m \qquad \frac{1}{\varepsilon_0} \to \mu_0 \qquad \mathbf{E} \to \mathbf{B}.$$

However, if we want to have a closer look at the features of the dipoles, the two models can produce significant results.

4.1.3 Effect of a Magnetic Field on Atomic Orbits



[11] A visualisation of the electron orbit.

Electrons **NOT** only **spin** as they also **revolve** around the nucleus. To give a better understanding of the aforementioned statement, let's assume the orbit is a circle[11] of radius R.

While, under the strictest technical definitions, this orbital motion does **NOT** constitute a steady current, in practice the period:

$$T = 2\pi R/v$$

is imperceptibly short that we might as well assume it is **steady-current**:[12]

[12] The minus sign accounts for the negative charge of the electron.

$$I = \frac{-e}{T} = -\frac{ev}{2\pi R}.$$

Following this chain of thought, the orbital dipole moment $\left(I\pi R^2\right)$ is:

$$\mathbf{m} = -\frac{1}{2}evR\hat{\mathbf{z}}. \tag{4.2}$$

Similar to any other magnetic dipole, this one is subject to a torque ($\mathbf{m} \times \mathbf{B}$) when we turn on a magnetic field.

> But it's a lot harder to tilt the entire orbit than it is the spin, so the orbital contribution to paramagnetism is small.

There is, however, a more significant effect on the orbital motion which is worth mention and it is that the electron speeds up or slows down, depending on the orientation of **B**. For whereas the centripetal acceleration v^2/R is ordinarily sustained by electrical forces alone,

> To avoid confusion with the magnetic dipole moment m, we will write the electron mass with subscript: m_e.

$$\frac{1}{4\pi\epsilon_0}\frac{e^2}{R^2} = m_e\frac{v^2}{R},$$

in the presence of a magnetic field there is an additional force, described as:

$$-e(\mathbf{v} \times \mathbf{B})$$

For the sake of argument, let's say that **B** is perpendicular to the plane of the orbit, as shown in Fig. 6.10; then

$$\frac{1}{4\pi\epsilon_0}\frac{e^2}{R^2} + e\bar{v}B = m_e\frac{\bar{v}^2}{R}.$$

Under these conditions, the new speed $\bar{v}$ is **greater** than v:

$$e\bar{v}B = \frac{m_e}{R}(\bar{v}^2 - v^2) = \frac{m_e}{R}(\bar{v} + v)(\bar{v} - v),$$

or, assuming the change $\Delta v = \bar{v} - v$ is small,

$$\Delta v = \frac{eRB}{2m_e}.$$

When **B** is turned on, then, the electron speeds up.[13] A change in orbital speed means a change in the dipole moment (Eq. (4.2)):

$$\Delta \mathbf{m} = -\frac{1}{2} e(\Delta v) R \hat{\mathbf{z}} = -\frac{e^2 R^2}{4 m_e} \mathbf{B}. \tag{4.3}$$

Notice that the change in **m** is opposite to the direction of **B**. (An electron circling the other way would have a dipole moment pointing upward, but such an orbit would be slowed down by the field, so the **change** is still opposite to **B**.) Ordi- narily, the electron orbits are randomly oriented, and the orbital dipole moments cancel out. But in the presence of a magnetic field, each atom picks up a little "extra" dipole moment, and these increments are all **antiparallel** to the field. This is the mechanism responsible for diamagnetism. It is a universal phenomenon, affecting all atoms. However, it is typically much weaker than paramagnetism, and is therefore observed mainly in atoms with even numbers of electrons, where paramagnetism is usually absent.

In deriving Eq. (4.3), I assumed that the orbit remains circular, with its original radius R. I cannot offer a justification for this at the present stage. If the atom is stationary while the field is turned on, then my assumption can be proved- this is not magnetostatics, however, and the details will have to await Chapter 7 (see Prob. 7.52). If the atom is moved into the field, the situation is enormously more complicated. But never mind-I'm only trying to give you a qualitative account of diamagnetism. Assume, if you prefer, that the velocity remains the same while the **radius** changes-the formula (Eq. (4.3)) is altered (by a factor of 2), but the qualitative conclusion is unaffected. The truth is that this classical model is fundamentally flawed,[14] so there's not much point in refining the details. What is important is the empirical fact that in diamagnetic materials the induced dipole moments point opposite to the magnetic field.

[13] I said (Eq. 5.11) that magnetic fields do no work, and are incapable of speeding a particle up. I stand by that. However, as we shall see in Chapter 7, a changing magnetic field induces an electric field, and it is the latter that accelerates the electrons in this instance.

[14] diamagnetism is really a **quantum** phenomenon.

4.1.4 Magnetization

In the presence of a magnetic field, matter becomes **magnetized**; that is, upon microscopic examination, it will be found to contain many tiny dipoles, with a net alignment along some direction. We have discussed two mechanisms that account for this magnetic polarization:

1. paramagnetism (the dipoles associated with the spins of unpaired electrons experience a torque tending to line them up parallel to the field), and

2. diamagnetism (the orbital speed of the electrons is altered in such a way as to change the orbital dipole moment in a direction opposite to the field)

Whatever the cause, we describe the state of magnetic polarization by the vector quantity

$$\mathbf{M} \equiv \text{magnetic dipole moment per unit volume}$$

where **M** is called the magnetization and plays a role analogous to the polarization **P** in electrostatics. In the following section, we will not worry about how the magnetization **got** there-it could be paramagnetism, diamagnetism, or even ferromagnetism-we shall take M as given, and calculate the field this magnetization itself produces.

Incidentally, it may have surprised you to learn that materials other than the famous ferromagnetic trio (iron, nickel, and cobalt) are affected by a magnetic field at all. You cannot, of course, pick up a piece of wood or aluminum with a magnet. The reason is that diamagnetism and paramagnetism are extremely weak:

> It takes a delicate experiment and a powerful magnet to detect them at all.

If you were to suspend a piece of paramagnetic material above a solenoid, as in Fig. 6.3, the induced magnetization would be upward, and hence the force downward. By contrast, the magnetization of a diamagnetic object would be downward and the force upward. In general, when a sample is placed in a region of nonuniform field, the paramagnet is attracted into the field, whereas the diamagnet is repelled away. But the actual forces are pitifully weak-in a typical experimental arrangement the force on a comparable sample of iron would be 10^4 or 10^5 times as great. That's why it was reasonable for us to calculate the field inside a piece of copper wire, say, in Chapter 5, without worrying about the effects of magnetization.

4.2 The Field of a Magnetised Object

4.2.1 Bound Currents

Let us assume we have a piece of magnetized material and the magnetic dipole moment per unit volume ($\mathbf{M}$) is given.

What field does this object produce?

Well, the vector potential of a single dipole $\mathbf{m}$ is given by Eq. 5.85:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathrm{r}\}^2}.$$

In the magnetized object, each volume element $\mathrm{d}\tau'$ carries a dipole moment $\mathbf{M}\,\mathrm{d}\tau'$, so the total vector potential is (Fig. 6.11)

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{M}(\mathbf{r}')\times\{^{\eta}\mathbf{r}\}}{\{^{\eta}\hat{\mathbf{r}}\}^2}\,\mathrm{d}\tau'. \tag{4.4}$$

That does it, in principle. But, similar to the electrical case, the integral can be cast in a more illuminating form by exploiting the following identity:

$$\nabla'\frac{1}{\{^{\eta}\mathrm{r}\}} = \frac{\{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathrm{r}\}^2}.$$

Using the aforementioned identity, we can re-express the equation in Eq. (4.4) as:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int\left[\mathbf{M}(\mathbf{r}')\times\left(\nabla'\frac{1}{\{^{\eta}\mathrm{r}\}}\right)\right]\mathrm{d}\tau'.$$

Integrating by parts, using product rule 7, gives

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\left\{\int\frac{1}{\{^{\eta}\mathrm{r}\}}[\nabla'\times\mathbf{M}(\mathbf{r}')]\,\mathrm{d}\tau' - \int\nabla'\times\left[\frac{\mathbf{M}(\mathbf{r}')}{\{^{\eta}\mathrm{r}\}}\right]\mathrm{d}\tau'\right\}.$$

Let's express the latter as a surface integral,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int\frac{1}{\lambda}[\nabla'\times\mathbf{M}(\mathbf{r}')]d\tau' + \frac{\mu_0}{4\pi}\oint\frac{1}{\lambda}[\mathbf{M}(\mathbf{r}')\times d\mathbf{a}'].$$

The first term looks just like the potential of a volume current,

$$\mathbf{J}_b = \nabla\times\mathbf{M}, \tag{4.5}$$

while the second looks like the potential of a surface current,

$$K_b = \mathbf{M} \times \hat{\mathbf{n}}$$

where $\hat{\mathbf{n}}$ is the normal unit vector. With these definitions,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{\mathcal{V}} \frac{\overrightarrow{J}_b(\mathbf{r}')}{\{\eta \mathrm{r}\}} \, \mathrm{d}\tau' + \frac{\mu_0}{4\pi} \oint_{\mathcal{S}} \frac{\overrightarrow{K}_b(\mathbf{r}')}{\{\eta \mathrm{r}\}} \, \mathrm{d}a'.$$

What this means is that the potential[15] of a magnetized object is the same as would be produced by a volume current $J_b = \nabla \times M$ throughout the material, plus a surface current $\overrightarrow{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$, on the boundary. Instead of integrating the contributions of all the infinitesimal dipoles, using Eq. (**4.4**), we first determine the **bound currents**, and then find the field **they** produce, in the same way we would calculate the field of any other volume and surface currents. Notice the striking parallel with the electrical case: there the field of a polarized object was the same as that of a bound volume charge $\rho_b = -\nabla \cdot \mathbf{P}$ plus a bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$.

[15] and hence also the field.

⁶It is no accident that the same factors appear in the "contact" term for the fields of electric and magnetic dipoles (Eqs. 3.106 and 5.94). In fact, one good way to model a perfect dipole is to take the limit ($R \to 0$) of a polarized/magnetized sphere.

4.2.2 Physical Interpretation of Bound Currents

In the last section, we found that the field of a magnetized object is identical to the field which would have been produced by a certain distribution of **bound** currents, $\mathbf{J}_b$ and $\mathbf{K}_b$.

It is now worth to have a look at how these bound currents arise physically. This will be a heuristic argument.[16] Figure 6.15 depicts a thin slab of uniformly magnetized material, with the dipoles represented by tiny current loops. Please observe all the **internal** currents cancel each other.

[16] A heuristic argument is an argument that reasons from the value of a method or principle that has been shown experimentally, usually through trial-and-error.

> Every time there is one going to the right, a contiguous one is going to the left.

However, at the edge there is no adjacent loop to do the canceling. The whole thing, then, is equivalent to a single ribbon of current I flowing around the boundary (Fig. 6.16).

What is this current, in terms of M?

Let's assume each of the tiny loops has area a and thickness t (Fig. 6.17). In terms of the magnetization M, its dipole moment is $m = Mat$. In terms of the circulating current I, however, $m = Ia$. Therefore $I = Mt$, so the surface current is $K_b = I/t = M$. Using the outward-drawn unit vector $\hat{\mathbf{n}}$ (Fig. 6.16), the direction of $\mathbf{K}_b$ is conveniently indicated by the cross product:[17]

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$$

[17] This expression also records the fact that there is no current on the top or bottom surface of the slab; here M is parallel to n̂, so the cross product vanishes.

This bound surface current is exactly what we obtained in Sect. 6.2.1.

It is a special **kind** of current, as *no single charge makes the whole trip*.

Each charge moves only in a tiny little loop within a single atom.

Nevertheless, the net effect is a **macroscopic** current flowing over the surface of the magnetized object. We call it a **bound current** as every charge is attached to a particular atom.

It's an actual physical current, producing a magnetic field in the same way any other current does.

When the magnetization is **non-uniform**, the internal currents no longer cancel. Figure 6.18(a) shows two adjacent chunks of magnetized material, with a larger arrow on the one to the right suggesting greater magnetization at that point. On the surface where they join, there is a net current in the x-direction, given by:

$$I_x = [M_z(y + dy) - M_z(y)]\,\mathrm{d}z = \frac{\partial M_z}{\partial y}\,\mathrm{d}y\,\mathrm{d}z, \quad \text{with volume current being} \quad (J_b)_x = \frac{\partial M_z}{\partial y}.$$

Similarly, a nonuniform magnetization in the y-direction would contribute an amount $-\frac{\partial M_y}{\partial z}$ (Fig. 6.18(b)), so

$$(J_b)_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}.$$

In general, then, $\mathbf{J}_b = \nabla \times \mathbf{M}$ is consistent, again, with the result of Sect. 6.2.1. Incidentally, like any other steady current, Jb should obey the conservation law in Eq. (**??**):

$$\nabla \cdot \mathbf{J}_b = 0,$$

given for the divergence of a curl is always zero.

The Magnetic Field Inside Matter 4.3 Information

Similar to the **electric field**, the actual **microscopic** magnetic field inside matter fluctuates wildly from point to point and instant to instant.

When we speak of magnetic field in matter, we mean the **macroscopic** field: the average over regions large enough to contains numerous atoms.[18] It is this macroscopic field that one obtains when the methods of Sect. 6.2.1 are applied to points inside magnetized material.

[18] The magnetization M is *smoothed out* in the same sense.

4.3 The Auxiliary H Field

4.3.1 Ampère's Law in Magnetized Materials

In the previous section, we found that the effect of magnetization is to establish bound currents $\mathbf{J}_b = \nabla \times \mathbf{M}$ within the material and $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ on the surface. The field due to magnetization of the medium is just the field produced by these bound currents.

It is time we put everything together:

> The field attributable to bound currents, plus the field due to everything else, which we will call the **free current**.

> The free current might flow through wires embedded in the magnetized substance or, if the latter is a conductor, through the material itself.

Regardless of any event, the total current can be written as

$$\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f. \tag{4.6}$$

There is nothing new in Eq. (4.6) as it is simply a **convenience** to separate the current into these two (2) parts,as they got there by quite different means:

- The free current is there because somebody hooked up a wire to a battery. It involves actual transport of charge.
- The bound current is there because of magnetization. It results from the collective behavior of many aligned atomic dipoles.

In view of Eq. (4.5) and Eq. (4.6), Ampère's law can be written as:

$$\frac{1}{\mu_0}(\nabla \times \mathbf{B}) = \mathbf{J} = \mathbf{J}_f + \mathbf{J}_b = \mathbf{J}_f + (\nabla \times \mathbf{M}) \quad \text{or tidying them up} \quad \nabla \times \left(\frac{1}{\mu_0}\mathbf{B} - \mathbf{M}\right) = \mathbf{J}_f.$$

The quantity in parentheses is designated by the letter $\mathbf{H}$:

$$\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}. \tag{4.7}$$

In terms of $\mathbf{H}$, then, Ampère's law reads

$$\nabla \times \mathbf{H} = \mathbf{J}_f, \qquad \text{in integral form} \qquad \oint \mathbf{H} \cdot d\mathbf{l} = I_{f_{\text{enc}}}, \tag{4.8}$$

where $I_{f_{\text{enc}}}$ is the total **free** current passing through the Amperian loop. $\mathbf{H}$ plays a role in magnetostatics similar to $\mathbf{D}$ in electrostatics:

Just as **D** allowed us to write *Gauss's law* in terms of the **free charge** alone, **H** permits us to express *Ampère's law* in terms of the **free current** alone.

Free current is what we control directly

Bound current, similar to bound charge, comes **naturally**. The material gets magnetized, and this results in bound currents.

We cannot turn them on or off **independently**, as we can free currents.

In applying Eq. (**4.8**), all we need to worry about is the **free** current, which we know about because we **put** it there. In particular, when symmetry permits, we can calculate **H** immediately from Eq. (**??**) by the usual *Ampère's law* methods.

Practicality of The Auxiliary Field

Turns out, **H** is a **more useful quantity** than **D**. In the laboratory, many scientist will be talking about **H**,[19] but will almost never speak of **D**, as most will work with **E** instead.

[19] Even more often than B.

The reason is this straightforward:

To build an **electromagnet** we run a certain current[20] through a coil. The **current** is the thing we read on our measuring equipment, and this determines **H**;[21] **B** depends on the **specific materials** we used and even, if iron is present, on the **history of the magnet**.

[20] As this is a current we are putting into the system we consider this **free current**.

On the other hand, if we want to set up an *electric field*, we don't plaster a known free charge on the plates of a parallel plate capacitor. Rather, we connect them to a battery of known **voltage**. It's the **potential difference** we read on your dial, and that determines **E**:[22]

[21] Or if we want to get pedantic, the line integral of H.

[22] Or rather, the line integral of E.

D depends on the details of the dielectric we're using.

If it were easy to measure charge, and hard to measure potential, then we'd find experimentalists talking more about **D** instead of **E**. So the relative familiarity of **H**, as contrasted with **D**, derives from purely practical considerations.[23]

[23] This is why majority of engineering applications use the H field in their calculation instead of using B [5]. Its just easier.

Theoretically, they're on an **equal footing**.

The Naming of the Field — 4.4 Information

Many authors call **H**, not **B**, the **magnetic field**. Then they have to invent a new word for **B**: the **flux density**, or **magnetic induction** .[24] **B** is indisputably the fundamental quantity, therefore we will continue to call it the **magnetic field**. **H** has no sensible name and therefore to keep it simple, just call it **H**.

[24] an absurd choice, given the term already has at least two other meanings in electrodynamics. This would do nothing but add more to the confusion.

4.3.2 A Deceptive Parallel

If we were to look at Eq. (4.8) closely, we will see that it looks just like *Ampère's original law* Eq. (**??**), with the exception that the total current is replaced by the free current, and **B** is replaced by $\mu_0\mathbf{H}$.

Similar to the case of **D**, however, We need to be careful, as it does **NOT** say that

> $\mu_0\mathbf{H}$ is just like **B**, only its source is $\mathbf{J}_f$ instead of **J**.

For the curl alone does not determine a vector field as we must also know the divergence.[25] And whereas $\nabla \cdot \mathbf{B} = 0$, the divergence of **H** is **NOT**, in general, zero.

[25] This is one of the key ideas of **Vector Field Theory** which we covered in Chapter 2.

In fact, from Eq. (4.7)

$$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}. \tag{4.9}$$

Only when the divergence of **M** vanishes is the parallel between **B** and $\mu_0\mathbf{H}$ faithful.

A Thought Experiment Consider the example of the bar magnet; a short cylinder of iron carrying a permanent uniform magnetization **M** parallel to its axis. In this case there is no free current anywhere, and a simple application of Eq. (4.8) might lead us to assume **H** being 0, and therefore assuming $\mathbf{B} = \mu_0\mathbf{M}$ inside the magnet and $\mathbf{B} = 0$ outside, which makes no physical sense. It is quite true that the $\nabla \times \mathbf{H}$ vanishes everywhere, but the divergence does **NOT**.

Approaching the Problem When we are asked to find **B** or **H** in a problem involving magnetic materials, first try look for symmetry. If the problem permits cylindrical, plane, solenoidal, or toroidal symmetry, then we can get **H** directly from Eq. (4.8) by the usual *Ampère's law* methods.[26] If the requisite symmetry is absent, we'll have to think of another approach, and in particular we cannotx assume that **H** is zero just because there is no free current in sight.

[26] Evidently, in such cases $\nabla \cdot \mathrm{M}$ is automatically zero, given the free current alone determines the answer.

4.3.3 Boundary Conditions

The magnetostatic boundary conditions of Sect. 5.4.2 can be rewritten in terms of **H** and the **free** current. From Eq. (4.9) it follows that

$$H^{\perp}_{\text{above}} - H^{\perp}_{\text{below}} = -(M^{\perp}_{\text{above}} - M^{\perp}_{\text{below}}),$$

whereas Eq. (**??**) says;

$$\mathbf{H}^{\parallel}_{\text{above}} - \mathbf{H}^{\parallel}_{\text{below}} = \mathbf{K}_f \times \hat{\mathbf{n}}.$$

In the presence of materials, these are sometimes more useful than the corresponding boundary conditions on **B** (Eq. (**??**) and Eq. (**??**)):

$$B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} = 0, \qquad \text{and} \qquad \mathbf{B}^{\parallel}_{\text{above}} - \mathbf{B}^{\parallel}_{\text{below}} = \mu_0(\mathbf{K} \times \hat{\mathbf{n}}).$$

4.4 Linear and Non-Linear Media

4.4.1 Magnetic Susceptibility and Permeability

For both **paramagnetic** and **diamagnetic** materials, the magnetisation (**M**) is sustained by the field (**B**).

> when **B** is removed, **M** disappears.

In fact, for most substances the magnetization is **proportional** to the field, provided the field is **NOT** too strong. To keep consistency with the electrical case given in Eq. (**3.12**), we will express the proportionality as follows:

$$\mathbf{M} = \frac{1}{\mu_0}\chi_m \mathbf{B}.$$

While the above expression is mathematically correct, scientific tradition prefers it be written in terms of **H**, instead of **B**:

$$\mathbf{M} = \chi_m \mathbf{H}. \tag{4.10}$$

The constant of proportionality (χ_m) is called the **magnetic susceptibility** and is a dimensionless quantity which varies from one substance to another.

> This is positive for **paramagnets** and negative for **diamagnets**.

Materials that obey Eq. (**4.10**) are called **linear media**. In view of Eq. (**4.7**), $B = \mu_0(H + M) = \mu_0(1 + \chi_m)H$, (6.30) for linear media. Thus **B** is **also** proportional to **H**:[27] $B = \mu H$, (6.31)

[27] Physically, therefore, Eq. (??) would say exactly the same as Eq. (4.10), [illegible] constant χ_m [illegible]

Table 4.1 Magnetic Susceptibilities (unless otherwise specified, values are for 1 atm, 20° C). Data from Handbook of Chemistry and Physics, 91st ed. (Boca Raton: CRC Press, Inc., 2010) and other references.

Diamagnetic		Paramagnetic	
Material	Susceptibility	Susceptibility	Material
Bismuth	$-1{,}7 \times 10^{-4}$	Oxygen (O_2)	$1{,}7 \times 10^{-6}$
Gold	$-3{,}4 \times 10^{-5}$	Sodium	$8{,}5 \times 10^{-6}$
Silver	$-2{,}4 \times 10^{-5}$	Aluminum	$2{,}2 \times 10^{-5}$
Copper	$-9{,}7 \times 10^{-6}$	Tungsten	$7{,}0 \times 10^{-5}$
Water	$-9{,}0 \times 10^{-6}$	Platinum	$2{,}7 \times 10^{-4}$
Carbon Dioxide	$-1{,}1 \times 10^{-8}$	Liquid Oxygen	$3{,}9 \times 10^{-3}$
Hydrogen ($H2$)	$-2{,}1 \times 10^{-9}$	Gadolinium	0,48

where

$$\mu \equiv \mu_0(1 + \chi_m).$$

μ is called the **permeability** of the material.[28] In a vacuum, where there is no matter to magnetize, the susceptibility χ_m vanishes, and the permeability is μ_0. That's why μ_0 is called the **permeability of free space**.

We might assume that linear media escape the defect in the parallel between **B** and **H**. given **M** and **H** are now **proportional** to **B**.

If this is true, doesn't this mean, their divergence, similar to **B**'s, must always vanish? Unfortunately, it does **NOT** at the boundary between two (**2**) materials of different permeability, the divergence of **M** can actually be infinite.

Divergence of Auxiliary-Field — 4.5 Information

Formally:

$$\nabla \cdot \mathbf{H} = \nabla \cdot \left(\frac{1}{\mu}\mathbf{B}\right) = \frac{1}{\mu}\nabla \cdot \mathbf{B} + \mathbf{B} \cdot \nabla\left(\frac{1}{\mu}\right) = \mathbf{B} \cdot \nabla\left(\frac{1}{\mu}\right),$$

so, in general, **H** is **NOT** divergenceless at points where μ is changing.

For instance, at the end of a cylinder of linear paramagnetic material, **M** is zero on one side but **NOT** on the other. For the "Gaussian pillbox" shown in Fig. 6.23, $\oint \mathbf{M} \cdot d\mathbf{a} \neq 0$, and hence, by the divergence theorem, $\nabla \cdot \mathbf{M}$ cannot vanish everywhere within it.

Incidentally, the volume bound current density in a homogeneous linear material is proportional to the free current density:

$$\mathbf{J}_b = \nabla \times M = \nabla \times \left(\chi_m \mathbf{H}\right) = \chi_m \mathbf{J}_f.$$

Particularly, unless free current actually flows through the material, all bound current will be at the surface.

4.4.2 Ferromagnetism

In a **linear medium**, the alignment of atomic dipoles is maintained by a magnetic field **imposed** from the outside. Ferromagnets, which are by nature **NOT** linear,[29] require no external fields to sustain the magnetization;

the alignment is **frozen in**.

Like paramagnetism, ferromagnetism involves the magnetic dipoles associated with the spins of unpaired electrons. The new feature, which makes ferromagnetism so different from paramagnetism, is the interaction between nearby dipoles:

In a ferromagnet, each dipole "wants" to point in the same direction as its **neighbours**.

[28] If you factor out μ_0, what's left is called the **relative permeability**: $\mu_r \equiv 1 + \chi_m = \mu/\mu_0$. By the way, formulas for H in terms of B (Eq. (??), in the case of linear media) are called constitutive relations, just like those for D in terms of E.

[29] In this sense, it is misleading to speak of the susceptibility or permeability of a ferromagnet. The terms are used for such materials, but they refer to the proportionality factor between a differential increase in H and the resulting **differential** change in M (or B); moreover, they are **NOT** constants, but functions of H.

The reason for this preference is essentially quantum mechanical and therefore significantly beyond the scope of an engineers need. It is enough to know that the correlation is so strong as to align virtually 100% of the unpaired electron spins. If we could somehow magnify a piece of iron and **see** the individual dipoles as tiny arrows, it would look something like **Fig.** 4.4.

But if that is true, why isn't every wrench and nail a powerful magnet?

The answer is that the alignment occurs in relatively small patches, called **domains**. Each domain contains billions of dipoles, all lined up,[30] but the domains **themselves** are randomly oriented. The a standard wrench contains an enormous number of domains, and their magnetic fields cancel, so the wrench as a whole is not magnetized.[31]

How, then, would you produce a **permanent magnet**?

If we were to put a piece of iron into a strong magnetic field, the torque $\mathbf{N} = \mathbf{m} \times \mathbf{B}$ tends to align the dipoles parallel to the field. Since they like to stay parallel to their neighbors, most of the dipoles will resist this torque. However, at the **boundary** between two domains, there are **competing** neighbors, and the torque will throw its weight on the side of the domain most nearly parallel to the field;

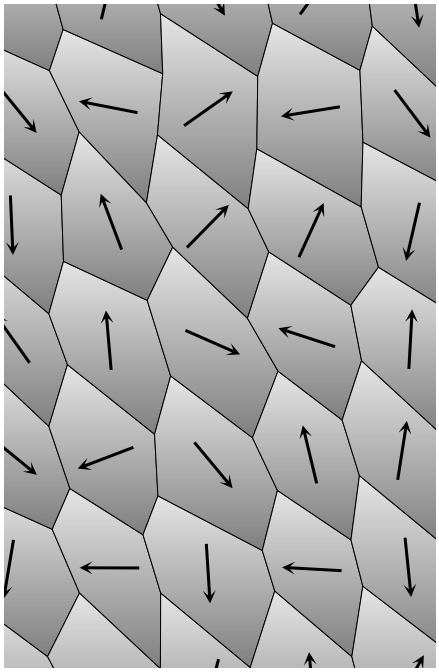


Figure 4.3
A closer look into the magnetic domains of a material. Here each domain has an orientation. As can bee seen, here there is no majority preference among the domains and this results in the macroscopic behavior of being non-magnetic. If there was an overall alignment, then the material would be magnetic. These domains are actually visible under a microscope, using suitable etching techniques, as an example shown in Fig. 4.4. As can bee seen, here there is no majority preference among the domains and this results in the macroscopic behavior of being non-magnetic. If there was an overall alignment, then the material would be magnetic. as many domains pointing one way as the other, so there is still no large-scale magnetization. Moreover, the crystals themselves are randomly oriented within any sizable chunk of metal.

This domain will win some converts, at the expense of the less favorably oriented one.

The net effect of the magnetic field, then, is to move the domain boundaries. Domains parallel to the field grow, and the others shrink. If the field is strong enough, one domain takes over entirely, and the iron is said to be saturated. It turns out that this process[32] is not entirely reversible: When the field is switched off, there will be **some** return to randomly oriented domains, but it is far from complete as there remains a preponderance of domains in the original direction.

[30] Actually, the orientation of domains is not completely random: within a given crystal, there may be some preferential alignment along the crystal axes. But there will be just

[32] the shifting of domain boundaries in response to an external field

We now have a permanent magnet.

A Simple Experiment

A simple way to accomplish this, in practice, is to wrap a coil of wire around the object to be magnetized. Run a current I through the coil; this provides the external magnetic field As we increase the current, the field increases, the domain boundaries move, and the magnetization grows. Eventually, we will reach the saturation point, with all the dipoles aligned, and a further increase in current has no effect on $\mathbf{M}$.

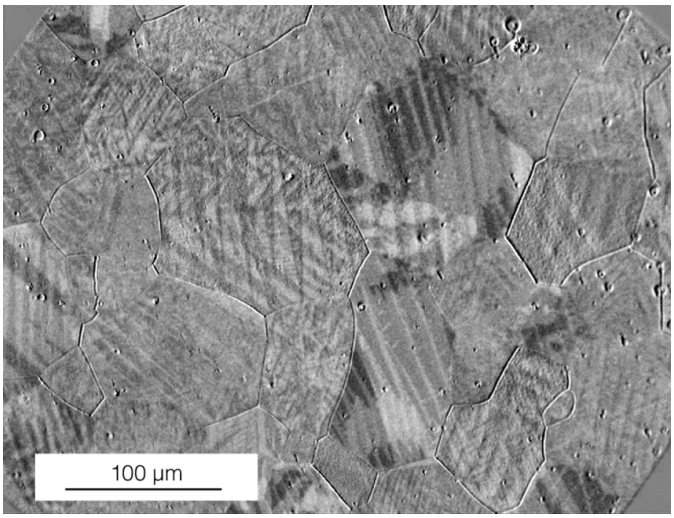


Figure 4.4
Illustration of the magnetic domains structure at the surface of a Fe-27%Co polycrystal [6] -Kerr microscopy.

Now suppose we **reduce** the current. Instead of retracing the path back to $\mathbf{M} = 0$, there is only a **partial** return to randomly oriented domains;

M decreases, but even with the current off there is some **residual magnetization**.

The object is now a **permanent magnet**. If we want to eliminate the remaining magnetization, we'll have to run a current backwards through the coil.[33] Now the external field points to the right, and as you increase I (negatively), **M** drops down to zero. If we turn I still higher, we soon reach saturation in the other direction;

all the dipoles now pointing to the right.

At this stage, switching off the current will leave the object with a permanent magnetization to the right. To complete the cycle, we turn I on again in the positive sense where **M** returns to zero, and eventually to the forward saturation point.

> The path we have traced out is called a **hysteresis loop**.

Notice that the magnetization of the object depends not only on the applied field,[34] but also on its **previous magnetic history**.[35] For instance, at three (**3**) different times in our experiment going

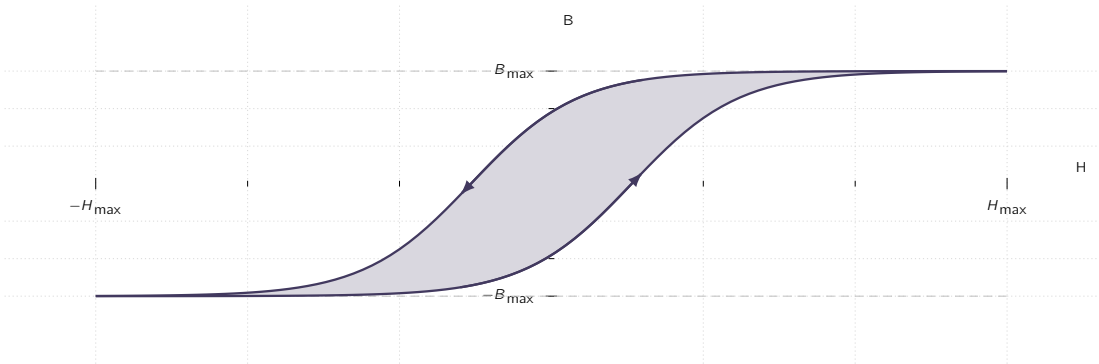


Figure 4.5
An example of a hysteresis curve showcasing relevant points of interest.

[33] Meaning we switch the polarity of the applied current.

[34] Basically meaning the current we have applied.

[35] Etymologically, the word *hysteresis* has nothing to do with the word *history* or *hysteria*. It derives from a Greek word meaning *lag behind*.

through the loop the current was zero and yet the magnetization was different for each of them.

Actually, it is customary to draw hysteresis loops as plots of B against H, rather than M against I.[36]

[36] If our coil is approximated by a long solenoid, with n turns per unit length, then $H = nI$, so H and I are proportional. Meanwhile, $B = \mu_0(H + M)$, but in practice M is huge compared to H, so to all intents and purposes B is proportional to M.

Relative Values of B and H — 4.6 Information

Generally, vertical scale is 3 to 4 order of times greater than the horizontal one.

A little current goes a long way, when we have ferromagnetic materials around. That's why anyone who wants to make a powerful electromagnet will wrap the coil around an iron core. It doesn't take much of an external field to move the domain boundaries, and when we do that, we have all the dipoles in the iron working together.

Curie Point

Random thermal motions compete with the ordering of magnetic domains, but as long as the temperature doesn't get too high, they cannot budge the dipoles out of line. It's not surprising, though, that very high temperatures can and will destroy the alignment. What is surprising is that this occurs at a precise temperature.[37]

[37] For example, to completely demagnatize iron we need to heat it up to 770 C.

Below this temperature, called the **Curie point**, iron is **ferromagnetic**; above, it is **paramagnetic**.

Interpretation The Curie point is similar to the *boiling point* or the *freezing point* in that there is no gradual transition from ferro- to para-magnetic behavior, any more than there is between water and ice. These abrupt changes in the properties of a substance, occurring at sharply defined temperatures, are known in statistical mechanics as **phase transitions**.

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Part IV

Electromagnetic Fields

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This velocity is so nearly that of light, that it seems we have strong reason to conclude that light itself is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field according to electromagnetic laws

(James Clerk Maxwell, in A Dynamical Theory of the Electromagnetic Field (1864) Introduction, p. 466.)

Chapter 5

Electrodynamics

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Before we start with thing that change with time[1] it is worthwhile to discuss some essential terminologies for understanding electrical circuits, electrical machines, and wide variety of electrical phenomena:

the Electro-motive Force (EMF).

[1] i.e., where time actually becomes a variable.

5.1 Electromotive Force

5.1.1 Ohm's Law

To make a current flow, we have to *push* on the charges. How *fast* they move, in response to a given push, depends on the nature of the material. For most substances, the current density **J** is proportional to the *force per unit charge*, **f**:

$$\mathbf{J} = \sigma \mathbf{f} \tag{5.1}$$

The proportionality factor σ, which should **NOT** to be confused with surface charge, is an empirical constant which varies from one material to another; it's called the *conductivity* of the medium. Actually, most resources usually list the *reciprocal* of σ, called the **resistivity**: $\rho = 1/\sigma$.[2] Some

[2] Which shouldn't be confused with charge density-I'm sorry, but we're running out of Greek letters, and this is the standard notation

Table 5.1
Resistivities, in ohm-meters (all values are for 1 atm, 20 C).

Conductive Materials	Resistivity	Semiconductor and Insulator Materials	Resistivity
Silver	$1{,}59 \times 10^{-8}$	Sea water	0.2
Copper	$1{,}68 \times 10^{-8}$	Germanium	0.46
Gold	$2{,}21 \times 10^{-8}$	Diamond	2.7
Aluminum	$2{,}65 \times 10^{-8}$	Silicon	2500
Iron	$9{,}61 \times 10^{-8}$	Water (pure)	$8{,}3 \times 10^{3}$
Mercury	$9{,}61 \times 10^{-7}$	Glass	$1{,}0 \times 10^{10}$ - $1{,}0 \times 10^{15}$
Nichrome	$1{,}08 \times 10^{-6}$	Rubber	$1{,}0 \times 10^{14}$ $1{,}0 \times 10^{16}$
Manganese	$1{,}44 \times 10^{-6}$	Teflon	$1{,}0 \times 10^{23}$ $1{,}0 \times 10^{25}$
Graphite	$1{,}6 \times 10^{-5}$		

typical values are listed in Table 7.1. Looking at the table we can see even *insulators* conduct slightly, though the conductivity of a metal is significantly greater.

In fact, for most purposes metals can be regarded as **perfect conductors**, with $\sigma = \infty$, while for insulators we can pretend $\sigma = 0$.

In principle, the force that drives the charges to produce the current could be anything:

chemical, gravitational, or pressure.

For our purposes, though, it's usually an electromagnetic force that does the job. In this case Eq. (5.1) becomes:[3]

$$\mathbf{J} = \sigma\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right)$$

Ordinarily, the velocity of the charges is sufficiently small that the second term can be ignored:[4]

$$\mathbf{J} = \sigma\mathbf{E}. \tag{5.2}$$

Eq. (5.2) is called Ohm's law;[5] though the physics behind it is really contained in Eq. (5.1), of which Eq. (5.2) is considered to be just a special case. x

If we recall from electrostatics, we said $\mathbf{E} = 0$ inside a conductor, which here we are contradicting. But that's for *stationary* charges $\mathbf{J} = 0$.

Moreover, for *perfect* conductors $\mathbf{E} = J/\sigma = 0$ even if current is flowing. In practice, metals are such good conductors that the electric field ($\mathbf{E}$) required to drive current in them is assumed **negligible**. Therefore we routinely treat the connecting wires in electric circuits as **equipotentials**. Resistors, by contrast, are made from *poorly* conducting materials.

[3] This equation should remind you of the *Lorentz force law*.

[4] However, in plasmas, for instance, the magnetic contribution to f can be significant and cannot be ignored.



[5] a German mathematician and physicist. As a school teacher, Ohm began his research with the new electrochemical cell, invented by Italian scientist *Alessandro Volta*. Using equipment of his own creation, Ohm found that there is a direct proportionality between the potential difference (voltage) applied across a conductor and the resultant electric current.

As these aforementioned examples show, the total current flowing from one electrode to the other is proportional to the potential difference between them:

$$V = IR$$

This, of course, is the more familiar, and more practical, version of *Ohm's law*. The constant of proportionality R is called the **resistance** and it's a function of the geometry of the arrangement and the conductivity of the medium between the electrodes.

Resistance is measured in ohms (Ω) where an ohm is a volt per ampere ($\mathrm{V} \cdot \mathrm{A}^{-1}$).

The proportionality between V and I is a direct consequence of Eq. (**5.2**):

> if we want to double V, we simply double the charge on the electrodes. This in turn doubles E, which, for an ohmic material, doubles **J**, which doubles **I**.

For steady currents and *uniform* conductivity,

$$\nabla \cdot \mathbf{E} = \frac{1}{\sigma} \nabla \cdot \mathbf{J} = 0,$$

and therefore the charge density is zero (**0**) and any unbalanced charge resides on the surface.[6]

It follows, in particular, that Laplace's equation holds within a homogeneous ohmic material carrying a steady current, so all the tools and tricks of Chapter 3 are available for calculating the potential.

[6] We proved this long ago, for the case of *stationary* charges, using the fact that $\mathrm{E} = 0$; evidently, it is still true when the charges are allowed to move.

5.1.2 Determining the Current Density

It's a little surprising that Ohm's law ever holds.

After all, a given field **E** produces a force $q\mathbf{E}$, acting on a charge q, and according to Newton's 2nd-law, the charge will accelerate.

> *But if the charges are accelerating, why doesn't the current increase with time, growing larger and larger the longer you leave the field on?*

Ohm's law implies, on the contrary,a constant field produces a constant current, which suggests a constant velocity.

> *Isn't that a contradiction to Newton's law?*

No, for we are forgetting the **frequent collisions** electrons make as they pass down the wire. If the length of the wire is λ and electron acceleration is a, the time it takes to go through a wire:

$$t = \sqrt{\frac{2\lambda}{a}} \qquad \text{which makes the average speed} \qquad v_{ave} = \frac{1}{2} a t = \sqrt{\frac{\lambda a}{2}}$$

While this is a better description we are still not finished. The aforementioned equation says that the velocity is **proportional** to the square root of the acceleration ($v \propto \sqrt{1/a}$), and therefore the current should be proportional to the square root of the field.

In practice, the charges are already moving very fast due to of their thermal energy. But the thermal velocities have **random directions, and average to zero**. The drift velocity we are concerned with is a tiny extra bit. So the time between collisions is actually much shorter than we supposed.

If we assume for the sake of argument that all charges travel the same distance λ between collisions, then:

$$t = \frac{\lambda}{v_{thermal}} \qquad \text{therefore} \qquad v_{ave} = \frac{1}{2}at = \frac{a\lambda}{2v_{thermal}}$$

If there are n molecules per unit volume, and f free electrons per molecule, each with charge q and mass m, the current density ($\mathbf{J}$) is:[7]

[7] This model is also known as the **Drude model** which was proposed in 1900.

$$\mathbf{J} = nfq_{ave} = \frac{nf q\lambda}{2_{thermal}}\frac{\mathbf{F}}{m} = \left(\frac{nf\lambda q^2}{2m_{thermal}}\right)\mathbf{E}.$$

Drude Model — 5.1 Information

The **Drude model** or the **DrudeLorentz** model of electrical conduction in metals was proposed in 1900 by Paul Drude, and refined by Hendrik Lorentz in 1905. The Drude model attempts to explain conduction in terms of the scattering of electrons, the carriers of electricity, by the relatively immobile ions in the metal that act like obstructions to the flow of electrons.

The model is an application of kinetic theory. It assumes that when electrons in a solid are exposed to the electric field, they behave much like a pinball machine. The sea of constantly jittering electrons bouncing and re-bouncing off heavier, relatively immobile positive ions produce a net collective motion in the direction opposite to the applied electric field.

While **NOT** perfectly accurate to modern standards, it does indicate the basic ingredients, and it correctly predicts that conductivity is proportional to the density of the moving charges and decreases with increasing temperature.

As a result of all the collisions, the work done by the electrical force is converted into heat in the resistor. Since the work done per unit charge is V and the charge flowing per unit time is I, the power delivered is

$$P = VI = I^2R.$$

This is the **Joule heating law**. With I in amperes (A) and R in ohms (Ω), P comes out in watts (W or $\mathrm{J \cdot s^{-1}}$).

5.1.3 Electromotive Force

Think of a simplest, practical electrical circuit;

> a battery hooked up to a light bulb.[8]



[8] Perhaps something like this which is a tantalum lamp with a 1 meter long filament.

An interesting question arises. In practice, the *current* is the same all *the way around the loop*;

> *why is this the case, when the only obvious driving force is inside the battery?*

Off hand, you might expect a large current in the battery and none at all in the lamp. Who's doing the pushing, in the rest of the circuit, and how does it happen that this push is exactly right to produce the same current in each segment? What's more, given that the charges in a typical wire move (literally) at a *snail's* pace,[9]

> *Why doesn't it take half an hour for the current to reach the light bulb? How do all the charges know to start moving at the same instant?*

[9] For a current of 1 ampere per millimeter squared, the velocity will be about $0{,}1\,\mathrm{cm}\cdot\mathrm{s}^{1}$ [1].

Interpretation If the current were **NOT** the same all the way around,[10] then charge would be piling up somewhere, and, here's the crucial point, the electric field of this accumulating charge is in such a direction as to **even out the flow**.

[10] for instance, during the first split second after the switch is closed.

Suppose, for instance, that the current *into* the bend is greater than the current *out*. Then charge piles up at the "knee", and this produces a field aiming *away* from the kink. This field *opposes* the current flowing in, slowing it down, and *promotes* the current flowing out, speeding it up, until these currents are equal, at which point there is no further accumulation of charge, and equilibrium is established.

> It's **automatically self-correcting** to keep the current uniform, and it does it all so quickly that, in practice, you can safely assume the current is the same all around the circuit, even in systems that oscillate at radio frequencies.

There are really two (**2**) forces involved in driving current around a circuit:

i. the source, $\mathbf{f}_s$, which is ordinarily confined to one portion of the loop,[11] and

ii. an *electrostatic* force, which serves to smooth out the flow and communicate the influence of the source to distant parts of the circuit:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E}.$$

[11] for example, a battery

The physical agency responsible for $\mathbf{f}_s$ can be of different things:

> in a battery it's a chemical force; in a piezoelectric crystal mechanical pressure is converted into an electrical impulse; in a thermocouple it's a temperature gradient that

does the job; in a photoelectric cell it's light; and in a Van de Graaff generator the electrons are literally loaded onto a conveyer belt and swept along.

Whatever the *mechanism*, its net effect is determined by the line integral of **f** around the circuit:

$$\mathcal{E} = \oint \vec{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}. \tag{5.3}$$

As we know $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for electrostatic fields, it doesn't matter whether we use f or f_s.

Here, $\mathcal{E}$ is called the **electromotive force**,[12] or **emf**, of the circuit. Within an ideal source of EMF (a resistanceless battery,4 for instance), the *net* force on the charges is *zero* (Eq. 7.1 with $\sigma = \infty$), so $\mathbf{E} = -\mathbf{f}_s$. The potential difference between the terminals (a and b) is therefore[13]

$$V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E}.$$

The function of a battery, then, is to establish and maintain a voltage difference equal to the electromotive force.[14] The resulting electrostatic field drives current around the rest of the circuit (notice, however, that *inside* the battery f_s drives current in the direction *opposite* to **E**).

Because it's the line integral of $\mathbf{f}_s$, $\mathcal{E}$ can be interpreted as the *work done per unit charge*, by the source-indeed, in some books electromotive force is *defined* this way. However, as you'll see in the next section, there is some subtlety in- volved in this interpretation, so I prefer Eq. 7.9.

Difference Between EMF and Voltage — 5.2 Information

EMF refers to the work per unit charge done by some force inside a conductor, which could be chemical, gravitational, magnetic, etc. The potential difference, or voltage, on the other hand, refers to the effect that other force has on the electric field inside the same conductor.

5.1.4 Motional EMF

In the previous section, we have listed several possible sources of EMF with batteries being the most familiar. But we did **NOT** mention the commonest one of all:

the electric generator.[15]

Generators exploit **motional** EMFs, which arise when you *move a wire* through a magnetic field. Figure 7.10 suggests a primitive model for a generator. In the shaded region there is a uniform magnetic field (**B**), pointing **into the page**, and the resistor R represents whatever it is[16] we're trying to drive current through. If the entire loop is pulled to the right with speed v, the charges in segment ab experience a magnetic force whose vertical component qvB drives current around the loop, in the clockwise direction.

[12] Some textbooks and resources use the term *electromance*.

[13] we can extend the integral to the entire loop because $f_s = 0$ outside the source anyway.

[14] a 6 V battery, for example, holds the positive terminal 6 V above the negative terminal.

[15] A significant application of a synchronous generator is covered by the lecturer of this course on **B.Sc Drive Technologies**.

[16] It basically represents anything which consumes power.

The EMF is

$$\mathcal{E} = \oint \mathbf{f}_{mag} \cdot d\mathbf{l} = vBh,$$

where h is the width of the loop.[17] Please observe the integral we perform to calculate $\mathcal{E}$ (Eq. 7.9 or 7.11) is carried out at one instant of time.

[17] The horizontal segments *bc* and *ad* contribute nothing, since the force there is perpendicular to the wire.

> We are basically taking a **snapshot** of the loop, so to say.

Therefore d**l**, for the segment *ab* in Fig. 7.10, points straight up, even though the loop is moving to the right.

> In particular, although the magnetic force is responsible for establishing the EMF, it is not doing any work-magnetic forces never do work.
>
> *Who, then, is supplying the energy that heats the resistor?*

Well, it's the person who's pulling on the loop. With the current flowing, the free charges in segment *ab* have a vertical velocity, let's call it **u**, in addition to the horizontal velocity **v** they inherit from the motion of the loop. Accordingly, the magnetic force has a component quB to the left. To counteract this, the person pulling on the wire must exert a force per unit charge

$$f_{\text{null}} = uB$$

to the *right* (Fig. 7.11). This force is transmitted to the charge by the structure of the wire. Meanwhile, the particle is actually *moving* in the direction of the resultant ve- locity w, and the distance it goes is $(h/\cos\theta)$. The work done per unit charge is therefore:[18]

[18] In the equation below, $\sin\theta$ is coming from the dot product.

$$\int \mathbf{f}_{pull} \cdot \mathrm{d}\mathbf{l} = (uB)\left(\frac{h}{\cos\theta}\right)\sin\theta = vBh = \mathcal{E}$$

As it turns out, then, the work done per unit *charge is exactly equal to the emf, though the integrals are taken along entirely* different paths (Fig. 7.12), and completely different forces are involved. To calcu- late the emf, you integrate around the loop at one instant, but to calculate the work done you follow a charge in its journey around the loop; f_{pull} contributes nothing to the emf, because it is perpendicular to the wire, whereas f_{mag} contributes nothing to work because it is perpendicular to the motion of the charge.⁶ There is a particularly nice way of expressing the EMF generated in a moving loop. Let Φ be the flux of **B** through the loop:

$$\Phi \equiv \int \mathbf{B} \cdot d\mathbf{a}.$$

$$\mathbf{f}_{\text{null}}$$

For the rectangular loop in Fig. 7.10,

$$\Phi = Bhx.$$

As the loop moves, the flux decreases:[19]

[19] The minus sign accounts for the fact that dx/dt is negative.

$$\frac{\mathrm{d}\Phi}{dt} = Bh\frac{dx}{dt} = -Bhv.$$

But this is precisely the emf ($Eq.7.11$); evidently the EMF generated in the loop is minus the rate of change of flux through the loop:

$$\mathcal{E} = -\frac{d\Phi}{dt}.$$

This is the flux rule for motional EMF. Apart from its delightful simplicity, the flux rule has the virtue of applying to nonrectangular loops moving in arbitrary directions through nonuniform magnetic fields; in fact, the loop need not even maintain a fixed shape.

There is a sign ambiguity in the definition of EMF in Eq. (**5.3**): Which way around the loop are you supposed to integrate? There is a compensatory ambiguity in the definition of $flux$ (Eq. 7.12): Which is the positive direction for da? In applying the flux rule, sign consistency is governed (as always) by your right hand: If your fingers define the positive direction around the loop, then your thumb indicates the direction of $d\mathbf{a}$. Should the emf come out negative, it means the current will flow in the negative direction around the circuit. The flux rule is a nifty short-cut for calculating motional emfs. It does not con- tain any new physics-just the Lorentz force law. But it can lead to error or ambi- guity if you're not careful. The flux rule assumes you have a single wire loop-it can move, rotate, stretch, or distort (continuously), but beware of switches, sliding contacts, or extended conductors allowing a variety of current paths. A standard "flux rule paradox" involves the circuit in Figure 7.14. When the switch is thrown (from a to b) the flux through the circuit doubles, but there's no motional EMF (no conductor moving through a magnetic field), and the ammeter (A) records no current.

Example 7.4 (the Faraday disk, or Faraday dynamo) involves a motional EMF that you can't calculate (at least, not directly) from the flux rule. The flux rule assumes the current flows along a well-defined path, whereas in this example the current spreads out over the whole disk. It's not even clear what the "flux through the circuit" would *mean* in this context.

Even more tricky is the case of **eddy currents**. Take a chunk of aluminum (say), and shake it around in a nonuniform magnetic field. Currents will be gen- erated in the material, and you will feel a kind of "viscous drag"-as though you were pulling the block through molasses (this is the force I called f_{pull} in the dis- cussion of motional emf). Eddy currents are notoriously difficult to calculate,⁷ but easy and dramatic to demonstrate. You may have witnessed the classic experiment in which an aluminum disk mounted as a pendulum on a horizontal axis swings down and passes between the poles of a magnet (Fig. 7.16a). When it enters the field region it suddenly slows way down. To confirm that eddy currents are re- sponsible, one repeats the demonstration using a disk that has many slots cut in it, to prevent the flow of large-scale currents (Fig. 7.16b). This time the disk swings freely, unimpeded by the field.

5.2 Electromagnetic Induction

We will now look at perhaps one of the more important laws governing the behavior of transformers and electric machines, which form the backbone of modern industry.

5.2.1 Faraday's Law

In 1831 Michael Faraday[20] reported on a series of experiments, including three (3) which can be characterized as follows:

Experiment 1

He pulled a loop of wire to the right through a magnetic field and observed a current flowing in the loop.

Experiment 2

He moved the *magnet* to the *left*, holding the loop still and again, observed a current flowing in the loop.

Experiment 3

With both the loop and the magnet at rest, he changed the strength of the field by using an electromagnet, and varied the current in the coil. Once again, he observed current flowing in the loop.

The first experiment, of course, is a straightforward case of motional EMF; according to the flux rule we have derived earlier:

$$\mathcal{E} = -\frac{d\Phi}{dt}.$$

It is worth noting that exactly the same EMF arises in Experiment 2.

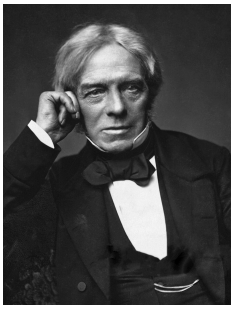
> All that really matters is the *relative* motion of the magnet and the loop

Indeed, in the light of special relativity it has to be so. But Faraday knew nothing of relativity, and in classical electrodynamics this simple reciprocity is a remarkable coincidence. For if the *loop* moves, it's a *magnetic* force that sets up the EMF, but if the loop is *stationary*, the force *cannot* be magnetic as stationary charges experience no magnetic forces.

In that case, what is responsible and what sort of field exerts a force on charges at rest?

Well, *electric* fields do, of course, but in this case there doesn't seem to be any electric field in sight.

Faraday had an ingenious inspiration:



[20] **Michael Faraday** (*1791 - 1867*)

An English chemist and physicist who contributed to the study of electro-chemistry and electromagnetism. His main discoveries include the principles underlying electromagnetic induction, diamagnetism, and electrolysis. Although Faraday received little formal education, as a self-made man, he was one of the most influential scientists in history [2]. It was by his research on the magnetic field around a conductor carrying a direct current Faraday established the concept of the electromagnetic field in physics. Faraday also established magnetism could affect rays of light and that there was an underlying relationship between the two phenomena. He similarly discovered the principles of electromagnetic induction, diamagnetism, and the laws of electrolysis. His inventions of electromagnetic rotary devices formed the foundation of electric motor technology, and it was largely due to his efforts that electricity became practical for use in technology. The SI unit of capacitance, the farad, is named after him.

A changing magnetic field induces an electric field.

It is this *induced* electric field which accounts for the EMF in Experiment 2. Indeed, if[21] the EMF is again equal to the rate of change of the flux,

[21] as Faraday found empirically.

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d[}{d\Phi}][t],$$

then **E** is related to the change in **B** by the equation

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}.$$

This is Faraday's law, in integral form and its differential form is derived using *Stokes' theorem*:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Note that Faraday's law reduces to the old rule $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ (or, in differential form, $\nabla \times \mathbf{E} = 0$ in the static case (constant **B**) as, of course, it should. In Experiment 3, the magnetic field changes for entirely different reasons, but according to Faraday's law an electric field will again be induced, giving rise to an emf $-d\Phi/dt$.

Indeed, one can subsume all three cases[22] into a kind of universal flux rule: Whenever (and for whatever reason) the magnetic flux through a loop changes, an emf

[22] and for that matter any combination of them.

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

will appear in the loop. Many people call this "Faraday's law." Maybe I'm overly fastidious, but I find this confusing. There are really two totally different mechanisms underlying Eq. 7.17, and to identify them both as "Faraday's law" is a little like saying that because identical twins look alike we ought to call them by the same name. In Faraday's first experiment it's the Lorentz force law at work; the EMF is *magnetic*. But in the other two it's an *electric* field (induced by the changing magnetic field) that does the job. Viewed in this light, it is quite astonishing that all three processes yield the same formula for the emf. In fact, it was precisely this "coincidence" that led Einstein to the special theory of relativity-he sought a deeper understanding of what is, in classical electrodynamics, a peculiar accident. But that's a story for Chapter 12. In the meantime, I shall reserve the term "Faraday's law" for electric fields induced by changing magnetic fields, and I do *not* regard Experiment 1 as an instance of Faraday's law.

Keeping track of the *signs* in Faraday's law can be a real headache. For in- stance, in Ex. 7.5 we would like to know which way around the ring the induced current flows. In principle, the right-hand rule does the job (we called Φ positive to the left, in Fig. 7.22, so the positive direction for current in the ring is counter- clockwise, as viewed from the left; since the first spike in Fig. 7.23b is negative, the first current pulse flows *clockwise*, and the second counterclockwise). But there's a handy rule, called Lenz's law, whose sole purpose is to help you get the directions right:10

Nature abhors a change in flux. The induced current will flow in such a direction that the flux *it* produces tends to cancel the change. (As the front end of the magnet in Ex. 7.5 enters the ring,

the flux increases, so the current in the ring must generate a field to the *right*-it therefore flows *clockwise*.) Notice that it is the *change* in flux, not the flux it- self, that nature abhors (when the tail end of the magnet exits the ring, the flux drops, so the induced current flows *counterclockwise*, in an effort to restore it). Faraday induction is a kind of "inertial" phenomenon: A conducting loop "likes" to maintain a constant flux through it; if you try to *change* the flux, the loop re- sponds by sending a current around in such a direction as to frustrate your efforts. (It doesn't succeed completely; the flux produced by the induced current is typi- cally only a tiny fraction of the original. All Lenz's law tells you is the *direction* of the flow.)

5.2.2 The Induced Electric Field

Faraday's law generalises the electrostatic rule, we have derived in previous chapters.[23] If **E** is a pure Faraday field,[24] then:

$$\nabla \cdot \mathbf{E} = 0. \qquad \text{and} \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

This is mathematically identical to **magnetostatics**,

$$\nabla \cdot \mathbf{B} = 0, \qquad \text{and} \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

[23] $\nabla \times \mathrm{E} = 0$ to the time-dependent régime. The *divergence* of E is still given by Gauss's law ($\nabla \cdot \mathrm{E} = \rho/\varepsilon_0$).

[24] due exclusively to a changing B, with $\rho = 0$

Conclusion Faraday-induced electric fields are determined by $-\partial \mathbf{B}/\partial t$ in exactly the same way as magnetostatic fields are determined by $\mu_0 \mathbf{J}$. The analog to *Biot-Savart* is[25]

$$\mathbf{E} = -\frac{1}{4\pi} \int \frac{(\partial \mathbf{B}/\partial t) \times \{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathrm{r}\}^2} \, \mathrm{d}\tau = -\frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}\mathrm{r}\}^2} \, \mathrm{d}\tau,$$

and if symmetry permits, we can use all the methods and simplifications associated with *Ampère's law* in integral form ($\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = \mu_0 I_{\mathrm{enc}}$), only now it's *Faraday's law* in integral form:

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}[\,]}{\mathrm{d}\Phi}[\,][t].$$

The rate of change of magnetic flux through the Amperian loop plays the role formerly assigned to $\mu_0 I_{\mathrm{enc}}$.

[25] Magnetostatics holds only for **time-independent** currents, but there is no such restriction on $\partial \mathrm{B}/\partial t$.

The Importance of Time 5.3 Information

We need to be aware of a small limitation which **restricts** applications of **Faraday's law**:

> Electromagnetic induction, of course, occurs only when the magnetic fields are **changing**, and yet we would like to use the apparatus of magnetostatics[26] to calculate those magnetic fields.

Technically, any result derived in this way is only approximately correct. But in practice the error is usually negligible, unless the field fluctuates extremely rapidly, or if we are interested in points very far from the source. Even the case of a wire snipped by a pair of scissors is considered *static enough*

Continues on next page →

[26] Ampère's law, the Biot-Savart law, and the rest.

← Continued from previous page

for *Ampère's law* to apply. This régime, in which magnetostatic rules can be used to calculate the magnetic field on the right hand side of *Faraday's law*, is called **quasistatic**.

Generally speaking, it is only when we come to electromagnetic waves and radiation that we must worry seriously about the breakdown of magnetostatics itself.

5.2.3 Inductance

Suppose we have two (2) loops of wire, at rest (Fig. 7.30). If were to run a steady current I_1 around loop 1, it produces a magnetic field $\mathbf{B}_1$. Some of the field lines pass through loop 2; let Φ_2 be the flux of $\mathbf{B}_1$ through 2. We might have a tough time actually *calculating* B_1, but a glance at the *Biot-Savart law*,

$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\mathbf{l}_1 \times \{^{\eta}\hat{\mathbf{r}}\}}{\{^{\eta}r\}^2}$$

reveals one significant fact about this field:

It is proportional to the current I_1.

Therefore, so too is the flux through loop 2:

$$\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 \qquad \text{therefore} \qquad \Phi_2 = M_{21} I_1$$

where M_{21} is the constant of proportionality; it is known as the **mutual inductance** of the two loops. There is a simple formula for the mutual inductance, which we can derive by expressing the flux in terms of the **vector potential**, and invoking *Stokes' theorem*:

$$\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 = \int (\nabla \times \mathbf{A}_1) \cdot d\mathbf{a}_2 = \oint \mathbf{A}_1 \cdot d\mathbf{l}_2.$$

Now, according to Eq. (**??**):

$$\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\mathbf{l}_1}{\{^{\eta}r\}},$$

We can use this equation to derive:

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\mathbf{l}_1}{\{^{\eta}r\}} \right) \cdot d\mathbf{l}_2. \qquad \text{therefore} \qquad M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\{^{\eta}r\}}. \tag{5.4}$$

This is the Neumann formula; it involves a double line integral;

one integration around *loop 1*, the other around *loop 2*.

It's not very useful for practical calculations, but it does reveal two (2) important things about mutual inductance:

i. M_{21} is a purely geometrical quantity, having to do with the sizes, shapes, and relative positions of the two loops.

ii. The integral in Eq. (**5.4**) is unchanged if we switch the roles of loops 1 and 2: it follows that

$$M_{21} = M_{12}.$$

This is an astonishing conclusion:

> Whatever the shapes and positions of the loops, the flux through 2 when we run a current I around I is identical to the flux through 1 when we send the same current I around 2. We may as well drop the subscripts and call them both M.

Suppose, now, we *vary* the current in loop 1. The flux through loop 2 will vary accordingly, and *Faraday's law* says this changing flux will induce an EMF in loop 2:[27]

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M\frac{dI_1}{dt}.$$

What a remarkable thing: Every time you change the current in loop 1, an induced current flows in loop 2-even though there are no wires connecting them! Come to think of it, a changing current not only induces an EMF in any nearby loops, it also induces an EMF in the source loop *itself* (Fig 7.33).

[27] In quoting Eq. 7.22-which was based on the Biot-Savart law-I am tacitly assuming that the currents change slowly enough for the system to be consid- ered quasistatic.

Once again, the field (and therefore also the flux) is proportional to the current:

$$\Phi = LI. \tag{5.5}$$

The constant of proportionality L is called the self inductance[28] of the loop. As with M, it depends on the geometry of the loop. If the current changes, the EMF induced in the loop is

$$\mathcal{E} = -L\frac{dI}{dt}. \tag{5.6}$$

[28] or simply the **inductance**.

Inductance is measured in henries (H); a henry is a volt-second per ampere.

Inductance, similarly to capacitance, is an intrinsically *positive* quantity. Lenz's law, which is enforced by the minus sign in Eq. (**5.6**), dictates that the EMF is in such a direction as to *oppose* any *change in current*. For this reason, it is called a back emf. Whenever you try to alter the current in a wire, you must fight against this back emf. Inductance plays somewhat the same role in electric circuits that *mass* plays in mechanical systems: The greater L is, the harder it is to change the current, just as the larger the mass, the harder it is to change an object's velocity.

5.2.4 Energy in Magnetic Fields

It takes a certain amount of energy to start a current flowing in a circuit. Here, were are **NOT** talking about the energy delivered to the resistors and converted into heat.[29] What we are concerned with, rather, is the work we must do **against the back emf** to get the current going. This is a fixed amount, and it is *recoverable*:

[29] that is **irretrievably** lost, as far as the circuit is concerned, and can either be large or small, depending on how long the current is left to run.

We get it back when the current is **turned off**.

In the meantime, it represents energy present in the circuit, and as we'll see in a moment, it can be regarded as energy stored in the magnetic field. The work done on a unit charge, against the back EMF, in one trip around the circuit is $-\mathcal{E}$.[30] The amount of charge per unit time passing down the wire is I. Therefore, the total work done per unit time is:

$$\frac{\mathrm{d}[\quad]}{\mathrm{d}W}][t] = -\mathcal{E}I = LI\frac{\mathrm{d}[\quad]}{\mathrm{d}I}][t].$$

If we start with zero current and build it up to a final value I, the work done[31] is:

$$W = \frac{1}{2}LI^2$$

It does not depend on how *long* we take to crank up the current, only on the geometry of the loop[32] and the final current I. There is a cleaner way to write W, which has the advantage that it is readily generalized to surface and volume currents. Remember that the flux Φ through the loop is equal to LI shown in Eq. (**5.5**). On the other hand,

$$\Phi = \int \mathbf{B}\cdot\,\mathrm{d}\mathbf{a} = \int (\nabla\times\mathbf{A})\,\mathrm{d}\mathbf{a} = \oint \mathbf{A}\cdot\,\mathrm{d}\mathbf{l}$$

where the **line integral** is around the perimeter of the loop. We can then make the conclusion that:

$$LI = \oint \mathbf{A}\cdot\,\mathrm{d}\mathbf{l} \quad \text{therefore} \quad W = \frac{1}{2}I\oint \mathbf{A}\cdot\,\mathrm{d}\mathbf{l} = \frac{1}{2}\oint (\mathbf{A}\cdot\mathbf{I})\,\mathrm{d}I$$

In this form, the generalization to volume currents is obvious:

$$W = \frac{1}{2}\int_{\mathcal{V}} (\mathbf{A}\cdot\mathbf{J})d\tau. \tag{5.7}$$

But we can do even better, and express W entirely in terms of the magnetic field: Ampère's law, $\nabla\times\mathbf{B} = \mu_0\mathbf{J}$, lets us eliminate $\mathbf{J}$:

$$W = \frac{1}{2\mu_0}\int \mathbf{A}\cdot(\nabla\times\mathbf{B})d\tau.$$

Integration by parts transfers the derivative from $\mathbf{B}$ to $\mathbf{A}$; specifically, product rule 6 states that

$$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B}).$$

which allows us to write:

$$\mathbf{A}\cdot(\nabla\times\mathbf{B}) = \mathbf{B}\cdot\mathbf{B} - \nabla\cdot(\mathbf{A}\times\mathbf{B}).$$

Consequently,

$$W = \frac{1}{2\mu_0}\left[\int B^2 d\tau - \int \nabla\cdot(\mathbf{A}\times\mathbf{B})d\tau\right] = \frac{1}{2\mu_0}\left[\int_{\mathcal{V}} B^2 d\tau - \oint_{\mathcal{S}}(\mathbf{A}\times\mathbf{B})\cdot\,\mathrm{d}\mathbf{a}\right], \tag{5.8}$$

where S is the surface bounding the volume V.

Now, the integration in Eq. (**5.7**) is to be taken over the *entire volume occupied* by the current. But any region larger than this will do just as well, for $\mathbf{J}$ is zero out there anyway. In Eq. (**5.8**), the

[30] The minus sign records the fact that this is the work done by us *against* the EMF, not the work done by the EMF itself.

[31] integrating the last equation over time

[32] Which this information is encoded in the inductance.

larger the region we pick the greater is the contribution from the volume integral, and therefore the smaller is that of the surface integral.[33] In particular, if we agree to integrate over all space, then the surface integral goes to zero, and we are left with

$$W = \frac{1}{2\mu_0}\int_{\text{all space}} B^2 d\tau.$$

In view of this result, we say the energy is **stored in the magnetic field**, in the amount $(B^2/2\mu_0)$ per unit volume. This is a nice way to think of it, though someone looking at Eq. (5.7) might prefer to say that the energy is stored in the *current distribution*, in the amount $1/2(\mathbf{A}\cdot\mathbf{J})$ per unit volume.[34]

[33] This makes sense, as the surface gets farther from the current, both A and B decrease.

[34] The distinction is one of bookkeeping as the important quantity is the total energy W, and we need not worry about where the energy is **located**.

We might find it strange that it takes energy to set up a magnetic field given magnetic fields *themselves* do no work. The point is that producing a magnetic field, where previously there was none, requires *changing* the field, and a changing **B**-field, according to Faraday, induces an *electric* field.

The latter, of course, **can do work**.

In the beginning, there is no **E**, and at the end there is no **E**; but in between, while **B** is building up, there is an **E**, and it is against *this* that the work is done. In the light of this, it is extraordinary how similar the magnetic energy formulas are to their electrostatic counterparts:

$$W_{\text{elec}} = \frac{1}{2}\int (V\rho)\,d\tau \quad = \frac{\varepsilon_0}{2}\int E^2 d\tau, \quad \text{(Energy Stored in Electric Field)}$$

$$W_{\text{mag}} = \frac{1}{2}\int (\mathbf{A}\cdot\mathbf{J})\,d\tau \quad = \frac{1}{2\mu_0}\int B^2 d\tau. \quad \text{(Energy Stored in Magnetic Field)}$$

5.4 Information

Superconducting Magnetic Energy Storage System

Superconducting magnetic energy storage systems[35] store energy in the magnetic field created by the flow of direct current in a superconducting coil that has been cryogenically cooled to a temperature below its superconducting critical temperature. This use of superconducting coils to store magnetic energy was invented by M. Ferrier in 1970.[2]

A typical SMES system includes three parts: superconducting coil, power conditioning system and cryogenically cooled refrigerator. Once the superconducting coil is energized, the current will not decay and the magnetic energy can be stored indefinitely.

The stored energy can be released back to the network by discharging the coil. The power conditioning system uses an inverter/rectifier to transform alternating current (AC) power to direct current or convert DC back to AC power. The inverter/rectifier accounts for about 23% energy loss in each direction. SMES loses the least amount of electricity in the energy storage process compared to other methods of storing energy. SMES systems are highly efficient; the round-trip efficiency is greater than 95%.[3]

Due to the energy requirements of refrigeration and the high cost of superconducting wire, SMES is currently used for short duration energy storage. Therefore, SMES is most commonly devoted to

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[35] An example of a SMES produced by FRAKO-TERM

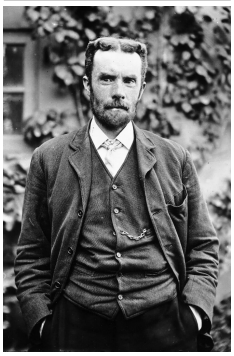
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improving power quality.

5.3 Maxwell's Equations

It is time we develop the bigger picture and derive all the equations needed to complete classical electromagnetism.

5.3.1 Electrodynamics Before Maxwell



[36] We can thank *Oliver Heaviside* for the simplification, who was a British mathematician and electrical engineer. He also invented a new technique for solving differential equations (equivalent to the Laplace transform), and independently developed vector calculus.

So far, we have encountered the following laws, specifying the *divergence* and *curl* of electric and magnetic fields. It is worth mentioning that they were **NOT** written in the compact form [3] we know today, but their physical content was familiar:[36]

$$\begin{aligned}
&\text{(i)} && \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} && \text{Gauss's law} \\
&\text{(ii)} && \nabla \cdot \mathbf{B} = 0 && \text{-} \\
&\text{(iii)} && \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} && \text{Faraday's law} \\
&\text{(iv)} && \nabla \times \mathbf{B} = \mu_0 \mathbf{J} && \text{Ampere's law}
\end{aligned}$$

These equations represent the state of electromagnetic theory in the mid 19th century, when *Maxwell* began his work. Now, it happens there is a major inconsistency in these formulas. It has to do with the old rule that **divergence of curl is always zero**. If were to apply the divergence operation $(\nabla \cdot)$ to number (iii), everything works out:

$$\nabla \cdot (\nabla \times \mathbf{E}) = \nabla \cdot \frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \cdot \mathbf{B})$$

The left side is zero given divergence of curl is zero and the right side is zero by virtue of equation (ii). But when we do the same thing to number (iv), we seem to hit a problem:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J}) \qquad (5.9)$$

the left side must be zero, but the right side, in general, is **NOT**. For steady currents, the divergence of **J** is zero, which we have seen in **Magnetostatics**, but when we go beyond magnetostatics *Ampère's law* cannot be right.

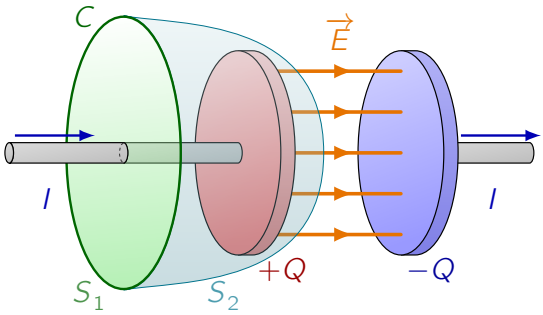


Figure 5.1 **The charging of the capacitor was one of the problem which the current knowledge of electrodynamics couldn't explain and was a pivotal point in developing the final element of electrodynamics.**

There's another way to see that *Ampère's law* is bound to fail for nonsteady currents. Suppose we're in the process of charging up a capacitor, shown in **Fig.** 5.1. To recall, in integral form, *Ampère's law* reads;

$$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = \mu_0 I_{\text{enc}}.$$

Let's apply it to the Amperian loop shown in the diagram. Before we do that, we need to ask the following question:

How do we determine I_{enc}?

Well, it's the total current passing through the loop, or, more precisely, the current piercing a surface that has the loop for its boundary. In this case, the **simplest** surface lies in the plane of the loop-the wire punctures this surface, therefore $I_{enc} = I$.

But what if we draw instead the balloon-shaped surface in **Fig.** 5.1?

No current passes through *this* surface, and we conclude that $I_{enc} = 0$. We never had this problem in magnetostatics because the conflict arises only when charge is piling up somewhere.[37] But for *non-steady* currents[38]

[37] in this case, on the capacitor plates.

[38] such as the charging of a capacitor.

The current enclosed by the loop is an ill-defined notion as it depends entirely on what surface we use.

This may sound like an unnecessary argument as one could say to use the plane surface, but remember, the Amperian loop could be some contorted shape that doesn't even lie in a plane.

Of course, we had no right to *expect* Ampère's law to hold outside of magnetostatics given we derived it from the *Biot-Savart law*. However, if we put all these knowledge in the time of *Maxwell's*, there was no *experimental* reason to doubt that *Ampère's law* was of wider validity.

The flaw was a purely theoretical one, and *Maxwell* fixed it by purely theoretical arguments.

5.3.2 How Maxwell Fixed Ampère's Law

The problem is on the right side of Eq. (**5.9**), which should be zero, but isn't. Applying the continuity equation and Gauss's law, the problematic term can be rephrased as:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t}\left(\varepsilon_0 \nabla \cdot \mathbf{E}\right) = -\nabla \cdot \left(\varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right).$$

If we were to combine:

$$\varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \qquad \text{with} \qquad \mathbf{J},$$

in *Ampère's law*, it would be just right to kill off the extra divergence:[39]

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{5.10}$$

Such a modification changes nothing, as far as magnetostatics is concerned:

when $\mathbf{E}$ is constant, we still have $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$.

In fact, *Maxwell*'s term is hard to detect in ordinary electromagnetic experiments, that's why *Faraday* and the others never discovered it in the laboratory.

[39] Maxwell himself had other reasons for wanting to add this quantity to Ampère's law. To him, the rescue of the continuity equation was a happy dividend rather than a primary motive. But today we recognize this argument as a far more compelling one than Maxwell's, which was based on a now-discredited model of the *ether* as at that time it was thought luminiferous aether (or ether), was a theorized medium for the propagation of light [4].

However, it plays a crucial role in the propagation of electromagnetic waves.

Apart from curing the defect in *Ampère's law*, *Maxwell*'s term has a certain aesthetic appeal:

> Just as a changing *magnetic* field induces an *electric* field,[40] so is a changing electric field induces a magnetic field.

[40] Faraday's law.

Of course, theoretical convenience and aesthetic consistency are only suggestive as there might be other ways to rephrase Ampère's law. The real confirmation of Maxwell's theory came in 1888 with Hertz's experiments on electro-magnetic waves [5].

Proof of Electromagnetic Waves — 5.5 Information

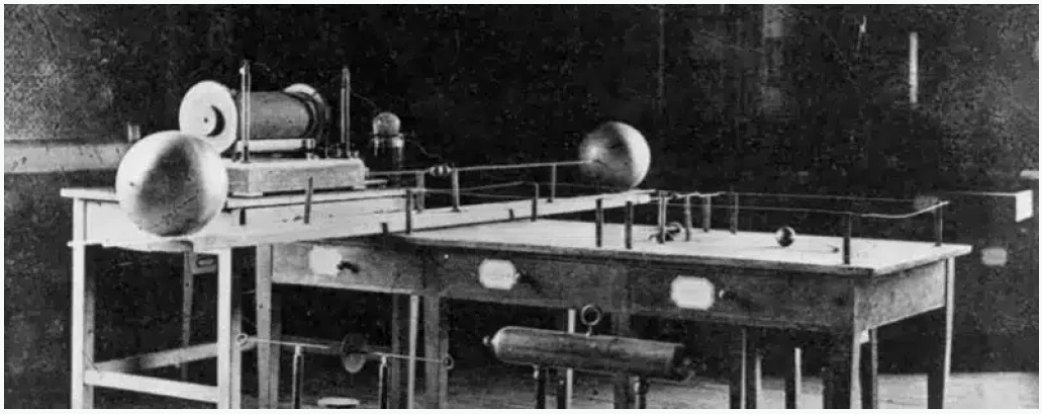


Figure 5.2
The experimental setup used to prove the existence of EM Waves.

Heinrich Hertz was a brilliant German physicist and experimentalist who demonstrated that the electromagnetic waves actually exist. Hertz is also the man whose peers honored by attaching his name to the unit of frequency; a cycle per second is one hertz.

Hertz used a simple homemade experimental apparatus, involving an induction coil and a Leyden jar (the original capacitor) to create electromagnetic waves and a spark gap between two brass spheres to detect them. The gaps were difficult to see, and required that the he perform his investigations in a darkened room.

In later experiments, he was able to calculate the speed of the radio waves he created, and found it to be the same as the speed of light.

A great number of subsequent developments, like radio and television, not to mention Wi-Fi, were spun out of his simple demonstrations. Hertz was well aware of the extent of his contribution.

> *We perceive electricity in a thousand places where we had no proof of its existence before. In every flame, in every luminous particle, we see an electric process. Even if a body is not luminous, provided it radiates heat, it is a center of electric disturbances. Thus the domain of electricity extends over the whole of nature.* [6]

Maxwell called his extra term the **displacement current** [3]:[41]

$$\mathbf{J}_d \equiv \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

[41] It's a misleading name given $\varepsilon_0(\partial \vec{E}/\partial t)$ has nothing to do with current, except that it adds to J in *Ampère's law*.

Let's see now how displacement current resolves the paradox of the charging capacitor in **Fig.** 5.1. If the capacitor plates are very close together,[42] then the electric field between them is:

[42] The calculation is simpler if we assume this.

$$E = \frac{1}{\varepsilon_0}\sigma = \frac{1}{\varepsilon_0}\frac{Q}{A},$$

where Q is the charge on the plate (C) and A is its area (m^2). Therefore, between the plates:

$$\frac{\partial E}{\partial t} = \frac{1}{\varepsilon_0 A}\frac{\mathrm{d}[\ ]}{\mathrm{d}Q}][t] = \frac{1}{\varepsilon_0 A} I.$$

Now, Eq. (**5.10**) reads, in its integral form as:

$$\oint \mathbf{B}\cdot \mathrm{d}\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0\varepsilon_0 \int \left(\frac{\partial \mathbf{E}}{\partial t}\right)\cdot \mathrm{d}\mathbf{a}.$$

If we were choose the *flat* surface, then $E = 0$ and $I_{\text{enc}} = I$. If, on the other hand, we use the balloon-shaped surface, then $I_{\text{enc}} == 0$, but:

$$\int \left(\frac{\partial \mathbf{E}}{\partial t}\right)\cdot \mathrm{d}\mathbf{a} = \frac{I}{\varepsilon_0}$$

So we get the same answer for either surface, though in the first case it comes from the conduction current, and in the second from the displacement current.

5.3.3 Maxwell's Equations

In the last section we put the finishing touches on Maxwell's equations:

$$\begin{aligned}
&\text{(i)} && \nabla\cdot\mathbf{E} = \frac{\rho}{\varepsilon_0} && \text{Gauss's law}\\
&\text{(ii)} && \nabla\cdot\mathbf{B} = 0 && \text{-}\\
&\text{(ii)} && \nabla\times\mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} && \text{Faraday's law}\\
&\text{(iv)} && \nabla\times\mathbf{B} = \mu_0\mathbf{J} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t} && \text{Ampere's law}
\end{aligned} \qquad (5.11)$$

Together with the force law,

$$\mathbf{F} = q\left(\mathbf{E} + \mathbf{v}\times\mathbf{B}\right).$$

summarise the entire theoretical content of classical electrodynamics.[43] Even the continuity equation,

[43] Except for some special properties of matter, which we have looked in the previous chapters.

$$\nabla\cdot\mathbf{J} = -\frac{\partial \rho}{\partial t}$$

which is the mathematical expression of conservation of charge, can be derived from Maxwell's equations by applying the divergence to number (iv).

In the aforementioned set of equations, we have written Maxwell's equations in the traditional way, which emphasizes that they specify the divergence and curl of **E** and **B**. In this form, they reinforce the notion that electric fields can be produced *either* by charges (ρ) or by changing magnetic fields ($\partial^{\mathrm{B}}t/\partial{-}NoValue{-}^{\mathrm{B}}$), and magnetic fields can be produced *either* by currents (**J**) or by changing electric fields ($\partial^{\mathrm{E}}t/\partial{-}NoValue{-}^{\mathrm{E}}$).

Actually, this is misleading, as $\partial\mathbf{B}/\partial t$ and $\partial\vec{E}/\partial t$ are *themselves* due to charges and currents, therefore it may be better to write these equations as:

$$\begin{aligned} &\text{(i)} \quad \nabla\cdot\mathbf{E} = \frac{1}{\varepsilon_0}\rho, && \text{(iii)} \quad \nabla\times\mathbf{E} + \frac{\partial\mathbf{B}}{\partial t} = 0, \\ &\text{(ii)} \quad \nabla\cdot\mathbf{B} = 0, && \text{(iv)} \quad \nabla\times\mathbf{B} - \mu_0\varepsilon_0\frac{\partial\mathbf{E}}{\partial t} = \mu_0\mathbf{J} \end{aligned}$$

with the fields (**E** and **B**) on the left and the sources (ρ and **J**) on the right. This notation emphasizes that all electromagnetic fields are ultimately attributable to charges and currents. Maxwell's equations tell you how *charges* produce *fields*; reciprocally, the force law tells you how *fields* affect *charges*.

5.3.4 Magnetic Charge

There is a pleasant symmetry to Maxwell's equations. It is particularly striking in **free space**, where ρ and **J** vanish:

$$\begin{aligned} &\text{(i)} \quad \nabla\cdot\mathbf{E} = 0, && \text{(iii)} \quad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii)} \quad \nabla\cdot\mathbf{B} = 0, && \text{(iv)} \quad \nabla\times\mathbf{B} = -\mu_0\varepsilon_0\frac{\partial\mathbf{E}}{\partial t} \end{aligned}$$

If we were to replace **E** by **B** and **B** by $-\mu_0\varepsilon_0 E$, the first pair of equations turns into the second, and vice versa. This symmetry[44] between **E** and **B** is broken, though, by the charge term in Gauss's law and the current term in Ampère's law.

[44] Dont be distracted by the constants μ_0 and ε_0; these are present only because the SI system measures E and B in different units, and would not occur, for instance, in the *Gaussian system*.

We can't help but wonder why the corresponding quantities are **missing** from $\nabla\cdot\mathbf{B} = 0$ and $\nabla\times\mathbf{E} = -\partial\mathbf{B}/\partial t$. What if we had (i) $\nabla\cdot\mathbf{E} = \frac{1}{\varepsilon_0}\rho_e$,
 (iii) $\nabla\times\mathbf{E} = -\mu_0\mathbf{J}_m - \frac{\partial\mathrm{B}}{\partial t}$,
 (ii) $\nabla\cdot\mathbf{B} = \mu_0\rho_m$,
 (iv)

$$\nabla\times\mathbf{B} = \mu_0\mathbf{J}_e + \mu_0\varepsilon_0\frac{\partial\mathbf{E}}{\partial t}$$

Then ρ_m would represent the density of magnetic "charge," and ρ_e the density of electric charge; J_m would be the current of magnetic charge, and J_e the current of electric charge. Both charges would be conserved:

$$\nabla\cdot\mathbf{J}_m = -\frac{\partial\rho_m}{\partial t}, \text{ and } \nabla\cdot\mathbf{J}_e = -\frac{\partial\rho_e}{\partial t}.$$

The former follows by application of the divergence to (iii), the latter by taking the divergence of (iv).

In a sense, Maxwell's equations strongly **imply** for magnetic charge to exist as it would fit just so perfectly. And yet, in spite of a diligent search, no one has ever found any [7]. As far as we know, ρ_m is zero everywhere, and so is J_m; **B** is *not* on equal foot- ing with E: there exist stationary sources for E (electric charges) but none for B.[45]

[45] This is reflected in the fact that magnetic multipole expansions have no monopole term, and magnetic dipoles consist of current loops, not separated north and south poles.

5.3.5 Maxwell's Equations in Matter

Maxwell's equations in the form given in Eq. (**5.11**) are complete and correct as they stand. However, when we are working with materials which are subject to **electric** and **magnetic** polarization there is a more convenient way to write them. For inside polarized matter there will be accumulations of **bound** charge and current, over which you exert no direct control. It would be nice to reformulate Maxwell's equations so as to make explicit reference only to the **free** charges and currents.

We have already learned, from the static case, that an electric polarization **P** produces a bound charge density

$$\rho_h = -\nabla \cdot \mathbf{P}$$

(Eq. 4.12). Similarly, a magnetic polarization[46] **M** results in a bound current:

[46] or simply **magnetisation**.

$$J_b = \nabla \times M$$

(Eq. 6.13). There's just one new feature to consider in the nonstatic case:

> Any *change* in the electric polarization involves a flow of (bound) charge, call it $\mathbf{J}_p$, which must be included in the total current.

For suppose we examine a tiny chunk of polarized material (Fig. 7.47). The polarization introduces a charge density σ_b = P at one end and $-\sigma_b$ at the other (Eq. 4.11). If P now *increases* a bit, the charge on each end increases accordingly, giving a net current

$$dI = \frac{\partial \sigma_b}{\partial t} da_\perp = \frac{\partial P}{\partial t} da_\perp.$$

The current density, therefore, is

$$J_p = \frac{\partial P}{\partial x} \tag{5.12}$$

This **polarization current** has nothing to do with the *bound* current $\mathbf{J}_b$. The latter is associated with *magnetization* of the material and involves the spin and orbital motion of electrons; J_p, by contrast, is the result of the linear motion of charge when the electric polarization changes. If **P** points to the right, and is increasing, then each plus charge moves a bit to the right and each minus charge to the left; the cumulative effect is the polarization current $\mathbf{J}_p$. We ought to check that Eq. (**5.12**) is consistent with the continuity equation:

$$\nabla \cdot \mathbf{J}_p = \nabla \cdot \frac{\partial \mathbf{P}}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{P}) = -\frac{\partial \rho_b}{\partial t}.$$

FIGURE 7.47

Yes: The continuity equation is satisfied; in fact, J_p is essential to ensure the conservation of bound charge. (Incidentally, a changing *magnetization* does not lead to any analogous accumulation of charge or current. The bound current $J_b = \nabla \times M$ varies in response to changes in M, to be sure, but that's about it.) In view of all this, the total charge density can be separated into two parts:

$$\rho = \rho_f + \rho_b = \rho_f - \nabla \cdot \mathbf{P},$$

and the current density into three (3) parts:

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p = \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}.$$

Gauss's law can now be written as

$$\nabla \cdot \mathbf{E} = \frac{1}{\varepsilon_0}\left(\rho_f - \nabla \cdot \mathbf{P}\right) \qquad \text{or} \qquad \nabla \cdot \mathbf{D} = \rho f \qquad \text{where, in static} \qquad \mathbf{D} \equiv \varepsilon_0 \mathbf{E} + \mathbf{P}$$

Meanwhile, Ampère's law (with Maxwell's term) becomes

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}\right) + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

or we can also write this expression as:

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \qquad \text{where, in static} \qquad \mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}.$$

Faraday's law and $\nabla \cdot \mathbf{B} = 0$ are not affected by our separation of charge and current into free and bound parts, since they do not involve ρ or $\mathbf{J}$. In terms of *free* charges and currents, then, Maxwell's equations read (i) $\nabla \cdot \mathbf{D} = \rho_f$, (iii) $\nabla \times \mathbf{E} = -\frac{\partial \mathrm{B}}{\partial t}$,
 (ii)

$$\nabla \cdot \mathbf{B} = 0 < /math >, (iv) < math > \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

Some people regard these as the "true" Maxwell's equations, but please under- stand that they are in no way more "general" than Eq. 7.40; they simply reflect a convenient division of charge and current into free and nonfree parts. And they 342 Chapter 7 Electrodynamics have the disadvantage of hybrid notation, since they contain both E and D, both **B** and **H**. They must be supplemented, therefore, by appropriate constitutive **relations**, giving D and H in terms of E and B. These depend on the nature of the material; for linear media

$$\mathbf{P} = \varepsilon_0 \chi_e \vec{E} < /math >, and < math > \mathbf{M} = \chi_m \mathbf{H}$$

_{so}

$$\mathbf{D} = \epsilon \vec{E} < /math >, and < math > \mathbf{H} = \frac{1}{\mu}\mathbf{B}$$

where $\epsilon \equiv \varepsilon_0(1+\chi_e)$ and $\mu \equiv \mu_0(1+\chi_m)$. Incidentally, you'll remember that D is called the electric "displacement"; that's why the second term in the Ampère/Maxwell equation (iv) came to be called the displacement current. In this context,

$$\mathbf{J}_d \equiv \frac{\partial \mathbf{D}}{\partial t} \tag{5.13}$$

5.3.6 Boundary Conditions

In general, the fields **E**, **B**, **D**, and **H** will be **discontinuous** at a boundary between two (2) different media, or at a surface that carries a charge density σ or a current density **K**. The explicit form of these discontinuities can be deduced from *Maxwell*'s equations given in Eq. (5.13), in their integral

form Applying (i) to a tiny, wafer-thin *Gaussian pillbox* extending just slightly into the material on either side of the boundary (Fig. 7.48), we obtain:[47]

$$\mathbf{D}_1 \cdot \mathbf{a} - \mathbf{D}_2 \cdot \mathbf{a} = \sigma_f a$$

Thus, the component of **D** that is perpendicular to the interface is discontinuous in the amount

$$D_1^{\perp} - D_2^{\perp} = \sigma_{\mathrm{f}}$$

Identical reasoning, applied to equation (ii), yields

$$B_1^{\perp} - B_2^{\perp} = 0.$$

Turning to (iii), a very thin Amperian loop straddling the surface gives

$$\overrightarrow{E}_1 \cdot \mathbf{l} - \overrightarrow{E}_2 \cdot \mathbf{l} = -\frac{d}{dt}\int_S \mathbf{B} \cdot \mathrm{d}\mathbf{a}.$$

But in the limit as the width of the loop goes to zero, the flux vanishes. (I have already dropped the contribution of the two ends to $\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l}$, on the same grounds.) Therefore,

$$\overrightarrow{E}_1^{\parallel} - \overrightarrow{E}_2^{\parallel} = \mathbf{0}.$$

That is, the components of E *parallel* to the interface are continuous across the boundary. By the same token, (iv) implies

$$\mathbf{H}_1 \cdot \mathbf{l} - \mathbf{H}_2 \cdot \mathbf{l} = I_{f_{enc}},$$

where $I_{f_{\mathrm{enc}}}$ is the free current passing through the Amperian loop. No *volume* current density will contribute (in the limit of infinitesimal width), but a surface current can. In fact, if $\hat{\mathbf{n}}$ is a unit vector perpendicular to the interface (pointing from 2 toward 1), so that $(\hat{n} \times l)$ is normal to the Amperian loop (Fig. 7.49), then

$$I_{f_{\mathrm{enc}}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \mathbf{l}) = (\mathbf{K}_f \times \hat{\mathbf{n}}) \cdot \mathbf{l},$$

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$$\hat{n} < /math >< math >\hat{}$$

T,

$$\langle 2 \rangle$$

$$\mathbf{K}_t$$

FIGURE 7.49 and hence

$$\mathbf{H}_1^{\parallel} - \mathbf{H}_2^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}.$$

So the *parallel* components of **H** are discontinuous by an amount proportional to the free surface current density. Equations 7.60-63 are the general boundary conditions for electrodynamics. In the case of *linear* media, they can be expressed in terms of E and B alone: (i) $\epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = \sigma_f$,
 (iii) $\overrightarrow{E}_1^{\parallel} - \overrightarrow{E}_2^{\parallel} = \mathbf{0}$,
 (ii) $B_1^{\perp} - B_2^{\perp} = 0$,
 (iv)

$$\frac{1}{\mu_1}\mathbf{B}_1^{\parallel} - \frac{1}{\mu_2}\mathbf{B}_2^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}$$

[47] The positive direction for **a** is *from* 2 *toward* 1. The edge of the wafer contributes nothing in the limit as the thickness goes to zero; nor does any *volume* charge density.

In particular, if there is no free charge or free current at the interface, then (i)

$$\epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = 0 < /math >, (iii) < math > \overrightarrow{E}_1^{\parallel} - \overrightarrow{E}_2^{\parallel} = \mathbf{0}$$

(ii) $B_1^{\perp} - B_2^{\perp} = 0$,
 (iv) $\frac{1}{\mu_1} B_1^{\parallel} - \frac{1}{\mu_2} B_2^{\parallel} = 0$. As we shall see in Chapter 9, these equations are the basis for the theory of reflec- tion and refraction. r

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Part V

Appendix

Part Contents

Acronyms **123**

The end of a melody is not its goal: but nonetheless, had the melody not reached its end it would not have reached its goal either.

A parable.

(Friedrich Nietzsche)

Acronyms

E

EMF Electro-motive Force

An energy transfer to an electric circuit per unit of electric charge, measured in volts. Devices called electrical transducers provide an EMF by converting other forms of energy into electrical energy. Other types of electrical equipment also produce an emf, such as batteries, which convert chemical energy, and generators, which convert mechanical energy. This energy conversion is achieved by physical forces applying physical work on electric charges. However, electromotive force itself is not a physical force

P

PDE Partial Differential Equation

Partial Differential Equation

PM Permanent Magnet

An object made from a material that is magnetized and creates its own persistent magnetic field



The document is designed with no intention of publication and has only been designed for education purposes.

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